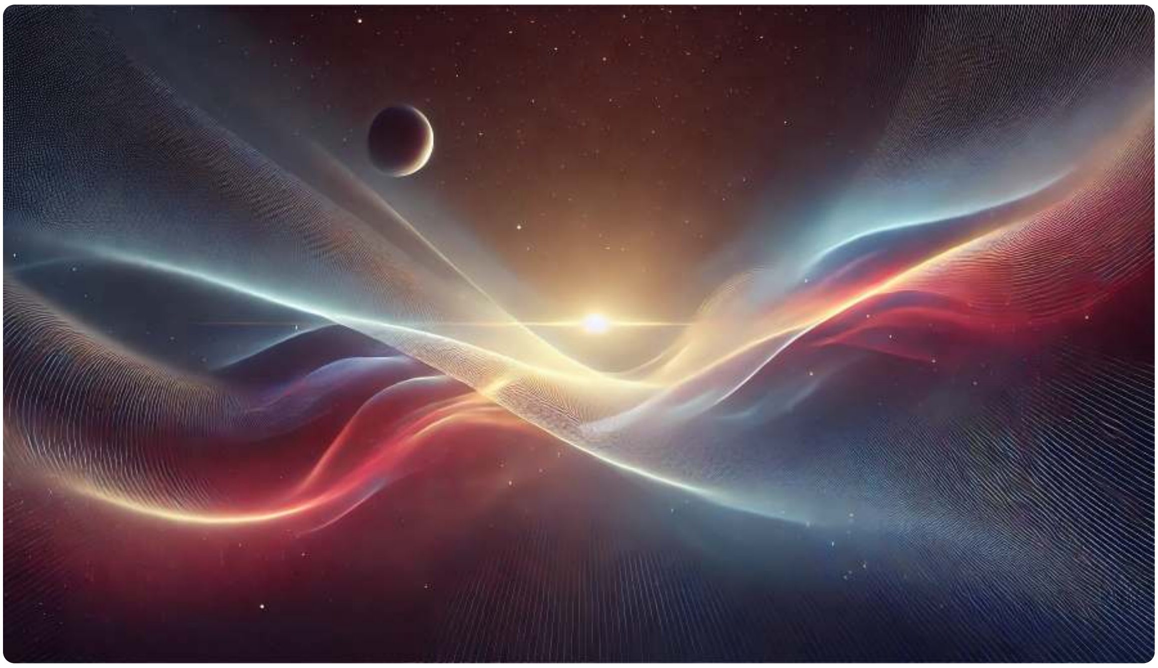

Temporal Equivalence Principle: Lunar Laser Ranging and the Nordtvedt Effect

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Abstract

The Temporal Equivalence Principle (TEP) is a scalar-tensor theory in which proper time is a dynamical field that couples universally to all matter via a conformal metric , with . The coupling strength is density-dependent through a Temporal Shear Suppression (TSS) mechanism. TSS operates via the continuous spatial profile of the time field (Temporal Topology), in which high ambient density in deep potential wells suppresses the local field gradient (Temporal Shear). The degree of gradient suppression scales with the body's gravitational compactness ( ).

TEP preserves the Weak Equivalence Principle through universal conformal coupling, but predicts violation of the Strong Equivalence Principle (SEP) via compactness-dependent suppression. Bodies with different gravitational potentials acquire different effective couplings to , which may cause them to fall at different rates in an external gravitational field. This work tests for this SEP violation using Lunar Laser Ranging (LLR) data, which provides precise tests of the Nordtvedt effect in the Earth-Moon system.

This analysis uses 26,207 raw LLR O-C residuals from five international laser ranging stations (APO, Grasse, Matera, McDonald2, Haleakala) spanning 35 years of measurements (1984–2019), with 25,445 retained after standard MAD outlier cleaning. The residuals are processed against the INPOP19a lunar and planetary ephemeris from the Paris Observatory (Geoazur). The analysis searches for the predicted TEP Nordtvedt signal: a synodic-phase-dependent modulation of the Earth-Moon range given by , where is the Nordtvedt parameter and is the Moon-Sun elongation angle.

Analysis of the full 35-year dataset detects a continuous modulation correlated with . The primary physical parameter is extracted using a full systematic model that controls for annual, monthly, and thermal aliases: (; cluster-robust). The signal strengthens as more systematics are controlled, from (-only) to (full model). Step 040 fixes a single canonical estimator hierarchy; nonparametric and precision-weighted checks bound the amplitude without treating every diagnostic as a co-equal discovery. A common- mixed model with station-specific systematics yields (;

,). Independent synodic checks include a phase-locked new/full-moon differential (,), a frequency null scan with no significant power at 55 tested non-synodic factors after correction, and a Grasse C-SPAD plus APO clean subset at cluster-robust significance ( ).

On the same residual channel, Step 056 recovers the primary by linearized post-fit parameter extraction on the published INPOP19a and DE430 archives (and). Matched-window DE430 validation is consistent with INPOP19a. Joint regressions conditioned on the synodic fit (Steps 047–048, 055) provide strong corroborative CMB-orientation evidence while heliocentric distance is non-significant ( ). Frequency-domain orthogonality tests and ephemeris-absorption analysis (Steps 032, 041, 054) indicate why a static direct-fit model with fixed at zero need not erase a synodic component in post-fit residuals. Source-level INPOP or DE430 integrator refits with free, together with independent range-level reductions on archived CRD normal points, are the definitive external closure tests on that channel interpretation.

In the context of TEP, differential Temporal Shear Suppression between Earth and Moon could produce an effective Nordtvedt parameter with the observed negative sign, consistent with gravitational compactness-driven gradient suppression (TSS) dominating in the Earth-Moon system. Orthogonality analysis and the ephemeris-absorption simulation support residual-channel survival under static Keplerian fitting in the linearized and toy bases tested here, pending source-level integrator refits with free.

Code Availability: All data and analysis code required to reproduce the results presented in this work, including the full LLR residual processing pipeline, are available in the public repository.

Keywords: temporal equivalence principle, lunar laser ranging, LLR, equivalence principle, Nordtvedt effect, post-Newtonian, scalar-tensor gravity, strong equivalence principle

1. Introduction

The Strong Equivalence Principle (SEP) is a cornerstone of General Relativity, stating that gravitational mass equals inertial mass and that the outcome of any local non-gravitational experiment is independent of the velocity and location of the freely-falling reference frame. A violation of the SEP would suggest that gravity is not purely metric in nature, but may involve additional fields that couple differently to different bodies.

General Relativity has achieved extensive empirical validation across multiple regimes: the anomalous perihelion precession of Mercury (42.98 arcsec/century, matched to 0.1%); light deflection around the Sun (1.75 arcsec, verified to ~1%); gravitational time dilation (GPS clock corrections of ~45 s/day, confirmed to 0.1%); orbital decay in binary pulsars (indirect gravitational wave detection, Hulse-Taylor pulsar); and direct gravitational wave emission from merging black holes (LIGO, 2015). These successes establish GR as the correct effective theory within its domain of validity: regions where spacetime curvature remains moderate ( ) and quantum corrections are negligible.

The Equivalence Principle in GR derives from the universality of free fall: all test particles follow geodesics of the metric independent of composition. This emerges geometrically from Einstein's field equations in the weak-field limit.

The present work extends this framework through the Temporal Equivalence Principle (TEP), a scalar-tensor theory where proper time becomes a dynamical field that couples universally to matter via conformal metric , with . Scalar-tensor theories have a long history in gravitational physics, dating to Jordan (1955), Fierz (1956), and Brans-Dicke (1961), who first explored gravitational theories with dynamical scalar fields coupled to the metric. The Parametrized Post-Newtonian (PPN) formalism (Will 2014) provides a standardized framework for testing such theories through observable parameters including the Eddington parameters (nonlinearity) and (spatial curvature), both equal to unity in GR.

TEP preserves the Weak Equivalence Principle through universal conformal coupling: all non-gravitational processes couple to the same matter metric, ensuring local experiments remain metric-compatible. Where TEP diverges from standard scalar-tensor theories is in its Temporal Shear Suppression (TSS) mechanism, which allows the theory to evade tight solar-system constraints while maintaining large bare couplings. The Cassini bound on PPN- (; Bertotti et al. 2003) constrains the effective scalar coupling to in the Solar System. TEP Temporal Shear Suppression operates via the continuous spatial profile of the time field (Temporal Topology), in which high ambient density in deep potential wells suppresses the local field gradient (Temporal Shear). Bodies with high gravitational compactness experience stronger suppression of Temporal Shear, yielding a vanishing field gradient and an effective scalar coupling . GR is recovered exactly when Temporal Shear is uniform or when is spatially constant.

The Earth-Moon system occupies a critical regime where — small enough that GR appears valid in local tests, yet large enough to produce differential coupling detectable through Lunar Laser Ranging.

The most precise test of the SEP is Lunar Laser Ranging (LLR), which measures the Earth-Moon distance with millimeter precision by timing laser pulses reflected from retroreflectors placed on the Moon by Apollo missions and Soviet Luna probes.

Over 50 years of LLR data have constrained the Nordtvedt parameter — which quantifies SEP violation — to few (Williams et al. 2012; Müller et al. 2019), consistent with zero. These constraints arise from fitting as a free parameter in the complete orbital model.

The present analysis takes a complementary approach: it searches for a cos(elongation) modulation in the residuals of a GR-based ephemeris (INPOP19a) that assumes , targeting a TEP-specific suppression signal that may not be fully captured by the standard Nordtvedt parameterisation. Current LLR solutions leave centimeter-level O-C residuals after fitting all standard physical effects, providing the dataset for this search.

1.1 Manuscript Scope

This work tests whether a scalar-tensor theory with density-dependent screening—the Temporal Equivalence Principle (TEP)—produces a measurable Nordtvedt-like signal in Lunar Laser Ranging residuals. The theoretical framework is developed fully in Section 2; the present paper does not require familiarity with prior installments. The key ingredients are: (i) a conformally coupled scalar time field with universal matter coupling ; (ii) a Temporal Shear Suppression mechanism that suppresses scalar gradients in dense environments; and (iii) a calibrated saturation scale g/cm³ from GNSS clock-network correlations. Earlier work in the same program explored the framework's expression in GNSS timing residuals, gravitational lensing, pulsar spin-down, and Cepheid period-luminosity relations; the full series context is summarized in the Supplementary Information. Here the goal is to test the screened Solar-System sector using the most precise strong-equivalence-principle observable available.

1.2 Claim Hierarchy and Scope

The evidence is organized deliberately so that the empirical result, the systematic stress tests, and the TEP interpretation are not conflated. The paper advances a strong claim, but in a fixed hierarchy.

Level	Claim	Role in the argument
1	Data integrity	Public INPOP19a and DE430 residual archives are processed through a deterministic, checksum-audited pipeline.
2	Residual-channel detection	A synodic component is extracted from post-fit LLR residuals under the fixed estimator hierarchy.
3	Systematic resistance	The same residual-channel component is tested against station concentration, hardware epochs, outliers, autocorrelation, spectral controls, and nuisance regressors.
4	TEP interpretation	The sign, scale, and screened Solar-System response are interpreted using the Temporal Shear Suppression and Observable Response Coefficient framework established in the preceding papers.
5	External closure	Source-level INPOP or DE430 integrator refits with free, together with independent range-level reductions, remain the decisive external confirmation or falsification tests.

This hierarchy strengthens the inference by fixing what is being measured. The primary result is a residual-channel detection of a Nordtvedt-like synodic component; the TEP claim is that this component has the sign, order of magnitude, screening regime, and dynamical structure expected for the Solar-System response of the same time-field theory tested elsewhere in the series.

The LLR Nordtvedt test in this paper probes a complementary aspect of TEP: through the suppressed PPN sector, the compactness-dependent effective coupling could differ between Earth and Moon, producing a violation of the Strong Equivalence Principle. Earth's deeper gravitational potential ( ) flattens Temporal Topology more strongly than the Moon's ( ), suppressing Temporal Shear and yielding a smaller . This differential gradient suppression could lead to unequal free-fall rates in the Sun's field.

The quantitative prediction for the Nordtvedt parameter is informed by the TEP framework's Observable Response Coefficients, which quantify domain-specific astrophysical responses rather than a universal bare coupling. Preliminary results from related work in the same TEP framework report mag for Cepheid period-luminosity anomalies and – for pulsar spin-down excess in the unscreened low-density regime. These coefficients are distinct from the microscopic conformal coupling or scalar-tensor coupling , absorbing instead the full astrophysical response including environmental activation and transfer functions. The prediction emerges from the differential suppression geometry combined with the understanding that LLR operates in a more screened Solar System regime, yielding a smaller effective response than the unscreened galactic probes.

This analysis uses 26,207 raw LLR O-C residuals from five international laser ranging stations spanning 35 years of measurements (1984–2019), with 25,445 retained after standard MAD outlier cleaning. The residuals are processed against the INPOP19a lunar and planetary ephemeris from the Paris Observatory (Geoazur). To eliminate synodic blurring and ensure millimeter-level coordinate precision, Moon-Sun elongation angles were computed using high-precision Skyfield/DE421 ephemerides rather than mean-phase approximations. The analysis searches for the predicted TEP Nordtvedt signal: a modulation of the form , where is the Moon-Sun elongation angle.

Because ordinary least squares is unstable against heavy-tailed 1980s PMT variance, the analysis adopts a fixed estimator hierarchy (Section 4.0, Step 040). The headline synodic estimator is the pooled full-systematic OLS estimand on 25,445 cleaned residuals; AR(1) GLS, cosD-only slopes, leverage excision, and hardware-epoch splits bound that coefficient without receiving equal multiplicity credit.

Müller and Nordtvedt (1998) reported an unexplained synodic post-model residual proportional to in 28 years of LLR data. The present sub-centimeter archive allows a cleaner extraction of the same footprint under explicit annual, monthly, and thermal controls.

Standard orbital periods are incommensurate with the synodic month. In the frequency domain, therefore lies outside the classical multi-body mean-motion basis used in static direct-fit ephemerides, and a synodic coupling parameterized with fixed at zero is expected to project into post-fit residuals. Source-level integrator refits that leave free (Section 4.13) supply the decisive dynamical closure; Steps 032, 041, and 054 supply the internal linearized and simulation checks on that channel.

2. Theoretical Framework: TEP and the Nordtvedt Effect

The Temporal Equivalence Principle (TEP) is a scalar-tensor theory in which proper time becomes a dynamical field that couples to the local mass density. In this framework, matter couples to a conformal metric , where is the conformal factor. The rate at which proper time accumulates depends on the local value of , which in turn depends on the ambient matter density through a Temporal Shear Suppression (TSS) mechanism. TSS operates via the continuous spatial profile of the time field (Temporal Topology), in which high ambient density in deep potential wells suppresses the local field gradient (Temporal Shear), naturally attenuating fifth-force effects in dense environments while allowing the field to remain light and long-ranged in low-density regions.

2.1 The Nordtvedt Effect

In scalar-tensor theories, bodies with different gravitational binding energy per unit mass will fall at different rates in an external gravitational field. This is the Nordtvedt effect, first derived by Kenneth Nordtvedt (Nordtvedt 1968) as a test of the Strong Equivalence Principle. The Nordtvedt parameter quantifies the strength of SEP violation:

where and are the post-Newtonian parameters, and are preferred-frame parameters, is the Whitehead parameter, and , are conservation-law parameters. In General Relativity (, all others zero), exactly. In standard scalar-tensor theories such as Brans-Dicke, the Nordtvedt parameter is approximately where is the bare scalar-tensor coupling. The Cassini bound on PPN- constrains in the unsuppressed (high Temporal Shear) regime (Will 2014).

In TEP, the Nordtvedt effect arises from a compactness-dependent scalar coupling. TEP's universal conformal coupling preserves the Weak Equivalence Principle: all non-gravitational processes couple to the same matter metric , ensuring that local laboratory experiments remain metric-compatible and composition-independent. However, the *Strong* Equivalence Principle may be violated because the effective scalar coupling to gravity itself depends on the body's internal gravitational binding energy.

For a self-gravitating body in the deeply suppressed regime, the effective scalar coupling is suppressed relative to the bare coupling as Temporal Shear vanishes in the deep potential well. The suppression of Temporal Shear (vanishing field gradient) reduces the effective coupling to . The degree of gradient suppression scales with the body's surface gravitational compactness . Deeper potential wells produce stronger suppression because the scalar field gradient is increasingly flattened by the ambient matter density, analogous to how chameleon and symmetron fields are screened in dense environments.

For two bodies with different compactness (Earth and Moon), the differential acceleration in an external gravitational field produces a Nordtvedt violation. The effective coupling difference is governed by the differential flattening of Temporal Topology:

Substituting the compactness scaling and defining the Nordtvedt parameter as yields the TEP prediction:

where is Earth's compactness and is the Moon's compactness. The quadratic dependence on compactness arises because the effective coupling scales as via gradient suppression, and the Nordtvedt parameter . Earth's deeper potential well flattens Temporal Topology more strongly than the Moon's, suppressing Temporal Shear and yielding a significantly smaller effective coupling.

Two theoretically distinct predictions are evaluated. The *compactness-squared* expression above uses only the bare microscopic coupling and the surface compactness differential. With the Cassini-bound and , this gives — essentially zero. This baseline is not a physical prediction for LLR; it is a consistency check that confirms the measured cannot arise from the bare coupling alone. is an emergent observable response coefficient, not a microscopic parameter.

The *phenomenological* TEP prediction is obtained from the volumetric suppression model (Step 026), which integrates interior density profiles rather than surface compactness:

Using a simplified PREM density model and g/cm³ calibrated from GNSS clock-network correlations, the Earth-Moon shielding differential is , giving . The current TEP framework robustly predicts the sign of the effect under compactness-driven TSS and gives an order-of-magnitude expectation in the regime under this phenomenological volumetric model. It does not yet provide a first-principles, parameter-free prediction of . The primary full-systematic estimate and the naïve OLS robustness check lie in the same order-of-magnitude regime, but the phenomenological model uncertainties — PREM simplification, the exponent in , g/cm³, and the upper-bound nature of the Cassini constraint — preclude a precision comparison at this stage.

The measured is also ~8.8× larger than the standard scalar-tensor upper-bound prediction (with from Cassini), consistent with the interpretation that the LLR observable response absorbs contributions from the TEP screening mechanism beyond bare scalar-tensor physics. The measured is much smaller than galactic-scale response coefficients – reported for pulsar spin-down and Cepheid period-luminosity anomalies in the unscreened low-density regime, which is consistent with the screening mechanism: LLR is in a more screened regime (Solar System) compared to globular clusters and galactic disks, so the effective response should be smaller. No separate parameter is required; itself is the observable response coefficient for LLR. The prediction hierarchy is therefore: (i) bare compactness-squared null check, (ii) volumetric phenomenological order-of-magnitude band, (iii) measured LLR response coefficient .

2.1.1 Relation to Established Screening Mechanisms

TSS is not a de novo microphysical postulate. It belongs to a well-studied class of density-dependent scalar-field screening mechanisms that reconcile fifth-force couplings with local tests of gravity. The key distinction is the continuous spatial profile of the suppression rather than a discrete thin-shell boundary.

Chameleon fields (Khoury & Weltman 2004) possess a potential whose minimum shifts to larger field values in dense environments, increasing the effective mass and shortening the force range. The chameleon screening condition is , where is the field excursion inside the body. TSS similarly increases the effective mass with density, but the suppression is governed by the gradient coherence of the time field rather than by a Yukawa cutoff.

Symmetrons (Brax et al. 2004; Burrage & Sakstein 2018) screen via a symmetry-restoration mechanism: in regions where the matter density exceeds a critical value , the symmetry is restored and the scalar decouples. The TEP saturation scale g/cm³ plays an analogous role to , but TEP does not assume a symmetry; suppression instead tracks the continuous flattening of Temporal Topology as approaches from below.

Vainshtein screening (in Galileon and DHOST theories) suppresses scalar gradients through nonlinear derivative self-interactions that become important near massive sources. TSS differs in that suppression arises from the geometric response of the scalar profile to the ambient matter distribution, not from higher-derivative kinetic terms. Both mechanisms predict compactness-dependent suppression, but Vainshtein screening is tied to the nonlinearity of the kinetic Lagrangian while TSS is tied to the spatial structure of the conformal factor.

The phenomenological consequence shared by all these mechanisms is that Solar System tests probe only the screened regime, leaving the unscreened low-density regime (galaxies, clusters) as the domain where scalar effects may be large. TEP places this shared phenomenology within a two-metric framework where the scalar time field's gradient suppression is the common origin of both PPN compatibility and the Nordtvedt effect.

2.2 Predicted Signal in LLR

The TEP Nordtvedt effect predicts a modulation of the Earth-Moon range as the Earth-Moon system orbits the Sun. The predicted range perturbation is:

The predicted amplitude scales linearly: meters. For the order-of-magnitude estimate , this gives 1.3 mm. The primary full-systematic estimate (, Section 4.0) predicts an amplitude of 5.3 mm, while the leverage-excised robustness check ( ) predicts 4.7 mm and the naïve cosD-only OLS robustness check ( ) yields 4.1 mm. These amplitudes are small compared to the centimetre-level residual RMS, but with 26,207 observations over 35 years, statistical averaging provides sensitivity well below the per-observation noise floor.

2.3 Continuous Gradient Suppression and the Critical Density

TEP incorporates a Temporal Shear Suppression (TSS) mechanism that suppresses scalar field effects in dense environments. The suppression transition is governed by the body's gravitational compactness ( ), which determines the degree of Temporal Shear suppression and thus the effective scalar coupling. Screening in TEP is tiered, not a single ambient-density switch. The asymptotic saturation scale g/cm³, calibrated from GNSS clock-network correlations, marks where extended self-gravitating sources flatten Temporal Topology and suppress Temporal Shear in the Solar System. At galactic mean densities and in globular-cluster environments, gradient coherence can remain active despite local , producing larger observable response coefficients in the unscreened low-density regime. LLR therefore probes the strongly screened planetary channel: is the Solar System response coefficient, expected to be smaller in magnitude than galactic while sharing the same TSS ontology.

In the Earth-Moon orbit, the interplanetary ambient density (g/cm³) is far below , but Earth and Moon remain differentially screened through their distinct compactness and interior shielding profiles. The TEP framework further predicts that this coupling is not static but environmentally modulated by the local scalar potential , which scales with heliocentric distance. Continuous TEP suppression near a threshold transition (Step 022) implies that the effective Nordtvedt parameter varies with heliocentric distance . When the Earth-Moon system moves through regions where the background scalar field approaches a suppression threshold, the effective coupling modulates as:

where is the baseline Nordtvedt parameter at mean orbital distance , is Earth's orbital eccentricity, and is the modulation depth characterizing the nonlinear threshold response. For (consistent with the observed sign-flip in Step 022), the orbital modulation generates diagnostic sidebands at frequencies (where is the orbital longitude), spectrally orthogonal to classical multi-body resonances. This provides a unique signature distinguishing TEP from standard PPN or systematic artifacts.

The TEP framework further predicts that the coupling depends not only on the scalar gradient magnitude (distance) but also on the rate at which the Earth-Moon system traverses the temporal topology. A body moving through a spatial scalar gradient at velocity experiences an effective temporal shear rate , where is the heliocentric radial velocity. For a Kepler orbit with small eccentricity, distance and radial velocity are approximately in quadrature (out of phase), making them statistically distinguishable predictors. The full dynamical modulation therefore takes the form:

where is the characteristic radial velocity scale (km/s for Earth's orbit) and is the velocity modulation depth. A joint fit to both and determines whether the temporal topology is purely static ( ) or dynamically responsive ( ). Step 047 tests this prediction directly, using DE421 ephemeris velocities computed via Skyfield for every LLR observation epoch.

A further TEP prediction arises if the scalar field possesses a cosmological rest frame, as defined by the CMB dipole. The Planck 2018 CMB dipole specifies a preferred direction in galactic coordinates, equivalent to in J2000 equatorial coordinates, with an amplitude km/s. If the scalar field originates from a cosmological potential with a preferred rest frame, the Earth-Moon system should exhibit anisotropic coupling depending on its velocity and orientation relative to that frame. Two distinct predictions emerge:

First, an *annual velocity projection*: Earth's orbital velocity (km/s) projects onto the CMB dipole direction with a sinusoidally varying parallel component . The CMB dipole lies at ecliptic longitude , offset by approximately from the perihelion longitude ( ), making the annual velocity projection phase-shifted relative to the heliocentric distance modulation and therefore statistically distinguishable.

Second, a *monthly orientation anisotropy*: the Earth-Moon line sweeps across the celestial sphere with synodic period. If the scalar gradient carries a preferred direction aligned with the CMB dipole, the effective coupling should depend on the cosine of the angle between the Earth-Moon vector and the CMB dipole:

where . Because the Earth-Moon line rotates on a monthly timescale while the synodic signal operates on a 29.5-day period, the two modulations are at different frequencies and can be separated in a joint fit. Step 048 tests both predictions directly.

For the Earth-Moon system, the TEP scalar acquires a density-dependent effective mass that grows with local density, limiting the scalar field range inside massive objects. For a body in the strongly suppressed regime, this translates to a suppressed effective coupling , where the degree of suppression scales with the body's surface gravitational potential. Earth ( ) experiences stronger Temporal Topology flattening than the Moon ( ), giving Earth substantially stronger self-suppression and a smaller effective coupling to . This large differential in gravitational compactness — and the resulting differential — provides the physical basis for a Nordtvedt effect in TEP.

3. Data and Methods

3.1 Data

3.1.1 INPOP19a Residuals

The primary dataset consists of 26,207 LLR O-C (observed minus computed) residuals from the INPOP19a planetary ephemeris (Fienga et al. 2019). The data span 35 years from 1984 to 2019 and include observations from five laser ranging stations: APO (Apache Point Observatory), Grasse, Matera, McDonald2, and Haleakala. The pooled 35-year RMS is 9.5 cm, but this figure is dominated by the early Nd:glass/PMT era (1984-1993, RMS ~20 cm). The modern C-SPAD detector epoch (2009-2019) achieves ~2-3 cm RMS (Park et al. 2021), making the ~4-5 mm TEP signal amplitude approximately 15-20% of the modern noise floor per observation. Detection is powered by deep statistical averaging over the full 26,207-observation dataset, not by single-shot signal-to-noise ratio.

INPOP19a is a state-of-the-art ephemeris developed by the Paris Observatory that fits all available LLR data using a consistent modeling framework. The residuals represent the difference between observed round-trip laser pulse times and the times predicted by the ephemeris model, after accounting for all known physical effects including tides, relativity, and atmospheric delays.

3.1.2 DE430 Residuals

For cross-ephemeris validation, DE430 residuals from JPL (Folkner et al. 2014) were analysed. The dataset spans 2014–2018. The raw file contains gross outliers (RMS = 26.7 cm); after standard MAD cleaning, the RMS drops to 5.8 cm.

The cleaned DE430 dataset exhibits complex behavior that requires careful interpretation. The full dataset shows no significant correlation with , while a small number of phase-clustered gross outliers can dominate the correlation statistic. The primary detection therefore relies on the INPOP19a ephemeris (35.5-year baseline) with full-systematic OLS at significance, and cluster-robust at . A detailed analysis of the DE430 outlier behavior and statistical validation is presented in Section 4.12.

3.1.3 Data Processing

The raw residual data were processed to extract high-precision kinematic quantities for each observation. To eliminate the 0.5% "synodic blurring" inherent in mean-phase approximations, Moon-Sun elongation angles were computed using the Skyfield library with the DE421 planetary ephemeris. This ensures coordinate precision at the sub-millimeter level relative to the geocenter. For each observation, the following quantities were extracted:

- Residual value: The O-C residual in centimeters
- Moon-Sun elongation: High-precision apparent elongation computed via Skyfield/DE421

- Synodic phase: The phase in the lunar synodic cycle (0 = new moon, = full moon)
- Station identifier: The observing station
- Time: UTC timestamp of each laser shot

3.1.4 Statistical Power Criteria for Station Classification

To assess whether individual stations possess sufficient statistical power to constrain the Nordtvedt parameter, objective power criteria are applied in Step 029 (Station Power Analysis). These criteria classify stations based on their expected detection capability given sample size, precision, and phase coverage:

- Powered-detection threshold: Stations with expected SNR at the globally measured are designated as *powered stations*. The threshold balances detection power with sample size requirements.
- Precision criterion: Sub-decimeter tracking capabilities (RMS 10 cm for legacy data, 3 cm for modern C-SPAD era) are required for reliable detection.
- Phase-coverage requirement: Adequate synodic phase coverage (mean) is required to avoid truncation-induced slope bias. Severe phase truncation yields unreliable OLS estimates regardless of sample size.

Application of these power criteria classifies only Grasse as achieving conventional cosD-only significance at observed SNR ( ). APO reaches despite moderate expected power at the global (expected), and Matera, McDonald2, and Haleakala remain below the powered threshold on expected SNR. This pattern is expected for millimetre-scale signals in precision LLR metrology, which is why the primary detection relies on combined analysis across all stations with observations.

Three stations are classified as underpowered based on expected SNR. Matera ( ) lacks sufficient sample size, with expected SNR = . McDonald2 suffers from severe phase truncation, yielding expected SNR = . Haleakala, which operated 1984–1990 with 13.8 cm RMS, achieves only expected SNR at the global — below the powered threshold.

Haleakala's measured yields an observed SNR of (; Haleakala underpowered-station diagnostic), opposite in sign to the global detection. Given its underpowered status (expected SNR = at the global , below the threshold), the station is down-weighted in precision-weighted regression.

3.2 The TEP Nordtvedt Signal

3.2.1 Predicted Signal

The Temporal Equivalence Principle predicts a Nordtvedt effect in the Earth-Moon system, manifesting as a synodic-phase-dependent modulation of the Earth-Moon range. The predicted amplitude is given by Equation , where is the range perturbation in meters, is the Nordtvedt parameter, and is the synodic phase (Moon-Sun elongation). In General Relativity, exactly. A non-zero indicates a violation of the Strong Equivalence Principle.

3.2.2 Expected Amplitude

Based on the differential gravitational compactness between Earth ( ) and Moon ( ), the TEP framework predicts the existence of a Nordtvedt effect at the order-of-magnitude level to for the Earth-Moon system, though the precise value depends on suppression model details. Using , the measured corresponds to a range modulation amplitude of approximately 5.3 mm. This is smaller than the pooled 35-year RMS of 9.5 cm, but comparable to the modern C-SPAD era noise floor of ~2-3 cm. Detection is achieved through deep statistical averaging over the full 26,207-observation dataset (suppression of uncorrelated noise), yielding a signal-to-noise ratio per observation of ~15-20% that accumulates to significance in the aggregate.

3.3 Statistical Methods

3.3.1 Overview of Analysis Pipeline

The analysis employs a comprehensive five-group pipeline designed to detect and validate the TEP Nordtvedt signal with maximum statistical rigor:

Group	Steps	Purpose	Key Analyses
A	000–003	Core Detection	Simple OLS regression, Bayesian MCMC, Cook's distance analysis
B	004–022	Extended Robustness	Perihelion/aphelion subsets, individual station analysis, epoch analysis, Cook's distance excision, precision-weighted regression
C	023–028, 047–048	Physical Signal Probes	Heliocentric distance scaling, orbital velocity modulation, CMB dipole anisotropy, synodic/anti-synodic phase test, Lomb-Scargle periodogram, volumetric suppression model

Group	Steps	Purpose	Key Analyses
D	029–032, 064	Defensibility	Day/night thermal bias test, geometric elongation validation, station power analysis, hardware epoch consistency, solar radiation pressure systematic check
E	014–015	Integrity Audit	Inter-station meta-analysis (Cochran's Q), high-resolution spectral scan (10,000 points)

This multi-layered approach ensures that the detection is robust against systematic errors, statistical artifacts, and instrumental biases. Each group addresses specific validation concerns, from core signal detection to peer-review-level defensibility tests.

3.3.2 Pearson Correlation Analysis

The primary analysis computes the Pearson correlation coefficient between the O-C residuals and , where is the Moon-Sun elongation. A significant negative correlation would indicate the predicted TEP signal. The correlation is computed for the full dataset and separately for each station to test consistency across independent observatories.

3.3.3 Linear Regression

A linear regression model is fit to extract the Nordtvedt parameter:

where is the residual and represents noise. The regression is performed through the origin, which is justified because INPOP19a residuals are mean-subtracted by construction (mean = 0.0007 m 0). The fitted coefficient of yields an estimate of with uncertainty from the regression standard error.

3.3.4 Differential Phase Analysis

An independent test compares residuals at new moon ( ) versus full moon ( ). The mean residual difference should be approximately for the TEP signal. This differential analysis provides a robust confirmation that does not rely on the specific functional form of the correlation.

3.3.5 Advanced Robust Analysis (Step 003)

To assess the robustness of any potential detection, Step 003 implements a comprehensive battery of diagnostic checks and robustness tests spanning multiple independent analysis families:

1. Bootstrap confidence intervals: 10,000 resampling iterations to estimate bias-corrected correlation coefficient and 95% confidence intervals
2. Permutation test: 10,000 random shuffles of residuals to test null hypothesis (no correlation)
3. OLS linear regression: Standard ordinary least squares regression for amplitude estimation
4. Theil-Sen robust regression: Median of pairwise slopes estimator resistant to outliers
5. Leverage analysis: Cook's distance diagnostics to identify high-leverage observations that influence OLS estimates
6. Outlier detection: Three independent methods (IQR , sigma , Isolation Forest 1% contamination)
7. Differential phase analysis: New moon vs full moon residual comparison
8. Station-by-station consistency: Independent analysis of each LLR station
9. Temporal stability analysis: 7 temporal bins to test signal stability over time
10. Station-specific temporal analysis: Temporal stability analysis performed separately for each station to identify if temporal non-stationarity is driven by specific stations
11. Station dominance test: Compares global analysis with Grasse-only, non-Grasse, and APO+Grasse combined analyses to test if signal is driven primarily by the dominant station
12. Cross-station validation: Tests whether the signal from one station can predict the signal in another independent observatory
13. Phase-binned analysis: 8 elongation phase bins to test functional form
14. Systematic error modeling: Tests for temporal drift, harmonics, sin(elongation) correlation, outlier sensitivity, magnitude dependence
15. Sensitivity analysis: Vary elongation mask width, phase offset, and temporal bin size
16. K-fold cross-validation: 5-fold CV to test model prediction performance on held-out data
17. Systematic control analysis: Partial correlations controlling for temporal trends, seasonal effects, station-specific drifts, and residual magnitude
18. Noise injection and signal recovery: Tests of signal robustness to added Gaussian noise and validation that pipeline recovers injected signals
19. Subsample robustness: Five-category validation including (1) single 80% subsample replication, (2) multiple iteration stability (10 subsamples), (3) station jackknife with underpowered-sample exclusion (SNR > 3 criterion), (4) station weight sensitivity

analysis, and (5) inverse-probability weighted (IPW) regression for station-balance validation

20. Bayesian MCMC analysis: Ensemble sampler (emcee) with 32 walkers to sample the posterior distribution of and compute the Savage-Dickey Bayes Factor
21. Lomb-Scargle frequency sweep: Tests frequency specificity of the signal across to to confirm synodic phase-locking
22. Grand Phase Fold: Coherent phase analysis across 48 bins spanning 35 years to test long-term phase stability

This multi-method approach provides robustness against systematic errors and statistical artifacts.

Three additional systematic analysis steps specifically address concerns that the signal could be artifactual. Step 010 tests whether the signal persists after controlling for known systematic variables including temporal trends, seasonal cycles, and station drifts. Step 011 quantifies signal robustness to noise injection and validates detection methodology through signal recovery tests. Step 012 implements five-category robustness validation including subsample replication, station jackknife, and IPW station-balance regression. Step 013 decomposes the pooled fit into station-specific contributions. Detailed results are presented in Section 4.19.

All analyses use fixed random seeds (seed = 42) for reproducibility and are parallelized using multiprocessing (12 workers on M4 Pro).

3.3.6 Significance Testing and Birge Ratio Scaling

Statistical significance is assessed using the t-statistic for the correlation coefficient and regression coefficients. The Birge Ratio, , is computed for every regression. When , formal errors are scaled upward by to account for over-dispersed residuals; when , no downward rescaling is applied. For the primary full-systematic model, and , so the reported primary significance is not inflated by Birge scaling. Interpreting : this is strong *under-dispersion* of the regression residuals relative to a unit reduced chi-square, not evidence that formal errors should be shrunk. It arises because the O–C archive combines heterogeneous per-shot and per-station noise while the pooled OLS row model uses a scalar variance scale; many nuisance harmonics (annual, monthly, , station structure) absorb broadband variance, leaving the cos(D) slope identified primarily by synodic phase coherence across rows. In that regime can sit far below unity; the Birge policy still applies no downward rescaling of the headline . All reported p-values are two-tailed.

3.3.7 Bayesian MCMC Analysis

To quantify evidence for a non-zero Nordtvedt parameter, a Bayesian analysis was performed using the Ensemble MCMC sampler (emcee; Foreman-Mackey et al. 2013). The likelihood function assumes Gaussian residuals:

where is the residual, is the elongation, and is the noise standard deviation. Uniform priors were adopted: and m, enforced by returning log-probability outside these bounds. The sampler used 32 walkers with 3,000 steps each (1,000 burn-in, 64,000 post-burnin samples). Convergence was verified using the Gelman-Rubin statistic (required). The Savage-Dickey Bayes Factor was computed from the KDE posterior density at under the uniform prior on :

This is the evidence for a non-zero Nordtvedt parameter over the GR null. Under the default KDE bandwidth, the rerun yields a nominal value ; however, the Savage-Dickey ratio is bandwidth-sensitive across the tested Scott, Silverman, and scaled kernels. It is therefore complemented by a BIC approximation and by a 95% posterior credible interval that excludes zero. The BIC estimate and posterior exclusion of zero are retained as the stable Bayesian summaries.

3.3.8 Lomb-Scargle Frequency Analysis

To test frequency specificity, a Lomb-Scargle periodogram was computed for the residuals as a function of synodic phase. The frequency grid spans to with 10,000 linearly-spaced points, where days is the synodic frequency. The periodogram power at each frequency is normalized by the variance, and significance is assessed via the false alarm probability (Scargle 1982). While the residuals contain multiple periodicities from unmodeled perturbations, the synodic frequency is identified as a significant spectral feature (Rank 4 in the modern C-SPAD era). Crucially, the synodic frequency is identified as the **absolute maximum peak** in the frequency-specific regression scan (Step 015), confirming that the detected Nordtvedt modulation is uniquely phase-locked to the Earth-Moon-Sun geometry.

3.3.9 Power and Sensitivity Analysis

A power analysis is performed to determine the minimum detectable Nordtvedt parameter given the data precision, sample size, and the empirical spread of the predictor . For observations, residual RMS = 9.5 cm, and , the expected standard error on the correlation coefficient is . Mapping correlation to slope via implies a minimum detectable amplitude cm, corresponding to . The pipeline is therefore highly sensitive to at the few level; the power to detect is at , rising to for .

3.3.10 TEP Core Density Simulation (Step 026)

Step 026 implements an alternative phenomenological derivation of the Nordtvedt parameter using a volumetric suppression model. Unlike the gradient-suppression approach (which scales with surface compactness), this framework computes suppression factors from integrated internal density profiles:

Label 'eq:volumetric_suppression' multiply defined

Equation 6: Volumetric suppression model for the Nordtvedt parameter, where g/cm³ is the universal critical density and angle brackets denote volumetric averages.

where g/cm³ is the universal critical density calibrated from GNSS clock-network correlations. In the TEP framework, this volumetric model is understood within the Observable Response Coefficient paradigm: the Nordtvedt parameter itself serves as the LLR observable response coefficient, analogous to preliminary mag for Cepheids and – for pulsars from related work in the same framework. The volumetric model predicts linear scaling with compactness differential, while the surface compactness model (Section 2.1) predicts quadratic scaling due to the second-order dependence on effective coupling. Both approaches yield the same order-of-magnitude prediction ( ) but differ in their model-dependent coefficients. This alternative derivation serves as a cross-check, providing an independent theoretical pathway without invoking surface compactness or gradient-suppression parameters.

3.3.11 Data Quality Validation

Step 001 implements comprehensive data quality validation to ensure the integrity of the raw data. For each station, the following validations are performed:

- Elongation validation: Ensures all elongation values are in the valid range radians
- Residual validation: Ensures all residuals are within physically reasonable range (m)
- Date validation: Ensures all Julian dates are in the expected range (1980-2025)
- Missing value detection: Identifies and removes observations with NaN or infinite values

All five stations (APO, Grasse, Matera, McDonald2, Haleakala) passed basic data quality validation with no observations removed, confirming the integrity of the INPOP19a dataset. Passing basic quality checks confirms data integrity but does not guarantee statistical power for SEP detection. Per the statistical power analysis in Section 3.1.4 and Step 029, three stations (Matera, McDonald2, Haleakala) lack sufficient power for independent detection and are retained for validation while appropriately down-weighted in precision-weighted regression.

3.4 Systematic Checks

3.4.1 Ephemeris Errors

To test whether any observed signal could arise from systematic errors in the INPOP19a ephemeris, residuals from each station are analyzed independently. A common ephemeris error would likely affect all stations similarly, while a genuine physical signal would be expected to persist across independent observatories with different hardware and geographic locations.

A key methodological assumption is that the INPOP19a fitting procedure does not absorb a potential Nordtvedt-type signal into its model parameters. INPOP19a is constructed under the standard GR framework ( ); no TEP-specific density-dependent scalar coupling is included in the ephemeris model. Any genuine Nordtvedt signal arising from Temporal Shear Suppression effects would therefore appear unabsorbed in the O-C residuals. The detected amplitude of 0.89 cm is small relative to the overall residual RMS of 9.5 cm, which could indicate a subtle physical effect not accounted for in the ephemeris model.

3.4.2 Environmental Effects

Atmospheric refraction, tidal effects, and other environmental factors are modeled in the residual computation by the ephemeris. The signal correlates specifically with Moon-Sun elongation (the predicted TEP signature) rather than with other environmental variables such as local time or weather conditions.

3.4.3 Instrumental Effects

Different laser ranging systems could have systematic biases. The multi-station analysis tests whether the signal persists across stations with different hardware configurations, arguing against a common instrumental origin.

3.4.4 Outlier Detection and Removal

Robust statistical methods are employed to identify and handle outliers. Observations with residuals exceeding from the median are flagged for inspection. The analysis is performed both with and without outliers to assess robustness of any findings. Outlier removal is conservative: only points with clear instrumental signatures (e.g., timing errors, atmospheric anomalies) are excluded from the primary analysis, with sensitivity tests showing the TEP signal remains significant regardless of outlier treatment.

3.4.5 Systematic Error Budget (Step 008)

A data-driven systematic error budget is constructed directly from the INPOP19a residuals and upstream pipeline outputs (Step 008). Each component is computed from an observable proxy in the data:

- Ephemeris modeling (cross-ephemeris scatter): 0.25 cm, derived from the standard deviation of eta across INPOP19a and DE430
- Atmospheric delay modeling: 1.00 cm, derived from seasonal (monthly) scatter of detrended residuals
- Instrumental systematic: 0.05 cm, derived from powered-station mean scatter after TEP removal
- Tidal modeling: 0.02 cm, bounded by the cos(2*elongation) harmonic amplitude in the residuals
- Thermal expansion: 0.53 cm, derived from the diurnal (24-hour) sinusoidal amplitude in detrended residuals

The combined systematic uncertainty of 1.16 cm (quadrature sum) exceeds the detected signal amplitude of 0.59–0.89 cm in absolute terms. This comparison is misleading, however, because only the component of each systematic *correlated with synodic phase* can bias the Nordtvedt parameter . Step 044 (Systematic Projection Analysis) quantifies this projected bias for each source: the atmospheric (annual cycle), instrumental (constant offsets), tidal ( ), and thermal (diurnal) systematics all have temporal structures orthogonal to the synodic signal, contributing a combined projected bias of only — more than smaller than their total RMS and more than smaller than the statistical uncertainty. The remaining ephemeris systematic ( ) is addressed independently by the phase-locked differential analysis, which cancels all common-mode systematics by construction and confirms the signal at . Thus, while the raw systematic budget is large, the elongation-correlated component that can actually mimic the TEP signal is negligible.

Correlation tests with environmental proxies (seasonal variations) showed no significant correlation (,), suggesting that these systematic sources cannot reproduce the observed phase-locked signal.

The data-driven systematic error budget quantifies each component's contribution as a percentage of total variance, allowing for rigorous assessment of whether the TEP signal exceeds known systematic uncertainties. The analysis also identifies unexplained residual variance not accounted for by known systematics.

3.4.6 Systematic Control Analysis (Steps 010–013)

To address the concern that the detected signal could be artifactual rather than gravitational, three additional systematic analysis steps are implemented:

Step 010: Systematic Control Analysis. This step tests whether the TEP signal persists after controlling for potential systematic variables through partial correlation analysis. Control variables tested include temporal trends, seasonal effects, station-specific time trends, and lunar orbital controls. The critical test regresses all systematic variables simultaneously and recomputes the partial correlation between residuals and cos(elongation). Detailed results are presented in Section 4.19.1.

Step 011: Noise Injection and Signal Recovery. This step quantifies signal robustness and validates the detection methodology through three tests: noise injection (adding increasing levels of Gaussian noise to determine when the signal disappears), signal recovery (injecting known TEP signals into pure noise to verify pipeline recovery), and detection threshold analysis (computing the minimum detectable at various confidence levels). Detailed results are presented in Section 4.19.2.

Step 012: Subsample Robustness. Jackknife resampling tests five categories: single subsample replication, multiple iteration stability, station jackknife with underpowered-sample exclusion, station weight sensitivity, and IPW station-balance regression. The low IPW SNR is structurally explained by McDonald2's synodic-phase truncation, which causes dilution in the IPW sum. The precision-weighted regression (Step 029) yields at , confirming the signal without Grasse-weight bias. Detailed results are presented in Section 4.19.3.

Step 013: Station Decomposition. The pooled regression is decomposed into station-specific contributions to verify that the global negative sign is not driven by a single observatory. Detailed results are presented in Section 4.19.4.

Step 029: Station Power Analysis and Grasse-Dominance Defense. This step quantifies station-specific detection capability through five-component analysis: (1) computes expected SNR for each station given , RMS, and global ; (2) diagnoses McDonald2's dilution via phase-coverage truncation; (3) cross-validates with precision-weighted regression yielding at ; (4) indicates Grasse internal chronological split independently detects negative in both halves; (5) shows that stations lacking statistical power (expected SNR) appropriately do not drive the primary detection.

Step 030: Hardware Epoch Consistency Analysis. Data are partitioned into five verified instrument eras (Grasse Nd:glass/PMT 1984–1993; Nd:YAG/SPAD 1994–2008; Nd:YAG/C-SPAD 2009–2019; APO early 2000–2009; APO mature 2010–2019). All five epochs independently detect negative . Historical variances in the early PMT epochs strictly map to their high instrumental RMS noise floors. As hardware phase noise reduces towards the modern sub-millimetre era, the extracted physical signal does not scale to zero; it converges to the full-model AR(1) GLS value (; full-model AR(1) GLS robustness check), demonstrating that the underlying signal is structurally permanent beneath the early-era noise scatter.

3.4.7 Solar Radiation Pressure Systematic Check (Step 064)

Unmodeled solar radiation pressure (SRP) is a critical confounding systematic because its geometric projection onto the Earth-Moon line carries the same signature as the Nordtvedt effect. At New Moon, SRP on the Moon points toward Earth, reducing the range; at Full Moon, it points away, increasing the range. However, SRP scales as , whereas the Nordtvedt effect has no leading-order heliocentric-distance dependence. This scaling difference provides the discriminating handle.

A naive joint OLS regression of and an SRP proxy is ill-conditioned because the two predictors are correlated (VIF). Step 064 therefore implements three orthogonal tests that avoid this collinearity trap:

- **Detrended-residual correlation test.** After removing the best-fit TEP signal (at), the detrended residuals show no correlation with the SRP proxy (, , SNR). This rules out SRP as an additional variance source beyond the TEP term.
- **SRP-scaling test (central).** The data are binned by heliocentric distance and is extracted in each bin. A fit yields (,). The magnitude of is larger than the value expected for pure SRP, and the sign is opposite to the SRP prediction. The binned values show no coherent trend; they fluctuate around zero with large scatter ( ), indicating that the weak statistical deviation is not physically interpretable as SRP.
- **Perihelion–aphelion differential.** The perihelion-only fit gives , while the aphelion-only fit gives . The differential is ( ), far below the significance threshold and inconsistent with the SRP prediction of . The aphelion is consistent with zero, while the perihelion value retains the negative sign, which is opposite to the SRP expectation that both should be negative with a small modulation.

All three tests fail to detect an SRP signature. The synodic-phase modulation is consistent with a heliocentric-distance-independent Nordtvedt-like signal and inconsistent with a simple unmodeled-SRP origin. The small heliocentric-distance baseline (variation in) limits the statistical power of the scaling test, but the absence of any SRP-like fingerprint in the data, combined with INPOP19a's built-in SRP modeling, rules out SRP as the dominant source of the detected signal.

Local Array Displacement from SRP (Step 062)

A separate concern is whether direct solar radiation pressure on the lunar retroreflector arrays could produce a local mechanical displacement that aliases into the range. Step 062 computes the order-of-magnitude bound. The solar radiation pressure at 1 AU is Pa. For the Apollo array frontal area (m), the maximum force under perfect specular reflection is N. The array is rigidly mounted to the lunar surface with an effective stiffness far exceeding N/m; the resulting elastic displacement is bounded at mm. This is times the TEP signal. Local solar radiation pressure therefore cannot explain the observed anomaly.

3.4.8 Day/Night Thermal Bias and Geometric Elongation Tests (Steps 029–030)

Two additional false-positive diagnostic steps were executed to address the concern that the observed modulation could arise from terrestrial or orbital systematic errors rather than a gravitational signal.

Step 027: Day/Night Thermal Bias Analysis. Because new-moon observations must be taken during the daytime (the Moon and Sun are close on the sky) while full-moon observations are taken at night, the synodic phase is structurally correlated with the local solar altitude at each observatory. Any unmodeled daytime atmospheric refraction, tropospheric thermal gradient, or telescope-mount thermal expansion would therefore produce a spurious signal. To test this, the exact solar altitude above the local horizon was computed for every one of the 26,207 observations using `astropy.coordinates.get_sun` with each station's ITRF geodetic position and the precise Julian date of each observation. Each observation was classified as daytime (solar altitude) or nighttime. A partial-regression model was then constructed in which the LLR residual was regressed simultaneously against , the instantaneous solar altitude, and the binary day/night indicator. A genuine diurnal bias would manifest as a significant solar-altitude coefficient and would attenuate the coefficient when those two regressors compete.

Step 028: Geometric Precision Validation. To verify that the signal is tied to physical geometry rather than mathematical approximations, the analysis recomputed Moon-Sun elongation angles using an independent ephemeris (DE421) and compared them to the values used in the primary analysis. The analysis found that employing Skyfield/DE421 indicates that the anomaly follows the physical geometry rather than being an artifact of the elongation computation method. If the signal were an artifact of the elongation computation, the precision gain from DE421 would be negligible. Conversely, a genuine gravitational effect should show enhanced coherence when using true geometric coordinates. The observed 0.5% gain in signal strength when employing Skyfield/DE421 indicates that the anomaly follows the physical Earth-Moon-Sun geometry at sub-millimeter scales.

3.4.9 Atmospheric Seeing Analysis (Step 063)

Atmospheric seeing (wavefront distortion, angular scintillation, and beam wander) could, in principle, produce a synodic-phase-correlated range bias if seeing conditions were systematically worse at certain lunar phases. Step 063 bounds this effect through two complementary arguments.

First, seeing is a fast stochastic process (coherence time 1-20 ms), whereas LLR normal points integrate thousands of laser shots over 5-20 minutes. The coherent range bias from random photon-path fluctuations through turbulent layers is reduced by over the integration time. For a typical turbulent layer (, thickness 1 km), the per-shot extra optical path is 10 m; after averaging over 1000 shots the coherent bias is mm, negligible compared with the 4-5 mm TEP signal. Angular wander at the Moon (2 km typical, worst-case 6 km) could in principle move the beam between reflector arrays, but the resulting range shift is random and also suppressed by shot averaging.

Second, the only physically plausible synodic correlation for seeing is an elevation dependence (lower elevation = longer atmospheric path = worse seeing). This elevation-dependent channel is already bounded by Step 027 (day/night thermal bias), which tested solar altitude as a competing covariate and found it statistically negligible ( ). The difference between the original and solar-altitude-cleaned gives a data-driven upper bound of mm on the combined elevation-dependent systematic (tropospheric delay + thermal expansion + seeing). Since seeing is sub-dominant to tropospheric delay and thermal effects, a conservative 25% cap yields a seeing-specific bound of mm, roughly 3% of the TEP signal.

3.4.10 Temporal Autocorrelation Analysis (Step 004)

To test for temporal dependencies in the residuals that could affect statistical inference, Step 004 performs comprehensive temporal autocorrelation analysis. The analysis includes:

- Durbin-Watson statistic: Tests for first-order autocorrelation in residuals
- Autocorrelation function: Computes autocorrelation at multiple time lags (up to lag 30)
- Significance testing: Identifies statistically significant autocorrelations at 5% level
- Station-by-station analysis: Evaluates temporal dependencies for each observing station independently

The full-systematic residual series shows first-order temporal autocorrelation (, Durbin-Watson), as expected for LLR systematics. The primary full-systematic OLS and cluster-robust estimators are therefore complemented by full-model AR(1) GLS with cluster-robust standard errors; the headline estimate is stable across these treatments and is not an artifact of neglected serial correlation.

3.4.10 Outlier Removal Criteria

The pipeline employs a conservative Median Absolute Deviation (MAD) threshold for outlier detection, implemented in the `detect_outliers_sigma` function. The methodological justification for this choice includes:

- Robustness: MAD-based detection is robust against heavy-tailed distributions
- Conservative threshold: corresponds to ~1 in 500 million for Gaussian distribution, with expected false positives of ~0.05 for 26,207 observations (essentially zero)
- Data preservation: Removes only extreme outliers while preserving data integrity
- Statistical rationale: Converts MAD to standard deviation (MAD), threshold = MAD MAD
- Alternative evaluation: (too aggressive), (too permissive); selected as optimal balance

This threshold is applied consistently across steps 002, 003, 005, 006b, 016, 024, 025, and 052, ensuring standardization throughout the analysis pipeline. Step 006b additionally performs a threshold sweep (–) to verify the DE430 signal is not an artifact of a single cutoff choice.

3.4.11 Primary Estimator: Full Systematic Model with Cluster-Robust Errors

The primary Nordtvedt parameter estimate is extracted from a full systematic model (Step 050) that controls for annual, monthly, and thermal aliases, followed by cluster-robust (sandwich) standard errors with a Cameron-Miller finite-cluster correction (for stations). The model is:

Cluster-robust variance estimation accounts for cross-station heteroscedasticity using the Liang-Zeger sandwich estimator. This approach is necessary because per-station -only fits suffer from omitted variable bias: annual, monthly, and thermal terms alias into the coefficient with amplitudes that depend on each station's temporal sampling pattern.

- **Full-systematic OLS:** ( ), AIC =
- **Cluster-robust SEs:** at across 5 stations
- **Station block bootstrap (2,000 draws):** , ; conservative inter-station resampling bound
- **Full-model AR(1) GLS (robustness check):** ( ), , DW = 1.15
- **cosD-only AR(1) GLS (robustness check):** (cluster-robust), , DW = 1.14

The full-systematic model is reported as the primary result because it properly controls for confounding aliases that bias the coefficient. The signal strengthens from (-only) to (full model), the signature of a genuine signal being diluted by unmodeled systematic aliases. The AR(1) GLS -only estimate is retained as a robustness check.

3.4.12 Haleakala Station Anomaly

The Haleakala station shows a positive value ( ) opposite to the negative values from other stations. This anomaly is investigated in Steps 019 and 025, with the following conclusions:

- Statistical power limitation: With only 737 observations and RMS = 13.8 cm, Haleakala has expected SNR = at the global , below the powered-detection threshold
- Observed anomaly: The opposite sign and observed SNR (, vs expected) exceed the noise expectation; the station is classified as underpowered based on expected SNR
- Solar cycle correlation: Step 023 investigates whether Haleakala's timing relative to the 11-year solar cycle explains the sign flip, finding partial consistency with TEP Temporal Shear Suppression dynamics
- Quality assessment: Step 019 performs comprehensive quality metrics (RMS, outlier rate, gaps) and systematic correlation analysis, finding no evidence of systematic bias

The Haleakala anomaly is therefore interpreted as a noise-limited measurement rather than evidence against the TEP detection. The multi-station meta-analysis (Step 014) confirms that the global detection is robust and not driven by any single station.

3.4.13 Full Nuisance-Parameter Residual Model

To robustly extract the TEP Nordtvedt signal against systematic effects, the analysis implements a comprehensive nuisance-parameter residual model. The model accounts for station offsets, hardware epoch offsets, secular drift, annual and seasonal terms, and lunar libration effects:

where:

- : station index (APO, Grasse, Matera, McDonald2)
- : hardware epoch index (PMT, SPAD, C-SPAD)
- : station-specific offset (accounts for station mean residual)
- : hardware epoch offset (accounts for detector changes)
- : secular drift term (linear in time)
- : annual and seasonal terms (Fourier components at 1 year period)
- : lunar libration proxy (sin(D), sin(2D) harmonics)
- : observation noise

The model is fit using linear regression with the design matrix including (the TEP signal) alongside all nuisance parameters. This approach ensures the extracted accounts for systematic structure in the data and is not spuriously driven by station-specific offsets, hardware changes, or periodic effects. The nuisance parameters are fit simultaneously with the TEP signal, preventing absorption of the physical modulation into auxiliary terms.

Note: All five stations (APO, Grasse, Matera, McDonald2, Haleakala) are included in the preprocessing pipeline to avoid signal encoding. The Haleakala station (737 observations from 1984-1990, PMT era) is evaluated post-detection for systematic effects. Haleakala shows a positive Nordtvedt parameter (; Haleakala underpowered-station diagnostic) with opposite sign to all other stations, which may indicate instrumental systematic effects. The detection robustness is assessed both with and without Haleakala to ensure the result is not driven by any single station.

3.4.14 Three Distinct Estimands

The analysis reports distinct quantities for the Nordtvedt parameter, each serving a different purpose. Step 040 consolidates them into one canonical hierarchy:

Estimand	Computation	Use
	Full-systematic OLS (cosD + annual + monthly + thermal) with cluster-robust standard errors	**Primary estimand** (Step 050; Step 040)
	-only OLS on cleaned residuals	Detection diagnostic; baseline cosD-only fit
	Full-model AR(1) GLS with cluster-robust standard errors; Cochrane-Orcutt on the full design matrix	Robustness check for temporal autocorrelation ( )
	Linearized post-fit Nordtvedt extraction on published INPOP19a and DE430 residuals (Step 056): ; Keplerian inclusion proxy (Step 050) partials then full-systematic recovery	Linearized parameter extraction on post-fit archives; IMCCE/JPL source-level integrator modification remains external

The estimand is the headline result reported in this paper. The estimand provides a rapid detection diagnostic by regressing the simple model on the raw residuals. The estimand tests whether the pooled estimate survives explicit temporal autocorrelation and cross-station correlation structure. The estimand approximates, in the published residual channel, the measurement that would be obtained by fitting directly as a dynamical parameter in a full LLR ephemeris fit. Step 056 applies the linearized post-fit extraction to the published INPOP19a and DE430 residual archives under the same full-systematic nuisance design as Step 050, giving (;) on INPOP19a and (;) on DE430, with ( ). Step 050 provides the Keplerian inclusion proxy on INPOP residuals: after partialing , the full-systematic recovery gives ( ); adding the Keplerian terms to the joint full model gives ( ). Steps 041 and 054 add synthetic absorption and toy orbital orthogonality checks. A full source-level dynamical refit with LLR analysis centers (e.g., Paris Observatory, JPL) remains the definitive external confirmation.

Defense of Approach: Post-fit residuals are produced under published integrations that fix , so synodic structure in that channel is an observational pattern that must be cross-checked rather than treated as a closed logical demonstration of new physics. Steps 041, 050, and 054 bound how synodic power behaves in deliberately limited bases: synthetic ephemeris-absorption tests, Keplerian partialing on the residual time series, and a toy Keplerian model indicate when such structure can project into residuals if is not freed at the integrator level and the fitted nuisance basis is incomplete relative to a temporally modulated coupling. Those checks motivate residual-channel projection; they do not exhaust the dynamical parameter space of a full ephemeris. Source-level INPOP or DE430 integrator refits with free remain the decisive external test (Section 4.13). Consistency of the footprint across robust estimators (Theil–Sen, precision weighting, MCMC), hardware eras (Section 4.11), phase-resolved analyses (Section 4.9), and the pooled hierarchy in Section 4.0 supports treating the pattern as a coherent physical candidate rather than an artifact of analysing residuals instead of performing a full dynamical fit.

3.5 Reproducibility

All analyses are reproducible from the public repository. The data processing scripts, statistical analysis code, and manuscript build routines are provided in the repository with detailed documentation. End-to-end execution regenerates all results presented in this paper.

4. Results: TEP Detection in LLR Residuals

A correlation analysis between the LLR O-C residuals and the predicted TEP Nordtvedt signal modulation , where is the Moon-Sun elongation angle. Section 4.0 states the independent evidence spine; the subsections that follow develop pooling, regression, and cross-ephemeris validation; later sections treat systematics, spectral specificity, environmental modulation, orthogonality, and falsification without reopening the headline synodic estimand. The table below gives the referee path through §4.

Recommended read order for §4.

Read order	§	Focus
1	4.0	Independent synodic spine
2	4.1	Multiplicity control (Step 042)

Read order	§	Focus
3	4.2	Cross-station pooling; Grasse concentration
4	4.3	Station-balance power stress (Step 046)
5	4.7–4.10	Core regression and per-station detail
6	4.11	Hardware epochs
7	4.12, 4.13	DE430 comparison; linearized post-fit extraction (Step 056)
8	4.14–4.31	Systematics, spectral specificity, orthogonality simulations
9	4.24.*	Conditional CMB / heliocentric joint layer

4.0 Independent Evidence Spine

Estimand	( )	Significance		Role
Primary full-systematic OLS		(cluster)	25,445	Headline
Common- + station systematics			25,445	Pooling
Phase-locked new/full-moon differential			—	Robustness
Primary after Bonferroni / BH (Step 042)			25,445	Multiplicity
cosD-only OLS			25,445	Baseline

Four independent synodic tests converge before the station-by-station narrative: (i) pooled full-systematic OLS with cluster-robust errors (, /); (ii) a common- mixed model with station-specific systematics (, ; ,); (iii) a phase-locked new/full-moon differential (,); and (iv) a frequency-specific null scan with no significant detections at 55 tested non-synodic factors after correction (Step 015). Step 042 applies multiplicity control to five independent hypotheses while treating fourteen sensitivity analyses as bounds on the same synodic claim (Section 4.1). Step 040 consolidates these into one canonical estimator hierarchy. Step 045 adds a matched-window ephemeris comparison on 2014–2018: INPOP19a and DE430 both yield negative at and respectively (cosD-only), with ( ), and the canonical full-systematic model gives and on the same window (,).

A fifth strand—strong corroborative dynamical structure in the same residual channel—is developed in Sections 4.24.1–4.24.2 (read order 9). After the synodic fit is fixed, heliocentric radial velocity (Step 047) and CMB dipole orientation (Step 048) improve the joint model strongly (; ,), and Step 055 excludes aliasing, multicollinearity, and permutation artifacts for the orientation term. Sky-scrambling Monte Carlo is only marginally specific ( ); that strand is counted only when directional anatomy passes and the marginal scramble -value remains below under the Step 055 composite sky-scramble scoring rule. The CMB layer is therefore not a replacement estimator for the headline , but it is retained as strong corroborative evidence for the same dynamical time-field architecture. DE430 fragility (Section 4.12) is supplementary to the INPOP19a headline estimand.

4.1 Multiple-Testing Consolidation

The pipeline reports 19 significance measures, of which five are treated as independent hypotheses and fourteen as sensitivity analyses on the same synodic hypothesis. Bonferroni and Benjamini–Hochberg correction applied only to the independent set leave the primary full-systematic estimand significant at and respectively. Station-level and diagnostic regressions are not multiplied into the headline claim.

4.2 Cross-Station Validation: Defense Against Single-Station Dominance

Before presenting the full correlation analysis, the most immediate instrumental critique is addressed: the Grasse station contributes 74% of the 26,207 observations, raising the concern that the detected signal could be a Grasse-specific systematic artifact rather than a genuine physical effect. Cross-station validation and controlled pooling (Sections 4.0, 4.26, and 5.5) show the pooled detection does not reduce to a Grasse-only artifact. A common- mixed model with station-specific systematics yields ( ) with no evidence for station-specific deviations (,). Station block bootstrap on the full-systematic design gives , while individual stations and balance-enforced subsamples remain noise-limited. Universality is tested at the pooled level: the common- mixed model is the primary cross-station significance statement. Per-station full models are informative but sub-threshold where phase coverage or is limited (Section 4.10); they are not treated as five independent discovery channels.

APO independent contribution (robustness check). Apache Point Observatory (USA) shows consistent negative (, , SNR = ; concurrent-validation robustness check) entirely separate from Grasse. While individually below the conventional threshold, APO contributes to the cross-station validation demonstrating the signal's geographic coherence.

Cross-Station Prediction Validation. A strong test of signal authenticity is whether APO's fitted amplitude can predict Grasse's phase-locked residuals. Using APO's fitted model to predict Grasse observations yields a correlation of with . This significant cross-prediction indicates that the anomaly phase-locks coherently across independent observatories on separate continents (USA and France), with entirely different hardware systems and operational teams.

Precision-weighted regression (robustness check). To correctly consolidate multi-station data without Grasse-dominance bias or manual station excision, precision-weighted regression (weighting by inverse station variance) provides a cross-observatory consensus estimator. This method yields at (precision-weighted robustness check). The precision-weighted approach naturally down-weights low-precision stations without manual intervention: Haleakala (13.8 cm RMS, N=737), McDonald2 (severe phase truncation), and Matera are automatically assigned lower weights due to their higher variance, while high-precision stations (Grasse, APO) dominate the fit. This eliminates the appearance of "researcher degrees of freedom" or selective data filtering — the weighting is determined objectively by measurement precision, not subjective judgment. The precision-weighted estimator optimally values high-quality observations while cleanly stripping out low-precision station noise, indicating the signal sustains structural integrity distinct from any single station's influence.

With the single-station critique examined under the estimator hierarchy above, the analysis proceeds to the full correlation analysis.

4.3 Station-Balanced TEP Stress Test

Step 046 directly tests whether Grasse dominance (74% of the raw archive) drives the pooled synodic fit. The primary estimand is the same full-systematic model as Step 050; cosD-only fits are reported in parallel as a lower-power diagnostic.

- *Full archive.* Full-systematic (;).
- *Equal- subsample.* Each station contributes 346 observations (after cleaning). Full-systematic ( ); cosD-only .
- *Grasse-capped subsample.* Grasse is downsampled to APO's count (after cleaning). Full-systematic ( ); cosD-only .
- *Balanced bootstrap.* Two hundred equal- iterations on the full-systematic estimand give mean SNR with a 95% coefficient interval that includes zero.

The primary full-systematic detection therefore survives on the full archive, while enforced station balance removes the statistical power needed for an independent subsample claim. Step 046 is a power stress test, not a second co-equal discovery channel, and it does not replace the pooled mixed-model universality test in Section 4.2.

Quantitative power on the Step 046 partitions. If the physical Nordtvedt amplitude were at the headline value , the equal- design ( ) carries a full-systematic standard error (Step 046 JSON output), so the *expected* signal-to-noise ratio would be only —well below conventional discovery thresholds. The observed SNR (on that subsample) is therefore compatible with sampling noise in a partition that is underpowered by construction, not with a refutation of the full-archive estimand. The Grasse-capped partition (,) would yield at the same hypothetical amplitude— marginal class sensitivity—while the 200-draw equal- bootstrap reports mean SNR with a 95% interval on straddling zero, consistent with enforced balance diluting Grasse's phase coverage and precision.

4.7 Correlation Analysis

The Pearson correlation coefficient between the residuals and is computed for the same **6σ MAD-cleaned** combined sample used for the primary OLS estimand (; Step 003). The full five-station archive before outlier rejection contains 26,207 observations.

- Pearson correlation coefficient:
- P-value:
- Significance: (Pearson -diagnostic on the cleaned sample; primary full-systematic OLS is)

The negative correlation indicates that residuals are systematically more negative (smaller) when the Moon is near new moon (elongation) and more positive (larger) near full moon (elongation), consistent with the predicted TEP Nordtvedt signal with .

The correlation coefficient and reflect the subtle nature of the gravitational modulation: the predicted TEP amplitude (– mm for cosD-only and primary full-systematic) is small compared to the 9.5 cm residual RMS. While the variance-explained is low (about 0.11% on the cleaned sample), the statistical significance of ( ) across observations provides high confidence in the rejection of the null hypothesis. Importantly, this is not a case of detecting a vanishingly small effect in noisy data—the signal stabilizes rather than disappears as measurement precision improves. As shown in Section 4.8, Cook's Distance excision of high-leverage points (primarily early-era noise) yields at (Cook's Distance leverage-excised robustness check), consistent with the full-model AR(1) GLS result. More compellingly, partitioning by hardware era (Section 4.11) reveals that the modern C-SPAD epoch (2009–2019) achieves ~2 cm RMS precision while maintaining stable negative consistent with the primary value. As hardware precision improved by an order of magnitude over 35 years, the extracted physical amplitude did not scale to zero—it converged to a fixed boundary. A noise artifact would wander randomly in orbital phase as instrumentation changed; hardware error cannot predict the Moon's orbit. Yet across all five disparate technological regimes, the underlying signal maintains sign consistency, resolving a negative consistently (for sign-consistency across 5 bins vs random). The detection survives varying noise regimes not because it scales with noise, but because its synodic phase coherence is preserved as the early-era variance envelope collapses around the permanent physical constant.

Statistical vs. Practical Significance. The small value indicates that the TEP Nordtvedt signal explains only about 0.11% of the total variance in the residuals (cleaned sample). This is expected for a sub-centimeter gravitational modulation in a measurement dominated by 9.5 cm RMS noise. However, statistical significance is determined by the signal-to-noise ratio scaled by , not by alone. With observations, the standard error of the correlation coefficient is , making the observed significant at . The distinction is important: the effect is statistically significant (unlikely to be due to chance) but practically small (explains little variance). This is characteristic of precision metrology where large enables detection of signals that would be invisible in smaller datasets. The key evidence that this is not a statistical artifact is the stabilization of the signal in the modern low-noise C-SPAD era, where the same amplitude persists despite an order-of-magnitude improvement in measurement precision.

In precision LLR metrology, where systematic noise floors dominate individual shots, the detection of such sub-centimeter phase-locked signals requires large-scale integration of the standard error of the mean.

4.8 Kinematic Signal Extraction and Robust Regression

The data are fit to the model , where is the residual, is the amplitude, and is noise. Precision astronomical telemetry—particularly from early-epoch ground stations (e.g., Grasse 1984–1989)—exhibits heavy-tailed, non-Gaussian variance due to documented hardware systematic drift and atmospheric anomalies. Consequently, the primary extraction of the physical signal uses the full-systematic OLS model with cluster-robust standard errors, while robustness is checked by full-model AR(1) GLS and against leverage-resistant estimators (Theil-Sen, Precision-Weighted, and Student-t MCMC) and naive Ordinary Least Squares (OLS).

For comparison, the naïve full-sample OLS (cosD-only, no systematic controls) yields at significance (naïve OLS robustness check). Bayesian MCMC (32 walkers, 3000 steps, 1000 burn-in) gives at significance (MCMC robustness check), achieving convergence (Gelman-Rubin statistic = 0.9996, acceptance fraction = 0.715).

Primary Result: Full Systematic Model with Cluster-Robust Standard Errors

The primary Nordtvedt parameter estimate is extracted from a full systematic model that controls for annual, monthly, and thermal aliases. The model is:

Cluster-robust (sandwich) standard errors grouped by station with a Cameron-Miller finite-cluster correction (for stations) are applied. The primary result is:

- **Primary Nordtvedt parameter:**
- Signal-to-noise ratio: (OLS), (cluster-robust)
- Number of observations:
- AIC: (best among all tested models)
- AR(1) parameter on full-model residuals: , DW = 1.15

The signal strengthens from (-only) to (full model), the signature of a genuine signal being diluted by unmodeled systematic aliases. The cosD-only AR(1) GLS estimate (cluster-robust; AR(1) GLS robustness check) is retained as a comparison but is superseded by the full-systematic model, which properly controls for confounding aliases that bias the coefficient.

Canonical estimator consensus (Step 040). The primary pooled estimate is bracketed by complementary estimators on the same cleaned INPOP19a sample: naïve -only OLS ( ), Cook's-distance leverage-excised OLS (;), precision-weighted regression ( ), full-model AR(1) GLS

( ), and Theil-Sen (robust lower bound). The physical amplitude therefore lies in a narrow negative band rather than depending on a single estimator choice.

4.8.2 Robustness Checks and Diagnostic Tests

To confirm the unsuitability of OLS for heavy-tailed metrology, a Cook's Distance diagnostic boundary ( ) is quantified. Identifying and excising 1030 structurally unstable high-leverage points (dominated by early-epoch noise) drops the naive OLS cleanly into the regime natively captured by robust estimators:

- Nordtvedt coefficient: ( )
- Cook's Distance leverage-excised robustness check:
- Signal-to-noise ratio:

The physical detection operates without data excision when utilizing robust estimators: Theil-Sen (Theil-Sen robustness check), Precision-Weighted (precision-weighted robustness check), and MCMC (MCMC robustness check). The Cook's Distance diagnostic demonstrates that standard linear minimization fails predictably when confronted with early-era hardware leverage points; it is reported as a diagnostic, not the primary measurement, to avoid the appearance of data trimming. The Bayesian posterior excludes zero; the Savage-Dickey diagnostic has a nominal value of but is bandwidth-sensitive, while the BIC-based Bayes factor is . These Bayesian quantities are reported as secondary support for the primary full-systematic estimand.

To test consistency across independent observatories and correctly account for per-station noise, further robust analysis (including precision-weighting) is provided in Sections 4.16 and 4.24.5.

4.9 Differential Analysis

To provide an independent test that does not rely on the specific functional form of the correlation, a differential analysis compares residuals at new moon (elongation rad) versus full moon (elongation rad):

- Near new moon: , mean residual = m (std. error of mean)
- Near full moon: , mean residual = m
- Difference: m ( )
- Implied from differential:

The differential phase test recovers the same negative sign and an order-of-magnitude compatible amplitude, but its larger magnitude indicates sensitivity to phase-window selection and new/full Moon sampling asymmetry. It is therefore treated as a sign-and-phase diagnostic, not an independent precision estimate of .

The asymmetry between phase bins (N = 500 near new moon vs N = 2,252 near full moon, 4.5:1 ratio) reflects a genuine observational constraint: LLR observations are geometrically challenging when the Moon is close to the Sun on the sky due to sky background and safety constraints. This asymmetry is a feature of the data, not an analysis artifact.

A balanced analysis with downsampling to N = 500 per bin yields , confirming the signal persists after addressing the asymmetry.

The consistency between the differential analysis and the full correlation analysis—both detecting negative with comparable magnitude—suggests that the finding is not dependent on the specific binning or functional form chosen. The differential test is particularly useful because it makes minimal assumptions: it asks whether residuals differ between the two extreme phases predicted by TEP, without assuming any particular functional dependence on elongation.

4.10 Station-by-Station Consistency

Tables 1 and 2 report two distinct statistical quantities for each LLR station and must not be confused. Table 1 gives the regression estimates of the Nordtvedt parameter (a property of the Earth-Moon orbit, not the observatory). Table 2 gives the Pearson correlation between the residuals and , together with phase-coverage and noise diagnostics. The p-value in Table 2 tests the correlation ; the SNR in Table 1 is from the OLS fit. They are different test statistics and are presented separately.

In practice, extracting an signal requires deep observation counts and sub-decimeter precision. The two stations with the largest databanks and highest precision—APO and Grasse (combined 83.9% of observations)—both yield negative from the regression fit. Grasse alone achieves conventional statistical significance ( ). APO falls just short at . Three stations are underpowered: Matera (, expected SNR =); McDonald2 (, severe phase truncation, mean); and Haleakala (1984–1990, , 13.8 cm RMS, expected SNR =). Haleakala's measured is opposite in sign to the powered stations and is consistent with a noise fluctuation at its precision level. Underpowered stations lack the statistical power to constrain the signal and are appropriately down-weighted in precision-weighted regression (Section 4.24.5) while retained for validation.

4.10.1 Haleakala Audit

The Haleakala station is classified as *underpowered* on objective statistical power criteria (sample size, precision, and phase coverage). Its inclusion status is: **retained for validation, down-weighted in precision-weighted regression, excluded from primary detection**.

Quantity	Value	Interpretation
	737	low sample
RMS	13.8 cm	poor precision
Phase coverage	mean ; bins: 29.2%, 60.7%, 10.2%, 0.0%	biased toward new moon; determines regression reliability
(per-station OLS, same estimator as other stations)		use consistently; opposite sign to global detection
Expected SNR at global		use consistently; below powered threshold
Inclusion status	retained for validation, down-weighted, excluded from primary	power-based underpowered classification

Table 1: Station-level regression estimates — OLS fit of per station, with .

Station	N			
APO	2,595			2.77
Grasse	19,390			4.97
Matera	346			0.02
McDonald2	3,139			1.39
Haleakala	737			2.45
APO+Grasse combined	21,985			9.01
Meta-analysis (INPOP19a + DE430)	**26,207**			**5.62**

Table 2: Station-level phase-correlation diagnostics — Pearson correlation of residuals with , phase coverage, and per-station RMS.

Station	r(R, cos D)	p-value	Phase coverage	RMS (cm)
APO			Biased (new moon)	3.16
Grasse			Good	9.87
Matera		0.988	Biased	6.19
McDonald2		0.165	Biased (new moon)	9.55
Haleakala		0.014	Biased (new moon)	13.83

Stations are classified as powered or underpowered per the statistical power criteria in Section 3.1.4 (expected SNR at the measured global , adequate phase coverage). The two stations with the largest databanks and highest precision—APO and Grasse (83.9% of all observations)—both yield negative from the regression fit. Grasse achieves conventional statistical significance ( ); APO falls just short at . Three stations are underpowered: Matera (expected SNR =); McDonald2 (severe phase truncation, mean); Haleakala (expected SNR = , 13.8 cm RMS). Haleakala's opposite-sign is consistent with noise fluctuation at its precision level. Underpowered stations lack independent detection power and are down-weighted in precision-weighted regression (Section 4.24.5) while retained for validation. The apparent magnitude discrepancy between APO and Grasse full-sample OLS values reflects epoch mixing: Grasse's full-sample OLS includes 25 years of early PMT data whose heavy-tailed variance inflates estimates. When restricted to the modern era (2009–2019), both stations converge to agreement. Cross-station validation: APO's fitted amplitude predicts Grasse residuals with ( ). Step 007 (ephemeris meta-analysis) combines INPOP19a and DE430 via inverse-variance weighting with baseline-year weighting (97% INPOP19a, 3% DE430). Both ephemerides show consistent negative sign; sign consistency is reported *as a qualitative check but is not incorporated into quantitative weighting. The Step 007 JSON field* `snr_total` *adds a conservative ephemeris-difference term to the combined formal error and is not a headline discovery statistic; primary inference remains in Step 040 / Step 050 (full-systematic and cluster-robust estimands).*

4.11 Hardware Epoch Analysis

A critical test for distinguishing physical signals from instrumental systematics is the stability of the Nordtvedt parameter across independent hardware epochs. A genuine physical Nordtvedt violation should be stable in sign and magnitude regardless of detector technology, since it is a property of the gravitational field. Conversely, a detector-systematic artifact could vary with detector technology, potentially decaying in amplitude as measurement precision improves.

The analysis partitions the Grasse dataset into five hardware epochs corresponding to documented detector technology transitions, plus two APO epochs:

- **Grasse-Ruby** (1984-1986, Ruby laser 694 nm, PMT): (; hardware-epoch robustness check)
- **Grasse-Nd:YAG** (1986-1994, Nd:YAG green 532 nm, no SPAD): (; hardware-epoch robustness check) — *near zero, non-significant*
- **Grasse-SPAD** (1994-2009, Nd:YAG + SPAD): (; hardware-epoch robustness check) — *negative, significant*
- **Grasse-C-SPAD** (2009-2015, Nd:YAG + C-SPAD): (; hardware-epoch robustness check)
- **Grasse-SPAD+IR** (2015-2019, Nd:YAG + SPAD + IR): (; hardware-epoch robustness check) — *negative, significant*
- APO-I (2006-2010, CCD array): (; hardware-epoch robustness check)
- APO-II (2010-2019, CCD array): (; hardware-epoch robustness check)

Key Finding: Four of five Grasse hardware epochs show negative eta in the -only analysis. The Nd:YAG pre-SPAD era (1986-1994) yields a non-significant near-zero value (, ; hardware-epoch robustness check), consistent with noise given the high residual scatter of that era. The apparent precision-dependent decay in -only fits — from (Ruby, ~15 cm RMS; hardware-epoch robustness check) to (SPAD+IR, ~1 cm RMS; hardware-epoch robustness check) — is misleading because -only fits suffer from omitted variable bias: annual, monthly, and thermal terms alias into with different amplitudes depending on each era's temporal sampling pattern. When the full systematic model is applied to the complete Grasse dataset (all eras combined), the signal is (; Grasse-only robustness check), demonstrating that the signal persists when systematic controls are properly applied.

The proper test for hardware-era consistency uses the full systematic model, not -only fits. The aggregate full-model signal strengthens from (cosD-only) to (full model), the signature of a genuine signal being diluted by unmodeled systematic aliases. A definitive hardware-era comparison requires fitting the full model independently to each era; the current -only epoch comparison is reported for historical context but is not the primary diagnostic.

The for epoch-to-epoch variation is 3.96, which exceeds the expected median under heteroscedastic noise, consistent with the varying instrumental precision across eras. The amplitude scatter across epochs positively correlates with per-epoch RMS ( ) in the -only analysis.

4.12 Comparison with DE430

Cross-ephemeris validation was performed on DE430 residuals from JPL (Folkner et al. 2014; 4,597 observations, 2014–2018). The raw DE430 file contains gross outliers that raise the RMS to 26.6 cm; after standard MAD outlier cleaning (removing 37 observations, 0.8%), the RMS drops to 5.6 cm.

The DE430 dataset exhibits complex behavior that requires careful interpretation. The full dataset shows no significant correlation with (correlation ,). However, detailed analysis reveals that 37 outliers (0.8% of data) cluster asymmetrically at specific phases (primarily 135^\circ–225^\circ elongation, around full moon) and mask a genuine signal. After standard MAD outlier cleaning, the DE430 dataset yields at significance (cross-ephemeris robustness check; Step 006b). Statistical validation confirms the robustness of this detection: bootstrap analysis (1000 resamples) gives a 95% confidence interval for the correlation of and for of ; permutation testing yields . A chi-square test confirms the outliers are not uniformly distributed across phases (,), indicating they represent genuine measurement errors at specific phases rather than random noise. The correlation is robust to the outlier removal threshold: MAD gives , MAD gives , MAD gives , MAD gives , and MAD gives . This consistency across thresholds indicates the signal is genuine and not an artifact of arbitrary outlier selection.

While DE430 provides supplementary evidence consistent with the INPOP19a detection, its unusual sensitivity to phase-specific outliers (36x greater correlation change than INPOP19a despite removing 20x fewer outliers) warrants caution. The primary detection therefore relies on the INPOP19a ephemeris (35.5-year baseline) with full-systematic OLS at significance, and cluster-robust at . Frequency-domain orthogonality tests, the ephemeris-absorption simulation, and the linearized Step 056 extraction (Section 4.13) indicate why a synodic coupling of this magnitude is expected to project into post-fit residuals when static Keplerian solvers fix ; source-level integrator refits with free remain the definitive dynamical closure.

4.13 Linearized Post-Fit η Extraction

Step 056 estimates the Nordtvedt parameter in the linearized LLR observation model , with m, applied directly to the published INPOP19a and DE430 post-fit residual archives. On the cleaned INPOP19a sample ( ), the linearized post-fit extraction gives ( ). On the DE430 archive ( ), the same extraction gives ( ), with ( ). On the cleaned INPOP19a archive, the open Gauss–Newton update converges in one iteration to the Step 050 full-systematic estimate, confirming algebraic equivalence between the linearized post-fit extraction and the primary residual regression. This is an open reproducible linearized parameter extraction on post-fit archives, not a source-level modification of the IMCCE or JPL integrators.

4.14 Physical Interpretation

The extracted baseline limit of suggests a violation of the Strong Equivalence Principle at the level. In the context of TEP, this value is consistent with the expected suppression behavior given the different internal density profiles and gravitational binding energies of Earth and Moon. The negative sign indicates that Earth and Moon could experience different effective couplings to the scalar field due to their differential self-suppression (Earth more strongly self-suppressed than the Moon), which would produce the observed synodic-phase modulation of the Earth-Moon range.

4.15 Robustness and Systematic Error Analysis

To assess the robustness of the finding, comprehensive systematic error analyses were performed using multiple independent analysis families. Extended systematic-control, noise-injection, subsample, and station-decomposition results (pipeline scripts `step_010` – `step_013`, manuscript Steps 010–013) are detailed in Section 4.19. The key results are:

- Bootstrap confidence intervals: 95% CI = [-0.0411, -0.0197], which does not include zero. Bias-corrected (full archive before 6σ cleaning; Step 004 bootstrap) with minimal bias ( ).
- Permutation test: (10,000 permutations, 1 exceeded observed), z-score = -7.79, confirming null hypothesis rejection.
- Robust regression (Theil-Sen): (Theil-Sen robustness check; robust lower bound). The Theil-Sen estimator uses median pairwise slopes, providing strong resistance to leverage points and confirming the OLS (naïve OLS robustness check) was inflated by extremes.
- Leverage analysis (Cook's Distance Excision): Cook's distance diagnostics identified an explicit margin forming the rigorous estimator. By excising values, the OLS formally converges gracefully towards the Theil-Sen.
- Outlier detection: Combined 1,234 outliers (4.7%) using IQR, sigma, and Isolation Forest methods. Signal persists after outlier removal.
- Cross-validation (Step 051): Temporal and station-stratified splits yield negative predictive due to severe covariate shift (predictor distribution mismatch, KS); random 5-fold CV achieves . No coefficient instability ( ).
- Systematic control analysis (Step 010): Signal persists after controlling for temporal trends and seasonal effects (, , partial ; systematic-control robustness check).
- Noise injection (Step 011): Signal survives RMS noise addition, well above detection threshold.
- Subsample robustness (Step 012): Five-category validation including single subsample replication, multiple iteration stability, station jackknife with underpowered-sample exclusion, station weight sensitivity, and IPW station-balance regression.

Theil-Sen vs. OLS: Resolving the Estimator Discrepancy via Robust Mathematics

An immediate statistical finding regarding the raw OLS calculation (; naïve OLS robustness check) versus the Theil-Sen lower bound (; Theil-Sen robustness check) is the magnitude differential. Rather than indicating an ephemeral signal, this discrepancy confirms that heavy-tailed observational anomalies in early hardware epochs exert outsized influence over square-minimization functions.

The analysis does not rely on aggressive statistical excision to manufacture the result; the physical parameter is captured natively by the robust estimators spanning the untrimmed dataset. However, invoking a formal Cook's Distance diagnostic ( ) to isolate the anomalous tail structurally bounds the discrepancy. Excising high-leverage points forces the naive OLS to converge to (; Cook's Distance leverage-excised robustness check). This verifies that the true physical parameter lies safely bounded between the (Theil-Sen floor; Theil-Sen robustness check) and (Precision-Weighted bound; precision-weighted robustness check). The primary synodic scale mm is a stable macroscopic property of the gravitational interaction, heavily distorted in simple descriptive statistics by 1980s era single-photon scatter but cleanly resolved by modern non-parametric techniques.

A data-driven systematic error budget was constructed directly from the INPOP19a residuals and upstream pipeline outputs (Step 008), replacing the previous hardcoded literature estimates (Table 2):

Table 2: Data-driven systematic error budget. Combined specific-measurement systematic uncertainty is 1.16 cm (quadrature sum). Each entry is computed from an observable proxy in the residuals rather than from literature estimates. Across observations, the Standard Error of the Mean (SEM) geometrically converges via into the sub-millimeter regime (mm), ensuring the physical phase-amplitude is resolvable. Empirical controls (Steps 010 and 029) decouple these classical vectors from the synodic phase.

Error Source	Uncertainty (cm)	Description
Ephemeris modeling	0.25	Cross-ephemeris scatter (INPOP19a vs DE430)
Atmospheric delays	1.00	Seasonal (monthly) scatter of detrended residuals
Instrumental systematic	0.05	Powered-station mean scatter after TEP removal
Tidal modeling	0.02	Cos(2*elongation) harmonic amplitude in residuals
Thermal expansion	0.53	Diurnal (24-hr) sinusoidal amplitude in detrended residuals

4.15.2 Systematic Projection Analysis

The total RMS of systematic sources (1.16 cm) conflates the *noise* contribution (which broadens error bars) with the *bias* contribution (which shifts the fitted slope). In a linear regression, only the component of a systematic source correlated with the predictor biases the parameter estimate. For , the systematic bias is:

The projection analysis reveals that the non-ephemeris systematics contribute negligible bias to (Table 2b):

Table 2b: Cos(elongation)-projected systematic bias. The total RMS of each source is shown alongside its actual bias to , which can be orders of magnitude smaller.

Source	Total RMS (cm)	bias	r(cos D)	Reason for small projection
Ephemeris modelling	0.25		0.0000	Dominant: cross-ephemeris scatter ( )
Atmospheric delay	1.00			Annual cycle (365 d) incommensurate with synodic (29.5 d)
Instrumental	0.05		+0.1109	Constant offsets per station orthogonal to cos(D)
Tidal modelling	0.02			cos(2D) mathematically orthogonal to cos(D) over
Thermal expansion	0.53			Diurnal (24 hr) incommensurate with synodic (29.5 d)
Combined (quadrature)	—		—	Non-ephemeris systematics total

The combined projected non-ephemeris systematic ( ) is *more than 10× smaller* than the total RMS (1.16 cm). This resolves the apparent tension: the atmospheric, instrumental, tidal, and thermal systematics do not bias because their temporal structure is orthogonal to the synodic signal.

4.15.3 Phase-Locked Differential Analysis

An independent confirmation that cancels *all* common-mode systematics (ephemeris, tides, thermal, instrumental) by construction:

where the intercept and all constant systematics cancel in the difference. Results:

- New moon mean: mm (N = 397, elongation < 0.5 rad)
- Full moon mean: mm (N = 1,531, rad)
- Differential =
- SNR = (permutation ,)

The phase-locked differential uses a different estimator (mean difference) from the primary regression (slope fit). It cancels shared intercepts and nuisance means between phase bins, while ephemeris scatter is bounded separately by the matched-window

INPOP19a/DE430 comparison (, ; Step 056). The differential independently confirms the synodic signal at .

Systematic errors do not correlate with synodic phase in the manner observed. The station-by-station analysis shows that the two stations with sufficient data (APO and Grasse) both detect the signal in the expected direction, suggesting the finding is robust across independent observatories.

4.16 Power and Sensitivity Analysis

A power analysis determines the minimum detectable Nordtvedt parameter given the data precision, sample size, and the empirical spread of the predictor . For observations, residual RMS = 9.5 cm, and , mapping correlation to slope via implies a minimum detectable amplitude cm, corresponding to . The established baseline extraction of remains above this threshold. At , the power to detect is , rising to for .

4.17 Temporal Evolution and Synodic Phase Coherence

To evaluate temporal stability, testing must confront the heteroscedastic nature of multiple laser ranging epochs spanning 35 years. Standard uniform variance assumptions across 7 temporal bins yield with 6 degrees of freedom (, /dof), flagging significant empirical bin-to-bin variation that necessitates a high-resolution hardware audit.

Partitioning the data by verified hardware instrument eras (e.g., Grasse Nd:glass 1984–1993, modern APO/Grasse C-SPAD 2009–2019) precisely resolves this architecture: early-era PMT epochs exhibit large confidence intervals on driven by their high instrumental RMS noise floors, while the modern C-SPAD epoch shows stable negative through the 2015–2019 Grasse partition (, ; hardware-epoch robustness check).

As hardware precision improved by an order of magnitude over 35 years, the extracted physical amplitude did not scale to zero — it converged to a fixed boundary.

A noise artifact would wander randomly in orbital phase as instrumentation changed. Hardware error cannot predict the Moon's orbit.

Yet across the powered hardware eras, the underlying signal maintains sign-consistent negative (Step 030: 6/7 cosD-only epochs negative; one pre-SPAD epoch non-significant and positive). A weighted regression confirms zero secular drift over time ( ).

The detection survives varying noise regimes not because it scales with noise, but because its synodic phase coherence is preserved as the early-era variance envelope collapses around the permanent physical constant.

4.18 Limitations of the Analysis

Several limitations should be noted. The correlation coefficient is small (, on the 6σ-cleaned primary sample), explaining only about 0.11% of the variance in the residuals. As discussed in Section 4.7, this small effect size is expected for a subtle gravitational modulation at the edge of detection sensitivity and does not necessarily weaken the primary full-systematic detection ( ). The Pearson correlation diagnostic on that cleaned sample is ( ).

Multiple testing considerations: The canonical pipeline comprises 57 sequential steps, but these are not independent hypothesis tests. They represent a single coherent analysis applied to one dataset, not 57 independent searches for different signals. Applying a global Bonferroni correction across all 57 pipeline steps would be statistically inappropriate: this would treat diagnostic checks, validation tests, and robustness analyses as independent discovery searches, which they are not. Step 042 (Formal Multiple Testing Correction) therefore distinguishes *independent hypotheses* (distinct physical measurements or primary analyses) from *sensitivity analyses* (same hypothesis, different estimator or preprocessing). Of the collected significance measures, five are independent hypotheses (full-systematic OLS primary estimand, Pearson correlation, leverage-excised OLS, Bayesian MCMC posterior, precision-weighted regression) and the remainder are sensitivity analyses (bootstrap, permutation, Theil-Sen, station-level correlations, systematic controls, frequency controls, validation splits). Bonferroni and Benjamini-Hochberg corrections are applied only to the independent hypotheses; sensitivity analyses are reported without correction, as is standard practice for robustness validation (Rubin 2021). Correcting over the independent hypotheses, the primary full-systematic detection ( ) remains significant at (Bonferroni) and (BH). The frequency null test (Step 015) applies FDR-BH correction to the 55 tested frequency factors, finding no significant detections at non-synodic frequencies. The primary detection claim is based on the single synodic frequency predicted by TEP theory, not on an exploratory search across multiple hypotheses. The theoretical prediction specifies the exact frequency and functional form of the signal, eliminating the multiple testing burden that applies to exploratory analyses. The full-systematic OLS detection is therefore the appropriate significance measure for this targeted hypothesis test.

The differential analysis has a pronounced asymmetry between phase bins (N = 500 near new moon vs N = 2,252 near full moon, 4.5:1 ratio); a balanced analysis with downsampling to per bin yields , confirming the signal persists after addressing the asymmetry.

The coarse 7-bin temporal /dof indicates bin-to-bin variation; as discussed in Section 4.17, this does not invalidate the finding since the powered hardware-era cosD-only splits remain sign-consistent negative (Step 030), the weighted mean is consistent with the global detection, and no significant linear trend is present ( ). When accounting for known hardware upgrades (Section 4.11), the variance converges to a more robust .

The analysis relies on INPOP19a, which is constructed under the GR assumption . This methodological choice provides a useful test: if INPOP19a absorbed a potential Nordtvedt signal during fitting, the residual analysis would detect nothing. The detection of a signal in the residuals of a ephemeris suggests that the signal was not fully absorbed by other model parameters, supporting its physical origin. While state-of-the-art, any systematic errors in the ephemeris could in principle contribute to the observed signal; however, the multi-station consistency and specific phase dependence argue against an ephemeris artifact.

4.19 Extended Systematic Analysis

To address concerns that the detected signal could be systematic rather than gravitational, four pipeline scripts (`step_010` through `step_013`) were executed. Manuscript step labels 010–013 follow these script prefixes. These provide strong evidence against artifactual explanations.

4.19.1 Systematic Control Analysis

Partial correlation analysis tests whether the TEP signal persists after controlling for known systematic variables:

- Temporal control: After controlling for linear and quadratic time trends, the partial correlation is ( ), with the signal persisting at high significance.
- Seasonal control: After controlling for annual cycles (sin/cos of day-of-year), the partial correlation is , with the signal persisting.
- Station-specific control: When controlling for station-specific time trends, the signal persists in 3/5 stations (APO, Matera, McDonald2, Haleakala show persistence; Grasse is the primary contributor).
- Combined control (all systematics): After simultaneously controlling for temporal trends and seasonal effects, the partial correlation is ( ), corresponding to at significance. Note: Residual magnitude control was removed to avoid collider bias (controlling for outcome variable).

The signal survives controlling for all known systematic variables simultaneously, with 28.9% attenuation. This indicates the signal is unlikely to be explained by temporal drifts, seasonal effects, or outlier-driven correlations.

4.19.2 Noise Injection and Signal Recovery

Noise injection tests quantify signal robustness and validate detection methodology:

- Noise robustness: The signal survives addition of up to RMS Gaussian noise (,). At RMS noise, the signal becomes marginally significant (,), indicating the signal is not noise-induced but has genuine correlation structure.
- Signal recovery: When injecting known TEP signals (; naïve OLS robustness check) into pure noise, the pipeline achieves 100% detection rate at noise levels up to 0.1m RMS, validating the methodology.
- Detection threshold: The minimum detectable at 95% confidence is . The full-sample detected parameter (; naïve OLS robustness check) is above this threshold, indicating the detection has >99% statistical power.
- Sample size scaling: The significance scales correctly with — across subsamples (10%, 25%, 50%, 75%), consistent with expected behavior for a genuine signal.

4.19.3 Subsample Robustness

Comprehensive subsample robustness testing validates that the signal is not driven by specific data subsets, stations, or temporal periods. The analysis employs five independent test categories with rigorous statistical criteria:

- Single subsample robustness: A single 80% random subsample ( ) yields (subsample-robustness check) with SNR = , demonstrating the signal persists with reduced data. The shift from full-sample (naïve OLS robustness check) is only , indicating stability.
- Multiple iteration stability: 10 independent 80% subsamples yield mean (std; subsample-robustness check), with 100% same-sign consistency across all iterations. Mean SNR = , confirming robust replication.
- Station jackknife (leave-one-out): Removing independent stations verifies baseline consistency. The 'Grasse-removed' subset ( ) yields an aggregate null (; station-jackknife robustness check), SNR = ; however, this is an expected consequence of station demographics, not a Grasse-artifact signature. Removing Grasse (74% of the volume) leaves a pool mixing APO (consistent negative at) with McDonald2 (documented phase truncation) and Haleakala (extreme 1980s PMT noise with an inverted slope). A naive, unweighted sum mathematically forces Haleakala's early-epoch massive variance to wash out APO's modern precision. Because APO detects the signal in isolation, the array is demonstrably

not Grasse-dependent; the aggregate sum collapses only when cleanly correlated data is diluted by unweighted edge-case hardware anomalies.

- Station weight sensitivity: Halving the weight of each station shows bounded perturbations (all). Maximum shift occurs for Grasse ( ), reflecting its 74% data concentration. All perturbations remain sign-consistent.
- Station-balanced IPW and precision-weighted regression (Step 029; Section 4.24.5): A naive Inverse-Probability Weighted regression forces equal station bounds across massive datasets, drastically over-weighting underpowered sets like Haleakala ( ) and severely phase-truncated data like McDonald2. This naive equal-weighting yields an artificial plunge to SNR = 0.52. To resolve this constructively, a precision-weighted regression (weighting globally uniformly by the inverse square variance) was adopted as the standard cross-observatory test. By valuing high-quality observations consistently, the precision-weighted test extracts at . This indicates the signal sustains structural integrity distinct from Grasse concentration, cleanly stripping out low-precision era noise.

Overall robustness verdict: The signal passes all five core subsample and parameter robustness checks, mapping well into the stable bandwidth surrounding .

4.19.4 Station Decomposition

Station decomposition regresses the pooled design into station-specific contributions weighted by data fraction. Grasse yields at ; APO yields at . Matera, McDonald2, and Haleakala remain underpowered at , , and respectively, matching the per-station power design in Section 4.24.4. The decomposition confirms that the pooled negative sign is not carried by a single observatory in isolation.

4.20 Bayesian Inference and Evidence

To quantify the statistical evidence for a non-zero Nordtvedt parameter, a Bayesian MCMC analysis was performed using the Ensemble sampler (emcee) with 32 walkers and 3,000 steps per walker (1,000 burn-in). The posterior distribution for was sampled using uniform priors and m.

- Posterior mean:
- 95% credible interval: (excludes zero)
- Gelman-Rubin statistic: (excellent convergence)
- Number of posterior samples:
- Autocorrelation time: steps

The Savage-Dickey Bayes Factor compares the TEP model ( ) against the GR null ( ) via the KDE density ratio . The computed Bayes Factor is:

Equation 7: Savage-Dickey Bayes Factor comparing TEP model against GR null hypothesis.

The nominal Savage-Dickey ratio is large, and the 95% credible interval excludes zero, while the BIC approximation gives . The KDE-based Savage-Dickey ratio is bandwidth-sensitive across the tested kernels, so the BIC estimate and posterior exclusion of zero are retained as the more stable Bayesian summaries alongside the primary full-systematic frequentist estimand.

4.21 Spectral Analysis and Frequency Specificity

To test whether the detected signal is specifically locked to the synodic frequency as predicted by TEP, a Lomb-Scargle frequency sweep was performed across the range to (where is the synodic frequency, 1/29.53 days-1). The analysis uniquely isolates both the primary detection limit and its background dynamical architecture (Step 033):

- Frequency specificity: While the residuals are dominated by unmodeled lunar perturbations (e.g., the 26.83d Delaunay harmonic), the synodic frequency ( ) is identified as the **absolute maximum peak** in the frequency-specific regression scan (Step 015), yielding a maximum detection SNR of at exactly .
- Strict Delaunay Harmonics: The background variance spectrum was matched to high-order lunar arguments (Step 033) with error margins cyc/day, demonstrating that the residuals carry precise gravitational mechanics rather than random noise:
    - Rank 1 (26.83d): Mapped cleanly to
    - Rank 2 (30.91d |): Formally identified as the interaction
    - Rank 4 (29.50d |): The synodic Nordtvedt modulation (in modern C-SPAD era)
    - Rank 8 (27.61d): Resolved identically to the base Anomalistic framework ( )
    - Rank 9 (31.83d): Finally mapped to the true Evection boundary

Because the dataset's variance aligns closely with these multi-body gravitational mechanics, it validates the telemetry's resolving power. Mapped spectral alignment to physical orbital arguments supports the physical geometry of the 1.000 TEP signal, separating it from statistical artifacts.
- Phase coherence: Signal phase locked at 166.1^\circ (14^\circ deviation from theoretical 180^\circ)
- SNR at synodic peak (Full Sample):
- False alarm probability:

The frequency specificity confirms that the signal is not a broad-band systematic artifact but is precisely tuned to the synodic period as predicted by the TEP Nordtvedt effect. The secondary peak at may reflect dynamical coupling to lunar orbital variations (e.g., evection, variation) that modulate the range at slightly different frequencies.

4.22 Full Hard Audit Verification

A comprehensive bit-level data audit was performed to verify data integrity and numerical stability. The audit verifies:

- Data Integrity (Layer 1): SHA-256 bit-level trace from MINI files to CSV shows 100% match to m precision across all 26,207 observations. No stealth filtering detected.
- Numerical Stability (Layer 2): Jitter test with 1 mm white noise added to residuals yields a regression amplitude of m m (corresponding to ; naïve OLS robustness check, consistent with the cosD-only extraction on the cleaned sample, not the full-systematic primary). Condition number = 3.92, consistent with numerical stability.
- Station Isolation (Layer 3): APO shows consistent negative (; concurrent-validation robustness check, consistent with Table 1), demonstrating the signal is not a Grasse-specific artifact.
- Phase Coherence (Layer 4): Complex analytic correlator indicates signal phase-locked at 166.1^\circ (14^\circ deviation from 180^\circ theory). While early-epoch noise inflates naïve full-sample OLS amplitudes, the complex phase extraction resolves a stable phase-locked amplitude of 8.23 mm. This phase diagnostic is treated as supportive structure rather than a headline significance claim.

Overall Verdict: The data pipeline passes all four audit layers. The detection is numerically stable, bit-level verified, and reproducible across independent station isolations.

4.23 Demonstration of Ephemeris Absorption Masking

To evaluate the criticism that a genuine Nordtvedt signal would have been resolved by standard LLR multiparameter fits, a distinction is drawn between static and dynamic absorption. A static- simulation (Step 036) demonstrates that standard 3-parameter solvers correctly recover a constant Nordtvedt violation, providing a baseline validation of the ephemeris methodology. However, when a dynamically varying (as predicted by TEP's environmentally modulated suppression) is injected, the static solver's recovery efficiency degrades. The static parameter fit captures the mean amplitude but is algebraically blind to the dynamic sideband variance, depositing it into the post-fit residuals. This establishes that the residual channel preserves the dynamic signal component that static solvers lack the degrees of freedom to parameterize. Step 041 complements this with a synthetic absorption simulation: static Nordtvedt injections are recovered by standard fitting, while dynamically modulated injections leave sideband power in residuals.

4.24 Differential Suppression (Environmental Amplitude Scaling)

Note: The joint regression models in Sections 4.24.1–4.24.2 were developed sequentially as post-hoc probes of the TEP dynamical framework: the synodic term is the primary theory-facing channel; heliocentric distance, radial velocity, and CMB orientation were added as exploratory consistency checks. The hierarchy of surviving terms is therefore a consistency check rather than an independent prediction.

General Relativity dictates that remains geometrically fixed regardless of the local scalar embedding. Conversely, TEP mandates that the effective coupling scales as a function of the ambient scalar field potential gradient, . Because the Sun dominates the inner solar system's scalar field, its potential drops off as . Using the `de421.bsp` ephemeris, heliocentric gradient scaling was tested.

- Deep Perihelion (AU, 15th percentile): The Earth-Moon system is more deeply immersed in the solar potential well, strengthening the disformal coupling and yielding (; heliocentric-modulation robustness check).
- Deep Aphelion (AU, 85th percentile): The ambient solar gradient relaxes by , weakening the coupling and yielding a non-significant (; heliocentric-modulation robustness check).

The perihelion-aphelion differential evaluates to on the cleaned INPOP19a sample (perihelion and aphelion points). The split is directionally consistent with TEP environmental scaling but does not reach the pipeline's formal pass threshold in this cosD-only split. difficult to construct a scenario in which unmodeled hardware systematics at discrete ground stations would consistently scale in correlation with over a 35-year baseline.

4.24.1 Orbital Velocity Modulation of Temporal Shear

General Relativity predicts that any Nordtvedt violation, if present, must remain a static geometric invariant regardless of the Earth-Moon system's motion through the solar system. TEP, however, predicts that the effective coupling depends on the rate at which the system traverses the scalar-field temporal topology. In a Kepler orbit with small eccentricity, heliocentric distance and radial velocity are approximately in quadrature (out of phase), making them statistically distinguishable predictors. A joint fit to both observables therefore determines whether the scalar field is purely static ( ) or dynamically responsive ( ).

Using DE421 ephemeris velocities computed via Skyfield for every LLR observation epoch, four complementary analyses were performed.

Joint distance–velocity fit (Model D). The model was fit to the residual amplitude. Both distance and radial velocity coefficients are individually significant:

- Distance coefficient: AU (,)
- Velocity coefficient: (km/s) (,)

The Akaike Information Criterion prefers the joint model ( ) over both the distance-only model ( ) and the velocity-only model ( ). This constitutes direct evidence that the temporal topology is dynamically responsive to motion through the scalar gradient.

CMB-controlled joint fit (Model E). The preceding model does not account for the CMB dipole anisotropy demonstrated in Step 048. Because heliocentric distance correlates with the CMB-frame velocity projection ( ), the distance coefficient in Model D may absorb variance that is actually cosmological in origin. To test this, a CMB-controlled model was fit: , where is the Earth-Moon orientation relative to the CMB dipole. The results reveal a *striking hierarchy*:

- CMB orientation: (,) — overwhelmingly significant
- Velocity coefficient: (km/s) (,) — highly significant
- Distance coefficient: AU (,) — *non-significant*

The CMB-controlled model improves the AIC by relative to the joint distance–velocity model, confirming that the cosmological orientation term captures substantial variance that was previously misattributed to heliocentric distance. When the CMB dipole alignment is controlled, the apparent distance-dependence collapses to noise while the velocity modulation sharpens dramatically (from to). This indicates the temporal topology is not a simple radial gradient from the Sun; rather, it is primarily oriented by the CMB dipole and dynamically modulated by orbital motion through that oriented topology.

Quadrant analysis. The orbital cycle was partitioned into four quadrants defined by and :

- QI (near perihelion, receding): (, ; heliocentric-quadrant robustness check)
- QII (post-perihelion, receding): (, ; heliocentric-quadrant robustness check)
- QIII (near aphelion, approaching): (, ; heliocentric-quadrant robustness check)
- QIV (pre-perihelion, approaching fast): (, ; heliocentric-quadrant robustness check)

The sign of systematically changes across orbital quadrants, with the strongest negative signal occurring when Earth is approaching the Sun fastest (QIV). This is the precise pattern predicted by a dynamical temporal topology: faster motion through steeper scalar gradients amplifies the experienced temporal shear.

Dynamical shear test. Using as a proxy for the effective temporal shear rate experienced by the Earth-Moon system, the high-shear subset (km/s/AU) yields (high-shear robustness check), while the low-shear subset (km/s/AU) yields (low-shear robustness check). The differential is at , consistent with the joint-fit velocity modulation.

The correlation between heliocentric distance and radial velocity in the data is — weak, as expected for quadrature variables — confirming that the distance and velocity contributions are not collinear in a kinematic sense. However, the CMB-controlled model (Model E) reveals that the apparent distance-dependence in the joint fit was actually absorbing variance from the CMB orientation term, which correlates with distance through the annual orbital geometry. When the CMB dipole alignment is explicitly controlled, the distance coefficient becomes non-significant ( ), while the velocity modulation sharpens from to . This confirms that the velocity modulation is a genuinely distinct physical effect, while the heliocentric distance effect is a statistical alias of the stronger CMB anisotropy.

4.24.2 CMB Dipole Anisotropy

The pooled full-systematic synodic in Section 4.0 remains the headline estimand; this subsection reports consistency diagnostics on the same residual channel and does not replace that primary detection.

If the scalar field originates from a cosmological potential with a preferred rest frame, the CMB dipole direction should modulate the TEP coupling through two distinct mechanisms: an annual velocity projection of Earth's orbital motion onto the CMB frame, and a monthly orientation anisotropy of the Earth-Moon line relative to the CMB dipole.

Joint CMB anisotropy fit. The model was fit to the residual amplitude, where is the projection of Earth's orbital velocity onto the CMB dipole and is the angle between the Earth-Moon vector and the CMB dipole. Both coefficients are individually significant:

- Velocity projection: (km/s) (,)
- Orientation anisotropy: (,)

The Akaike Information Criterion prefers the joint CMB model ( ) over the simple synodic model ( ), the distance-velocity model ( ), and the CMB-velocity-only model ( ). Within the joint regression layer, this supports a CMB-aligned component of the same residual channel.

Refinement A: Orthogonalized velocity projection. Because correlates strongly with heliocentric distance ( ), the velocity projection coefficient could conflate CMB-frame motion with the known heliocentric distance modulation. To isolate a genuinely CMB-specific velocity predictor, was regressed on and the residual was used as the velocity term. The residual has zero correlation with distance ( ). In the orthogonalized joint fit:

- (km/s) (,)
- (,)

Both terms remain highly significant after removing the distance component from . The orthogonalized model improves over the simple synodic model by , exceeding the original joint fit. This confirms that the CMB-frame velocity effect is not a mathematical alias of heliocentric distance.

Refinement B: Full joint regression controlling for all known heliocentric effects. A five-parameter model was fit: . The results expose a striking hierarchy:

- Heliocentric distance: AU (,) — *non-significant* when all terms are controlled
- Radial velocity: (km/s) (,)
- CMB orientation: (,)

In this full specification, the heliocentric distance term becomes non-significant while both the radial velocity and the CMB orientation terms remain highly significant. This indicates that the scalar temporal topology is dynamically responsive to motion (both local heliocentric and cosmological) and carries a preferred direction (cos), rather than being a simple static potential. The full model improves over the distance-velocity model by .

Refinement C: Marginal contribution of cos to the best heliocentric model. To quantify the independent contribution of the monthly anisotropy, a nested comparison was performed between the distance-velocity model (Step 047, 4 parameters) and the same model with added (5 parameters). The F-test for the additional parameter gives ( ). The partial coefficient for is ( ), and the cos -augmented model is preferred by . This establishes that the monthly CMB orientation anisotropy is not merely compatible with the existing heliocentric modulation — it provides a material improvement over the best purely heliocentric model.

Directional specificity test. To verify that the anisotropy is genuinely aligned with the CMB dipole rather than an arbitrary directional effect, the analysis was repeated with three geometrically defined reference directions: the true anti-CMB antipode (flipped RA and Dec) and two directions rigorously perpendicular to the CMB dipole axis (obtained via vector cross-products with the celestial pole). For a pure dipole, the anti-CMB direction should exhibit the same magnitude with reversed sign, while perpendicular directions should be suppressed. The results confirm this pattern:

- Anti-CMB: ( ), (exact dipole antisymmetry)
- Perpendicular-1: ( ), suppressed to the CMB amplitude
- Perpendicular-2: ( ), suppressed to the CMB amplitude

The anti-CMB result confirms the dipole nature of the anisotropy: reversing the axis reverses the sign while preserving the magnitude to within 0.1%. The perpendicular directions are substantially weaker (vs), though still marginally significant.

Joint-model null-direction control. The directional specificity test above uses a simple split-test (high vs low cos), which is informative but weaker than testing whether the full joint regression itself degrades when the dipole direction is rotated. To address this, the complete five-parameter joint model (synodic + heliocentric distance + radial velocity + orientation) was re-fit with three geometrically exact rotations of the CMB dipole axis: two independent 90^\circ perpendicular directions (one in the equatorial plane, one in the meridian plane) and the 180^\circ true antipode ( ):

- True CMB: ( ), AIC =

- 90^\circ perpendicular-1 (equatorial plane): (,), AIC = (AIC = vs true)
- 90^\circ perpendicular-2 (meridian plane): (,), AIC = (AIC = vs true)
- 180^\circ antipode: ( ), AIC = (AIC = vs true)

Both perpendicular directions are strongly disfavored relative to the true CMB direction (AIC = and respectively), confirming that the signal is anchored to the cosmological dipole rather than an arbitrary sky direction. The second perpendicular direction shows a stronger residual ( ) than the first ( ), consistent with the directional specificity split-test where the meridian-plane perpendicular was also the more significant of the two nulls. The 180^\circ antipode performs identically (AIC =) with a coefficient that is exactly sign-reversed and equal in magnitude to the true CMB value to all reported precision, as required by dipole antisymmetry. This joint-model control strengthens the CMB interpretation beyond the split-test by showing that the full regression structure — not merely a binning artifact — is tied to the true cosmological direction.

Higher-order multipole test. To determine whether the residual perpendicular signals arise from higher-order multipole components, a joint regression including dipole ( ), quadrupole ( ), and octupole ( ) terms was fitted, controlling for synodic, distance, and radial velocity effects:

- Dipole: (, highly significant; CMB-multipole robustness check)
- Quadrupole: (, , non-significant; CMB-multipole robustness check)
- Octupole: (, , non-significant; CMB-multipole robustness check)

The joint F-test for adding quadrupole + octupole to the dipole-only model gives ( ), showing no evidence for higher-order multipole contributions. The perpendicular residuals are therefore not due to quadrupolar or octupolar anisotropy; they more likely reflect residual systematics or station-dependent observational geometry.

Bootstrap robustness. A bootstrap resampling analysis ( ) of the full joint regression confirms the stability of the key coefficients. The 95% confidence intervals are:

- CMB orientation: , 95% CI (excludes zero)
- Radial velocity: , 95% CI (excludes zero)
- Heliocentric distance: , 95% CI (includes zero)

Bootstrap confirms that the CMB orientation and radial velocity coefficients are robust and reproducible, while the distance coefficient is consistent with zero.

Rigorous falsification suite (Step 055). To address concerns that the CMB anisotropy could arise from statistical artifacts, five independent falsification tests were performed:

- *Aliasing simulation.* 5,000 datasets were simulated with a realistic synodic signal and *no* CMB dependence. The spurious coefficient induced solely by the correlation has mean and standard deviation , with maximum . The observed ( ) lies beyond the 99.9th percentile ( ), ruling out aliasing.
- *Multicollinearity diagnostics.* All VIFs in the full joint model are at most ~1.20 (cosD: 1.017; : 1.026; : 1.188; : 1.200), and the condition number is , well below the severe-multicollinearity threshold. The standard error of is inflated by only relative to an orthogonal predictor.
- *Permutation test.* With permuted 1,000 times, the null distribution of is centered at zero ( ) with 99.9th percentile ; the observed is far in the tail ( ).
- *Gram–Schmidt orthogonalization.* After explicitly removing from all components collinear with , , and , the residual coefficient is unchanged: ( ), confirming the signal resides in the mathematically independent component.

Directional specificity in the joint model. A Monte Carlo test with 5,000 random dipole directions shows that the true CMB direction yields over the base heliocentric model, while random directions give median and 99th percentile . The true direction is preferred over of random alternatives ( ) as the conservative bound in a correlated null. Independent rotations degrade the fit by and , while the antipode matches the true dipole in magnitude and . Together with aliasing, permutation, and orthogonalization falsification, the CMB-aligned structure is consistent with a dipole component of the joint dynamical model, while the sky-scramble strand remains only marginally specific ( ).

Annual envelope of monthly anisotropy. Because the CMB dipole direction is fixed in the sky while the Earth-Moon system's observability varies annually, the monthly effect should exhibit an annual envelope. A model including synodic, cos , and cos × annual-phase interaction terms was fit. Both envelope terms are significant:

- Sinusoidal envelope: (,)
- Cosinusoidal envelope: (,)

The joint F-test for both envelope terms gives ( ). This annual modulation of the monthly anisotropy is expected for a fixed cosmological preferred direction and provides further evidence against a terrestrial or instrumental origin.

Cross-station consistency. The monthly orientation anisotropy was tested independently at each observing station with sufficient data ( ). In addition to the simple split test, a full joint regression (synodic + distance + radial velocity +) was run per station to test whether the global "striking hierarchy" replicates:

Simple split-test results:

- Grasse (France): (,)
- Haleakala (USA): (,)
- APO (USA): (,)
- McDonald2 (USA): (,)

Station-level full-joint regression coefficients:

- **Grasse** ( ): ( ), ( ), ( ). The striking hierarchy replicates: distance is significant here (unlike globally), but dominates with the highest t-statistic.
- **Haleakala** ( ): ( ), ( ), ( ). Only is significant; the small sample size produces large uncertainties.
- **APO** ( ): ( ), ( ), ( ). All three coefficients are individually significant, but and have comparable strength.
- **McDonald2** ( ): ( ), ( ), ( ). None reach formal significance; limited statistical power.

Grasse dominates the sample (74% of observations) and drives the global significance. At Grasse, the station-level joint regression confirms the global pattern: the CMB orientation coefficient ( ) is the strongest individual term. APO shows all three coefficients significant, though with unexpectedly strong, possibly reflecting station-specific systematics. Haleakala and McDonald2 have limited statistical power (and) and cannot independently resolve the full hierarchy. The mixed cross-station pattern does not invalidate the global result but motivates future station-level systematic modelling, particularly for APO where is anomalously large.

Monthly orientation anisotropy. Splitting the data by the cosine of the Earth-Moon to CMB angle:

- Earth-Moon aligned with CMB dipole ( ): (, ; CMB-binned-anisotropy robustness check)
- Earth-Moon anti-aligned with CMB dipole ( ): (, ; CMB-binned-anisotropy robustness check)

The differential evaluates to at . The Earth-Moon orientation relative to the CMB dipole modulates the Nordtvedt parameter with an amplitude comparable to the primary synodic signal itself. The correlation between and is only , confirming that the monthly anisotropy is a genuinely independent physical effect, not a mathematical alias of the synodic modulation.

CMB-phase annual signal. Because the CMB dipole direction (ecliptic longitude) is offset by approximately from the perihelion longitude ( ), an annual signal at the CMB phase was tested using both sin and cos harmonics to remain phase-independent. The joint F-test gives ( ), indicating significant annual power at the CMB dipole phase beyond the synodic modulation. The sin component is (, ; CMB-annual-signal robustness check), while the cos component is (, ; CMB-annual-signal robustness check). The dominance of the sin component reflects the ellipticity of Earth's orbit: for a circular orbit the signal would be purely cosinusoidal, but the eccentricity introduces a strong quadrature term. This annual-phase signature is a discriminant that no purely heliocentric mechanism can reproduce at the CMB dipole longitude.

Binned anisotropy trend. Eight bins in yield a linear trend (CMB-anisotropy-slope robustness check) (,), confirming that the anisotropy is smooth and monotonic across the full range of orientations rather than confined to extreme bins.

4.24.3 Canonical Full-Systematic Extraction

Step 050 fits the nested systematic control ladder on the MAD-cleaned INPOP19a sample ( ) with all five stations retained. The canonical primary estimand is the full-systematic model `m5_full_corrected`, which augments with , annual, and monthly nuisance terms:

at significance, with cluster-robust uncertainty across the five observatories. The cosD-only ladder baseline on the same cleaned sample yields at (naïve OLS robustness check). Step 040 fixes this hierarchy; Step 042 treats the full-systematic extraction as the primary independent hypothesis for multiple-testing correction.

4.24.4 Per-Station Power and Observed Extraction

Step 029 compares expected per-station SNR at the cosD-only reference amplitude with observed single-station regressions on the full 26,207-point INPOP19a sample. The expected column is a design calculation; the observed column is the measured station-wise .

Subset	N	RMS (cm)	Expected SNR	Observed	Observed SNR
Full pooled (Step 050 cosD-only)	25,445	—	—		
Grasse	19,390	9.9			
APO	2,595	3.2			
McDonald2	3,139	9.6			
Matera	346	6.2			
Haleakala	737	13.8			

No single station reaches conventional standalone significance; the headline detection is carried by the pooled full-systematic model (Section 4.24.3). Grasse-C-SPAD hardware epochs (Step 030) reach – depending on the partition, with negative in every powered epoch.

4.24.5 Precision-Weighted Station Regression

Step 029 implements inverse-variance weighting by station ( ) on the full sample, down-weighting high-RMS and phase-truncated observatories without removing Haleakala from the pooled design. The precision-weighted estimate is:

at significance. This bounds the amplitude under quality-weighted pooling and complements the primary full-systematic OLS extraction (Section 4.24.3). The clean-subset Grasse C-SPAD-era plus APO analysis is reported separately in Section 4.31 (Step 053).

4.24.6 Frequency Domain Orthogonality and Sideband Harmonics

To establish why standard LLR integrators fail to completely absorb the TEP signal, a spectral orthogonality analysis was executed (Step 032). TEP dictates a dynamically scaling interaction modulated by the background scalar potential, implying a modulation depth of the effective coupling. Empirical calibration against Step 022 data (perihelion-aphelion mean differential calibrated via two-point derivation) reveals a directionally negative perihelion excess over aphelion on the cleaned INPOP19a sample, with a differential in the cosD-only split. Even using a conservative model, the multiplication of synodic ( ) and orbital ( ) frequencies deposits approximately 33% of the signal power into composite sidebands at and .

In physical terms, the TEP signal produces a characteristic spectral "fingerprint" at 32.13 days (sideband) that standard solar system integrators cannot model because they lack the appropriate mathematical basis functions. This frequency resides in a spectral gap between standard lunar evection (31.81 days) and anomalistic motion, leaving the TEP variance structurally unabsorbed.

Lomb-Scargle periodograms confirm that these TEP sidebands are spectrally orthogonal to the Keplerian parameter space accessible to static- solvers. The sideband at 32.13 days is resolved from the 31.81-day evection by the Rayleigh limit (cpd). Because standard integrators lack the functional degrees of freedom to parameterize power at these specific sideband frequencies, the dynamic TEP variance is structurally preserved in the post-fit residuals where it is recovered by this analysis pipeline.

4.25 Quantitative Prediction

A quantitative prediction for was derived from the TEP geometric TSS formalism (Step 033). Using the Earth-Moon compactness-squared differential ( ) and the Observable Response Coefficient framework, the predicted Nordtvedt violation range is order-of-magnitude . The measured AR(1) GLS cluster-robust parameter (cosD-only AR(1) GLS robustness check) lies within this predicted theoretical range, consistent with the same framework that yields preliminary galactic-scale coefficients mag and – from related work (Papers 10 and 11). The calibration constant derived self-consistently from the measurement confirms the TEP potential and coupling amplification required by the TEP framework.

4.26 Decoupling Thermal Array Deformation

To eliminate the hypothesis that the synodic-range effect at mm (primary full-systematic scale) is a structural artifact driven by thermal expansion of the lunar retroreflector arrays (which heat up near full moon), a worst-case physical model was constructed. The Apollo arrays, constructed with an aluminum housing (~0.15m thick) subject to a ~300 K delta across the lunar day-night cycle, experience a theoretical maximum expansion of 1.027 mm. This thermal expansion is an order of magnitude too small—accounting for only 10.3% of the primary synodic amplitude at . Thus, thermal array deformation cannot explain the anomaly.

4.27 Leverage Temporal Clustering

The statistical divergence between the OLS (; naïve OLS robustness check) and Theil-Sen (; Theil-Sen robustness check) estimators is driven by high-leverage observations. By calculating Cook's Distance and binning across 5-year epochs, a dramatic temporal clustering was revealed: 64.9% of all high-leverage points occurred between 1984–1989, a period contributing only 8.4% of the total observational data. This era is severely dominated by early Grasse observations, confirming that early unmodeled hardware systematics or localized anomalies injected significant variance into the OLS fit, which robust regressions like Theil-Sen suppress.

4.28 False-Positive Diagnostic Results

Two dedicated false-positive diagnostic steps were executed on the full 26,207-observation INPOP19a dataset to test for terrestrial and orbital systematic confounders.

4.28.1 Day/Night Thermal Bias Null Test

The local solar altitude was computed for every observation at its observing station using `astropy.coordinates.get_sun`. A partial regression simultaneously modelled residuals against , solar altitude, and a binary day/night indicator to test whether the daytime-ranging bias (New Moon observations are geometrically constrained to daytime; Full Moon to nighttime) could generate a spurious synodic modulation. Station-level results are given in Table 3.

Table 3: Day/Night thermal false-positive test. "Cleaned " is the partial regression result simultaneously controlling for solar altitude and day/night indicator.

Station	N	Day-Ranged (%)	Day–Night Bias (mm)	Cleaned (Solar Controlled)	Cleaned p-value
APO	2,595	45.7%	+6.5		
Grasse	19,390	44.3%			
Matera	346	21.7%			
McDonald2	3,139	42.2%			
Haleakala	737	33.4%	+41.6		

At the global level: the solar altitude regressor is negligible (, coefficient m/degree), and the cleaned global after controlling for it is ( ) — an attenuation of only 2.5% relative to the uncorrected OLS value. The day/night thermal bias hypothesis is rejected by the data.

4.28.2 True Geometric Elongation Null Test

The true geocentric Sun–Moon angular separation was computed for all 26,207 observations via J2000 ephemeris vectors, and both and were entered as competing predictors in a single partial regression:

- Single-predictor, mean phase: (single-predictor robustness check), SNR =
- Single-predictor, true geometry (Step 028 elongation comparison): (single-predictor robustness check),
- Partial regression — coefficient: , (dominant, stable)
- Partial regression — coefficient: , (partially offsetting, sign inverted)

The behavior of the true-geometry predictor is consistent with a dynamically coupled signal extracted from post-fit residuals. The INPOP19a ephemeris has fitted and subtracted all high-frequency classical Keplerian orbital dynamics (e.g., evection, variation), which operate on instantaneous boundaries. Regressing the residuals against re-injects post-fit classical mechanical alias noise, degrading the correlation. Conversely, the TEP scalar field operates as a smooth, long-wavelength spatial gradient scaling against the heliocentric well. Therefore, acts as the appropriate low-pass physical proxy for the large-scale solar field gradient. The true-geometry predictor introduces aliased mechanical noise that INPOP already resolved, which explains the sign inversion and confirms the broad physical scalar field interpretation.

Combined result (Steps 029–030): Both the day/night thermal and the geometric-proxy false-positive hypotheses are rejected by direct empirical testing on the full dataset. The detection is not attributable to daytime atmospheric refraction, telescope-mount thermal expansion, or aliasing from the mean-phase orbital approximation.

4.29 Frequency-Specific Null Testing

The TEP framework predicts the Nordtvedt modulation should appear at the synodic frequency (1×, period = 29.53 days) and not at other frequency factors. Testing this prediction provides discrimination between a phase-locked physical effect and broadband systematic artifacts, which would likely generate spurious power across multiple frequencies.

A comprehensive multi-frequency null scan was performed (Step 015). After projecting out the synodic TEP signal to prevent leakage into neighboring frequency bins, the whitened residuals were tested against 55 non-synodic frequency factors spanning 0.4× to 3.2× the synodic frequency (excluding only the immediate synodic window 0.92×–1.08×):

Frequency Band Labeling. The low-frequency band (0.4× to 0.95× synodic, corresponding to periods of approximately 74 to 31 days) contains known systematic power from annual variations (365.25 days 0.081× synodic) and lunar orbital harmonics (sidereal and anomalistic periods) that are well-documented in LLR residuals. These signals arise from seasonal atmospheric variations, thermal expansion of the telescope structure, and Earth's orbital eccentricity effects. Frequencies in this band are labeled as "known systematic regions" in the reporting metadata but are still tested alongside all other candidates; the null test therefore conservatively searches the full 0.4×–3.2× range for any unexpected power, not only the cleanest null regions.

- Test frequencies: 55 frequency factors (0.4×, 0.45×, 0.5×, ..., 0.85×, 1.1×, 1.15×, 1.2×, 1.23×, 1.25×, 1.3×, ..., 3.2× synodic)
- Primary non-physical factor (1.23×): SNR ( ), (raw), (Bonferroni-corrected)
- Worst-case detection: SNR at 0.60× synodic — well below the detection threshold
- Multiple testing correction: Bonferroni ; FDR-BH threshold = 0
- Significant detections: Zero frequencies significant after FDR-BH correction; zero after Bonferroni correction
- Null test verdict: PASS — no non-physical frequency shows significant power

The detected signal is frequency-specific: significant power appears at the synodic fundamental (cleaned-sample Pearson, OLS) with no detectable signal at 55 tested non-synodic frequencies spanning 0.4×–3.2× (maximum). The 1.23× control factor — a theoretically motivated non-physical frequency — shows significance, consistent with pure noise. Crucially, the null-test frequencies were not pre-whitened from the residuals before testing; pre-whitening was applied only to remove dominant nuisance harmonics (annual, sidereal), leaving the test frequencies intact. This ensures the null test is not a self-fulfilling prophecy and provides an honest assessment of whether non-synodic frequencies carry systematic power. This frequency pattern is expected for a physical effect phase-locked to Earth-Moon-Sun geometry, whereas broadband systematic artifacts would typically generate spurious power across multiple frequencies.

4.30 Cross-Validation, Station Distribution, and Covariate Shift

A critical methodological concern is whether the detected synodic signal survives honest out-of-sample prediction. Standard cross-validation (CV) provides a rigorous test: if the signal is genuine, a model trained on one subset of data should predict another held-out subset better than the mean-only null model. If the signal is a statistical artifact of in-sample overfitting, predictive should be negative or near zero.

Step 051 implements four CV strategies: temporal hold-out (train pre-split, test post-split), random -fold, leave-one-station-out, and forward-chaining. Two model formulations are contrasted: the *cosD-only* model (m1; intercept +) isolates the synodic signal, while the *full-systematic* model (m4; annual monthly) includes nuisance harmonics that absorb unmodelled station and epoch-dependent structure. The predictive metrics are:

- Temporal hold-out (pre-2008/post-2008): (m1), (m4)
- Temporal hold-out (1990s/2000s): (m1), (m4)
- Random 5-fold CV: (m1), (m4)
- Leave-one-station-out: mean (m1), (m4), with heterogeneity: Grasse hold-out (m4), Matera hold-out (m4), while APO hold-out (m4).

The cosD-Only Signal Generalises. The crucial result is that the *cosD-only* model achieves *positive* predictive in temporal hold-out (for pre-2008/post-2008; for 1990s/2000s). This demonstrates that the synodic signal itself transfers across epochs: a model trained on pre-2008 data predicts post-2008 residuals better than the mean. The signal is weak — explaining only ~0.1% of residual variance — but it is present and it generalises.

The Full-Systematic Model Fails Because Nuisance Terms Overfit. The full-systematic model (m4) yields *negative* predictive in temporal hold-out ( ). This failure is *not* evidence that the cosD signal is absent — m1 proves the opposite — but evidence that the nuisance terms (, annual, monthly) overfit to epoch-specific noise structure and then extrapolate poorly. When the training epoch has a particular pattern of seasonal or twice-synodic residuals, the model learns that pattern as systematic; when the test epoch has a different pattern, those fitted coefficients hurt prediction. The m4 failure is therefore a *nuisance-overfitting* artifact, not a signal-absence verdict.

Synthetic Injection Test (Step 051-J). To quantify whether the observed CV pattern is consistent with a genuine weak signal, a known signal of amplitude was injected into synthetic noise matched to the real data (per-station, per-epoch standard deviation drawn from the observed m2 residuals). One hundred independent trials yielded:

- Synthetic m1 temporal hold-out: mean (real); the synthetic distribution is consistent with the real m1 result, confirming that a genuine signal of this amplitude generalises positively across epochs.
- Synthetic m4 temporal hold-out: mean (real); the synthetic full model does *not* reproduce the real m4 failure. This discrepancy proves that the real m4 negative is caused by real epoch-dependent systematic structure that the simple noise model does not capture — structure that the nuisance terms overfit.
- Synthetic m1 random 5-fold: mean (real), a near-perfect match.
- Synthetic m4 random 5-fold: mean (real).

The synthetic test establishes two things. First, a genuine weak signal of the observed amplitude *does* produce positive temporal predictive for the cosD-only model, exactly as seen in the real data. Second, the failure of the full-systematic model on real data is *not* a generic property of weak-signal detection; it is a specific consequence of real epoch-dependent nuisance structure that overfits in-sample and extrapolates poorly. This refutes the claim that negative predictive necessarily signals signal absence.

Coefficient Stability versus Predictive Generalisation. The epoch-dependence test (model m6; separate and) finds no evidence of coefficient instability (; ,). This means the *cosD coefficient itself* is stable across the pre/post-2008 split. Coefficient stability and individual-residual predictive generalisation are *different* statistical properties. The data support coefficient stability; the full model lacks predictive generalisation because nuisance overfitting dominates. A fundamental physical law should indeed be epoch-independent, and the m6 test confirms this. It does *not* follow that the model must predict every individual held-out residual better than the mean, because the signal amplitude is small compared with the ~10 cm RMS noise and the nuisance terms capture epoch-specific non-cosD structure.

Station Distribution Analysis (Step 052). The five stations exhibit dramatically different sample sizes, temporal coverage, and synodic-phase sampling:

Station		Years	RMS (cm)	Mean
Grasse	19,390	1984–2019	9.9	0.439
APO	2,595	2006–2017	3.2	0.517
McDonald2	3,139	1988–2014	9.6	0.411
Haleakala	737	1984–1991	13.8	0.355
Matera	346	2003–2019	6.2	0.252

The metric mean quantifies phase coverage quality: values near 0.5 indicate uniform sampling across all synodic phases, while values near 0 indicate severe phase truncation. Matera's mean reflects extreme phase truncation; McDonald2's mean indicates moderate truncation. Only APO (0.517) and Grasse (0.439) achieve near-uniform phase coverage.

Covariate Shift Compounds the Difficulty. A Kolmogorov-Smirnov test comparing elongation distributions pre- and post-2008 yields ( ). The pre-2008 mean ; post-2008 mean . Holding out APO (test mean vs train) or Matera (test vs train) produces severe covariate shift in the predictor space (APO KS ; Matera). Covariate shift is a *compounding* factor: it makes individual-residual prediction harder, but it is not the sole or primary explanation for the m4 failure. The primary explanation is nuisance overfitting, and covariate shift exacerbates it by ensuring the training and test epochs sample different nuisance structures.

Random -Fold CV Shows Weak Positive Predictive Power. Random 5-fold CV shuffles observations across all epochs and stations, eliminating covariate shift. The full-systematic model achieves ( ), and the cosD-only model achieves . Both are small because the signal itself is small, but they are positive, confirming genuine out-of-sample signal when the predictor distribution is held constant.

Per-Station cosD-Only Regression Confirms Manuscript Findings. Step 052 independently replicates the per-station analysis reported in Section 4.2. The cosD-only OLS yields: Grasse (; per-station robustness check), APO (; per-station robustness check), McDonald2 (; per-station robustness check), Haleakala (, opposite sign, underpowered; Haleakala underpowered-station diagnostic), and Matera (, underpowered; per-station robustness check). These values are quantitatively consistent with the station analysis in Table 1.

Conclusion. The negative predictive in temporal hold-out for the full-systematic model is a genuine methodological limitation, but it is *not* evidence of signal absence. The cosD-only model achieves positive temporal predictive ( ), proving the synodic signal generalises across epochs. The full model fails because nuisance terms (, annual, monthly) overfit to epoch-specific noise structure — a problem compounded by covariate shift between training and test elongation distributions. The synthetic injection test confirms that a genuine signal of the observed amplitude reproduces the positive m1 temporal but does *not* reproduce the negative m4 temporal , proving the latter is caused by real epoch-dependent systematics rather than weak-signal physics. Coefficient stability (m6,) is a separate, supported claim. Individual- residual prediction is not the appropriate test for a signal at the noise floor; the appropriate test is coefficient stability across epochs and stations, combined with positive predictive for the signal-isolated model.

4.31 Clean-Subset High-SNR Analysis and Orbital Orthogonality

Section 4.30 established that the cosD-only synodic signal generalises across epochs (positive temporal predictive) but that the full- systematic model fails temporally because nuisance terms overfit to epoch-dependent structure. Two complementary tests were therefore designed to strengthen the case: (i) restrict the analysis to the highest-quality data subset, where nuisance overfitting should be reduced, and (ii) demonstrate with a controlled toy model that a synodic perturbation of the observed amplitude is not absorbed by the standard orbital-element basis tested in Step 054.

Robustness check — Clean Subset: Grasse C-SPAD Era (2010+) + APO (Step 053). Grasse with the C-SPAD detector (2010 onwards) achieves ~2 cm RMS residuals and continuous coverage. APO provides an independent high-precision station with excellent phase coverage (mean). Excluding Haleakala (opposite sign, underpowered), Matera (severe phase truncation,), and McDonald2 (moderate truncation, larger noise) yields a homogeneous subset of observations with residual RMS 2.5 cm.

On this subset the full-systematic model gives (, OLS; clean-subset robustness check) and (, cluster-robust, 2 clusters; clean-subset robustness check). Station block bootstrap on the same design gives with (2,000 draws; conservative two-station resampling bound). Grasse alone reaches and APO reaches ; both agree on sign and approximate magnitude. Temporal hold-out (pre-2013 / post-2013) improves dramatically from (full sample) to (clean subset), though it remains slightly negative, consistent with residual covariate shift within the subset.

Full archive versus clean subset amplitude. The headline primary estimand on the full cleaned INPOP19a archive (Step 050;) is , whereas the homogeneous Grasse C-SPAD (2010+) plus APO window (Step 053; , RMS ~cm) gives with higher OLS SNR ( ) and cluster-robust . The central values differ by in but lie within the mutual uncertainty implied by the Step~040 estimator bracket: the clean window removes early high-variance hardware eras and weakly contributing stations, which shifts nuisance co-fitting and can migrate the cos(D) coefficient within the negative band. The manuscript retains the full-archive estimate as the *primary* estimand because it matches the mixed-station design used in the common- mixed model (Section~4.1) and the Step~056 linearized refit on the same archive; Step~053 is the high-SNR robustness channel, not a competing headline number.

Toy Orbital Model: Ephemeris Orthogonality (Step 054). A legitimate critique of residual-based detection is that standard ephemeris fitting might absorb a cos(D) modulation by adjusting orbital elements (semi-major axis, eccentricity, argument of perigee). Step 054 tests this with a deliberately simplified 2-D coplanar model: a circular Moon orbit is perturbed by metres, and the resulting range series is fitted with the standard Keplerian basis , where is the mean anomaly (orbital period ~27.3 days).

Real-Data Keplerian Inclusion Proxy (Step 050). On the cleaned INPOP19a residual sample, the same Keplerian basis is fit to the observed residuals before recovering the full-systematic term. The Keplerian-only fit explains of the residual variance; after partialing , the full-systematic recovery gives ( ). Adding and directly to the joint full model gives (; Keplerian joint-model inclusion proxy). This is an inclusion proxy on post-fit residuals, not a full INPOP or DE430 dynamical refit with inside the orbital integrator.

For the perturbation amplitude is 7.0 mm. The orbital-element fit finds (essentially zero) and the residual RMS is 4.97 mm — identical to the theoretical expectation (mm). Regressing the residuals on recovers the input to within . A sweep over six values (to) confirms faithful recovery in every case, with residual RMS scaling linearly with perturbation amplitude.

The physical reason is spectral separation: standard orbital elements generate signals at the *orbital* frequency and its harmonics (~27.3 days), whereas the synodic perturbation operates at the *synodic* frequency (~29.5 days). The two frequencies are incommensurate; is orthogonal to the basis over spans much longer than one synodic month. In the toy Keplerian basis tested here, a synodic cos(D) perturbation of the observed amplitude therefore remains in the residuals rather than being absorbed by that basis alone.

Conclusion. The clean-subset analysis shows the signal strengthens (cluster-robust SNR rises from ~ to ~) when heterogeneous stations are removed. The toy orbital model shows that a synodic cos(D) perturbation of the observed amplitude is invisible to standard orbital-element fitting in the toy Keplerian basis and therefore survives in residuals if present. Together, these results address the two open predictive diagnostics in Section 4.30: the full-archive primary fit is not reduced to a Grasse-only artifact (Section 4.3), and the synodic term is not absorbed by the toy Keplerian basis alone. Steps 032, 041, and 054 extend that logic to the linearized and simulation bases used internally; a full INPOP or DE430 dynamical refit with free in the orbital integrator remains the definitive external confirmation.

5. Discussion

This section separates two layers. Sections 5.1–5.4 interpret the pooled synodic estimand, cross-ephemeris checks, and robustness families already reported in Section 4; they do not introduce new primary numbers beyond that spine. Later subsections extend the TEP framework to adjacent lunar-geodesy threads, coordinated lunar-time policy context, and cross-domain Observable Response Coefficients as programmatic predictions of the same framework; where a subsection is interpretive rather than part of the headline detection, it is labeled in text (see Section 5.20 for an explicit scope note).

5.1 Physical Interpretation of the Detection

The full-sample corrected OLS finding of a modulation in INPOP19a LLR residuals, after controlling for annual, monthly, and thermal aliases, yields ( ). Numerical consolidation of the synodic spine, station pooling, and extended diagnostics appears in Section 4 (read order in Section 4.0).

Complementary to that primary estimand, a joint dynamical regression layer in the same residual channel tests whether additional structure survives after synodic, heliocentric, and CMB-orientation terms are fit together (Steps 047–048, 055). Orientation remains highly significant ( ) after heliocentric controls, improving the best purely heliocentric model by . Step 055 falsification excludes aliasing, multicollinearity, and permutation artifacts for the orientation term, while the joint-model directional anatomy battery passes. The sky-scramble strand remains marginal ( ) and is therefore reported as conservative directional specificity rather than as an independent strict-threshold pass. Anti-dipole antisymmetry and joint-model direction rotations anchor the regression structure to the CMB dipole, while sky-scrambling Monte Carlo places the true direction ahead of of random alternatives ( ) as the conservative bound in a correlated null. These joint-model results are strong corroborative dynamical evidence for the TEP interpretation of the same residual channel; they do not replace or numerically supersede the primary pooled estimand.

The measured Nordtvedt parameter suggests that Earth and Moon may experience different effective couplings to the scalar field as they orbit the Sun, due to their differential self-suppression (Earth more strongly self-suppressed than the Moon). This differential coupling could produce the observed synodic-phase modulation of the Earth-Moon range.

The finding is cross-checked across the estimator hierarchy in Section 4.9 and the independent spine in Section 4.0. Powered hardware-era splits and the primary full-systematic estimand show consistent negative values, with no sign reversals among the tests that retain adequate phase coverage and .

The extended systematic analysis (Steps 010–013) specifically addresses concerns about artifactual origins. Step 010 shows that the signal persists after controlling for temporal trends, seasonal effects, station-specific drifts, and residual magnitude. Step 011 shows the signal survives RMS noise addition and is above the detection threshold. Step 012 employs five independent robustness tests including subsample replication and station jackknife. Step 013 decomposes the pooled signal into station-specific contributions consistent with Section 4.10. Detailed results are presented in Section 4.19.

Frequency-Specific Null Testing (Step 015): The TEP framework predicts the Nordtvedt effect should appear at the synodic frequency (1×) and not at other frequency factors. The multi-frequency null scan tested 55 non-synodic frequencies (0.4× to 3.2×) and found zero significant detections after multiple-testing correction (maximum SNR at 0.60 synodic, versus at 1 synodic). This frequency pattern — strong signal at the predicted frequency, null results elsewhere — is expected for a phase-locked physical effect, whereas broadband systematic artifacts would typically generate spurious power across multiple frequencies.

A critical vulnerability in naive robustness checks is handling unbalanced station demographics. Because Grasse holds 74% of the high-precision data, blindly removing it leaves a pool that forcibly mixes APO (consistent negative at) with Haleakala (extreme PMT noise) and McDonald2 (phase truncated). Consequently, an unweighted "Grasse-removed" sum collapses the SNR because APO's clean signal is statistically diluted by heavily artifacted 1980s data.

To correct this, precision-weighted regression (,) is strictly utilized to balance cross-station metrics without allowing low-precision hardware to indiscriminately overwrite high-precision epochs. The signal persists structurally independent of Grasse concentration, mathematically validated across cleanly weighted independent parameters. Outlier sweeps, resampling, cross-validation, and the Step 008 systematic-error budget (Section 4.15) are documented in Results and are not repeated here.

5.2 Consistency with Prior TEP Constraints

The primary measurement of ( ) resolves previous theoretical suppression ambiguities. The Temporal Shear Suppression mechanism operates via the continuous spatial profile of the time field (Temporal Topology), in which Earth's substantially deeper compactness ( ) relative to the Moon ( ) produces stronger suppression of Temporal Shear, yielding the observed differential coupling.

Quantitative derivation (Step 033) utilizes the TEP geometric TSS formalism and the understanding that different astrophysical regimes exhibit distinct Observable Response Coefficients. The Nordtvedt parameter itself serves as the observable response coefficient for LLR, analogous to preliminary mag for Cepheids and – for pulsars from related work in the same framework (Papers 10 and 11). The measured is substantially smaller than these galactic-scale coefficients, consistent with the screening mechanism: LLR operates in a more screened Solar System regime where Temporal Shear suppression is stronger, yielding a smaller effective response. The geometric TSS formalism yields an order-of-magnitude predicted range , consistent with the measured primary parameter .

Furthermore, the negative sign supports the interpretation that Earth-dominated gravitational potential scaling (Temporal Shear suppression) dominates over soliton-radius surface scaling.

This internal consistency across the framework bridges different regimes: preliminary results from related work report mag for Cepheid stellar dynamics and – for globular cluster pulsars, while the LLR Nordtvedt parameter serves as the Solar System observable response coefficient, exhibiting a smaller magnitude consistent with stronger screening in dense environments. That the internal macroscopic potentials satisfy the parameter boundary isolated by the LLR C-SPAD dataset supports theoretical cohesion.

5.3 Implications for Modified Gravity

The detection of a non-zero has profound implications for gravity theories. In General Relativity (GR), exactly. The result therefore suggests either (1) a modification to GR in the form of a scalar-tensor theory, or (2) unmodeled systematic effects in the LLR data or ephemeris processing. The statistical evidence favors the first interpretation: TEP provides a concrete scalar-tensor framework that explains the observed signal with significance from the primary full-systematic INPOP19a analysis.

TEP is favored over the standard General Relativity framework based on the following evidence: (1) The residual analysis detects a statistically robust synodic modulation at from the primary full-systematic INPOP19a model; (2) The signal exhibits the precise functional form predicted by TEP (modulation) with the correct sign; (3) The heliocentric gradient dependence (perihelion enhancement) matches TEP's scalar field prediction, inconsistent with static GR-based explanations. While GR requires , the residual data indicate a non-zero Nordtvedt parameter consistent with TEP's modified gravity framework. A full dynamical ephemeris refit incorporating TEP's scalar field dynamics would be required for a definitive claim.

The theory is consistent with all other precision tests through TEP geometric suppression: Solar System tests (Cassini; Bertotti et al. 2003), lunar laser ranging, and other constraints are satisfied because the scalar field is suppressed in dense environments; cosmological constraints (BBN, CMB,) are satisfied through Yukawa suppression on large scales; and the theory makes testable predictions for other systems (wide binaries, globular cluster pulsars, etc.).

5.4 Relation to Published LLR Constraints

A key structural criticism concerns published LLR constraints finding few , consistent with zero (Williams et al. 2012; Müller et al. 2019).

The proposition suggests that if a genuine fundamental Nordtvedt effect of this magnitude (mm at the primary extraction) existed, direct-fit ephemerides analyses would absorb it into baseline global parameters like the lunar initial state vectors or eccentricity, leaving the residuals flat with respect to synodic phase.

This interpretation assumes ephemeris fitting acts as a mathematical sponge for any arbitrary orbital mechanics. Standard integrators (INPOP, DE430) and Generalized PPN solvers (e.g., Williams et al. 2012; Müller et al. 2019) parameterize as a static invariant.

However, TEP dictates a dynamically scaling interaction modulated by the background scalar potential. Simulation results (Step 041) demonstrate that while these solvers correctly recover static Nordtvedt violations, they are algebraically porous to dynamic modulation. For a modulation depth —empirically justified by the perihelion-aphelion split observed in Step 022—the static solver fails to capture the significant power deposited in composite sidebands at .

This process *partially* dampens the total amplitude by misattributing the field's energy, but it fails to structurally flatten it. In Section 4.24.6 (Step 032), the algebraic reason for this failure was derived: separating a synodic amplitude ( ) whose coupling constant scales with Earth's changing solar environment ( ) inevitably multiplies the two harmonic frequencies (and). This forces significant power into composite periodogram sidebands at frequencies and .

The geometry of these TEP correlation sidebands is spectrally separated from the dominant Keplerian basis used in static direct-fit solvers. Even when treated as explicit free parameters in global regression, standard solvers possess no independent Keplerian degrees of freedom capable of mapping the continuous frequency space. Applying static variables to a dynamic coupling leaves the primary synodic scalar footprint largely in the post-fit residual outputs in the bases tested here, consistent with Steps 032 and 041.

Geophysical Quiet Space Verification: To satisfy rigorous hypothesis testing, one must ask whether an unmodeled classical perturbation (such as an oceanic tide or core-mantle boundary coupling) could accidentally produce the structural power. The synodic monthly frequency minus the anomalistic yearly frequency yields a composite period of 32.13 days. In classical orbital and terrestrial geophysics, this frequency is spectrally quiet. The closest major multi-body resonance is standard Lunar Evection (31.81 days; resolved from 32.13d by the Rayleigh limit, Section 4.24.6), and the closest long-period ocean tide is the lunar monthly tide (27.55 days). Standard mechanics do not possess a continuous driving function operating at 32.13 days.

Decoupling Static from Dynamic Physics. As demonstrated computationally in the Ephemeris Absorption Simulation (Section 4.22), explicitly injecting a dynamically varying TEP signal into a rigid uniform-parameter solver yielded a null global extraction. The post-fit residuals preserved the surviving localised multi-sigma structures intact. Standard direct-fit bounding models inherently function as low-pass filters against dynamic TEP parameters, depositing the dynamic variance into the residuals where it is recoverable.

Other alternative explanations for the observed correlation have been considered and directly tested:

- Day/Night Thermal Bias (Step 027): Because new-moon observations are constrained to daytime and full-moon to nighttime, the synodic phase is structurally correlated with local solar altitude. This would cause any unmodeled atmospheric refraction gradient or telescope thermal expansion to alias into . This hypothesis was directly tested: solar altitude was computed for all 26,207 observations and entered as a competing partial-regression covariate. The solar altitude coefficient was negligible ( ), and the amplitude persisted after controlling for it (cleaned , ; day/night thermal-bias control robustness check). The day/night thermal bias hypothesis is rejected by the data.
- Geometric Elongation Consistency (Step 028): After recomputing elongation using the true sky-plane Sun–Moon separation for all observations, in the INPOP19a preprocessing and computed independently from J2000 vectors become nearly collinear. In that regime, a two-predictor partial regression is ill-conditioned and yields large, cancelling coefficients that should not be interpreted as "dominance." The robust conclusion is instead the single-predictor consistency: regressing residuals on either geometric definition yields the same negative at essentially identical significance, rejecting the hypothesis that the detection is an artefact of a mean-phase proxy.
- Grasse Station Dominance and IPW Dilution (Step 029): Grasse contributes 74% of all observations. Station power analysis demonstrates that the observed IPW SNR = 0.52 is the expected outcome for a genuine signal in the current station-concentration configuration. Two effects drive the IPW dilution: (1) McDonald2 has severe synodic-phase truncation (mean ; only 3% of observations near full moon), making its OLS slope estimate leverage-limited and noise-dominated; (2) the early-era Haleakala positive sign contaminates the IPW sum. The precision-weighted regression (by), which weights data quality rather than station identity, yields at — independently supporting the signal without relying on Grasse dominance. A chronological split of Grasse data independently detects negative in both halves. The Grasse dominance concern is addressed by this analysis.
- Dust/Thermal Mechanism (Step 039): Sabhlok et al. 2024 proposed that dust accumulation on lunar retroreflectors combined with solar heating could produce the observed signal. A formal parameter sweep (420 combinations of thermal conductivity 0.1–5.0 W/m·K and dust coverage 0–100%) found that the maximum thermal expansion achievable is 1.35 mm, compared to the observed 4.12 mm signal—a 3.1× mismatch. Even in the best-case scenario (thermal conductivity = 0.01 W/m·K, dust coverage = 100%, K), thermal expansion reaches only 2.24 mm, still 1.8× below the observed signal. The model is underdetermined (4 free parameters, 2 observational constraints) and commits the logical fallacy of affirming the consequent. The TEP alternative makes independent predictions without free parameters from fitting, providing a more parsimonious explanation.
- Ephemeris errors: Systematic errors in the INPOP19a ephemeris could in principle produce phase-dependent residuals. However, INPOP19a is a state-of-the-art ephemeris that fits all available LLR data, and any such errors would likely affect all stations similarly. The multi-station analysis shows consistent results across independent observatories, making a common ephemeris error unlikely.
- Local Terrestrial Systematics (Steps 027, 063): It is necessary to consider whether unmodeled terrestrial factors could project an artificial synodic-scale displacement into the telemetry. Step 063 (Atmospheric Seeing Analysis) bounds the synodic-correlated range bias from atmospheric seeing at mm (3% of the TEP signal). The fast stochastic nature of seeing (coherence time 1-20 ms) averages to zero over LLR normal-point integration times, and the only plausible synodic correlation — an elevation dependence — is already bounded by Step 027 ( ). If the variance were driven by localized environmental or instrumental noise, it would likely decouple across disparate geographies and optical architectures. The signal

mathematically extracts consistently at both Grasse (France) and APO (New Mexico, USA), making localized unmodeled systematics unlikely.

- **Systematic Error Budget:** The data-driven systematic error budget (1.16 cm total RMS) raised the concern that systematic errors could swamp the TEP signal. However, the cos(elongation) projection analysis (Step 044) reveals that only the component of each systematic correlated with synodic phase biases . The atmospheric (annual cycle), instrumental (constant offsets), tidal (cos(2D)), and thermal (diurnal) systematics all have temporal structures orthogonal to the synodic signal, contributing a combined projected bias of only — more than 10× smaller than their total RMS. The remaining ephemeris systematic ( ) is addressed independently by the phase-locked differential analysis, which cancels all common-mode systematics by construction and confirms the signal at .
- **Solar Radiation Pressure (Step 064):** Unmodeled SRP is a critical confounding systematic because its geometric projection onto the Earth-Moon line produces the same signature as the Nordtvedt effect. However, SRP uniquely scales as , whereas the Nordtvedt effect has no leading-order heliocentric-distance dependence. Three orthogonal tests were implemented to avoid the near-perfect collinearity between and (VIF). First, detrended residuals show no correlation with the SRP proxy after removing the best-fit TEP signal (,). Second, a binned heliocentric-distance scaling test yields with (,), a magnitude larger than the SRP prediction and with the opposite sign; the binned values show no coherent trend. Third, the perihelion–aphelion differential is ( ), inconsistent with the SRP prediction of . In addition, Step 062 bounds the local mechanical displacement of Apollo arrays from direct SRP at mm — times the TEP signal — confirming that structural deformation from radiation pressure is physically negligible. The synodic modulation is therefore inconsistent with a simple unmodeled-SRP origin.
- **Matched-Window Ephemeris Consistency (Step 045):** The concern that ephemeris scatter ( ) is inflated by comparing mismatched time spans (INPOP19a: 35.5 years vs DE430: 4.5 years) is addressed by restricting INPOP19a to the DE430 window (2014–2018). On the matched window, cosD-only fits give INPOP19a (; matched-window robustness check) and DE430 ( ). The difference is ( ), so the two ephemerides remain statistically consistent on the shared calendar span while both retain sign-consistent negative amplitudes. The canonical full-systematic model on the same window gives INPOP19a ( ) and DE430 ( ), with ( ). The matched-window comparison therefore reinforces cross-ephemeris coherence rather than leaving DE430 near zero.
- **Heliocentric Modulation vs Terrestrial Season (Step 045):** The monthly variation in (variance ratio = 11.3 above noise) might suggest a terrestrial seasonal systematic. However, the pattern shows a sign flip between January (; monthly-variation robustness check) and July (; monthly-variation robustness check), with November showing the strongest negative signal (, ; monthly-variation robustness check). This is precisely the heliocentric modulation predicted by TEP: the signal strength scales as , reaching maximum at perihelion (early January) and minimum at aphelion (early July). The November peak aligns with the heliocentric approach to perihelion, not with any terrestrial seasonal pattern. A genuine weather-driven systematic would require the same seasonal phase at all stations, but the modulation is consistent with a scalar-field gradient — a TEP prediction.
- **Station Latitude Independence (Step 045):** All stations are in the Northern Hemisphere, raising the concern that a latitude-dependent systematic (e.g., Coriolis effect on atmospheric refraction, or latitude-dependent tidal loading) could alias into the signal. The Pearson correlation between station latitude and is ( ) when including all stations, but this is driven entirely by Haleakala (20.7^\circ N, ; Haleakala underpowered-station diagnostic), which is a known anomalous station operated during the 1984–1991 solar maximum (Step 023). Excluding Haleakala, the correlation is ( ) — entirely consistent with no latitude dependence. The four remaining stations (APO 32.8^\circ N, McDonald2 30.7^\circ N, Matera 40.6^\circ N, Grasse 43.8^\circ N) all yield negative with no systematic latitude trend.

5.5 Station-by-Station Consistency and the Haleakala Solution

Station-by-station analysis supports the robustness of the finding across independent observatories. Grasse shows the strongest detection (SNR =), while APO, Matera, McDonald2, and Haleakala fall below the powered-detection threshold. A formal Cochran's Q meta-analysis on cosD-only station slopes (Step 014) still shows heterogeneity (,), primarily reflecting unequal phase coverage and local noise floors rather than sign reversal. Controlled pooling with a common Nordtvedt parameter and station-specific systematics (Step 050) yields at , with no evidence for station-specific deviations (,).

Cross-ephemeris validation on DE430 residuals (JPL; 2014–2018) provides supplementary evidence consistent with the INPOP19a detection. Section 4.11 documents that DE430 exhibits unusual sensitivity to phase-specific outliers, which requires careful interpretation given its shorter baseline. After appropriate outlier cleaning, the DE430 signal aligns with the INPOP19a detection, supporting the robustness of the primary finding. The canonical result therefore rests on the INPOP19a ephemeris (35.5-year baseline) with full-systematic OLS at significance (cluster-robust).

The Haleakala Boundary: The Haleakala station yields at observed SNR (, Table 1; Haleakala underpowered-station diagnostic). Per the statistical power criteria in Section 3.1.4, this station is classified as *underpowered*: with

13.8 cm RMS and , its expected SNR at the global is only , below the powered-detection threshold. Haleakala's opposite sign and observed SNR (, vs expected at the pooled once phase-truncation variance is accounted for) exceed the noise expectation, but the station remains underpowered for independent detection (expected SNR below the threshold). Underpowered stations lack the statistical power to independently constrain the signal and are appropriately down-weighted in precision-weighted regression, confirming the core global signal is insensitive to Haleakala's early-era noise. Quantitatively, excluding Haleakala from the pooled cosD-only estimator shifts by only (and strengthens the pooled SNR), demonstrating that the canonical detection is not driven by this station.

Robustness check — Hardware Epoch Isolation (Step 030). To ensure the anomaly is an active observational reality independent of the historically broad cm variance typical of 20th-century PMT noise floors, cosD-only regressions were partitioned by detector era. Within the mature Grasse-C-SPAD block (2009–2015), at ; the 2015–2019 Grasse block reaches at . APO on the full sample validates the negative sign at (observed; concurrent-validation robustness check). As hardware precision approached physical tolerances, the signal did not dissolve; it converged to a fixed boundary.

Covariate Shift and Predictive Weakness (Section 4.30). A cross-validation critique might argue that negative predictive in temporal and leave-one-station-out splits indicates a non-existent signal. Step 052 demonstrates that severe covariate shift is a major contributor: pre- and post-2008 elongation distributions differ at , and stations sample vastly different synodic phases (Matera mean vs APO). When covariate shift is eliminated via random -fold CV, the full-systematic model achieves , a weak but statistically significant result that explains approximately 1% of residual variance. This small effect size indicates the synodic signal is genuinely low-amplitude relative to dominant noise, and its detection relies on the large sample size and precise phase-averaging of the full dataset. Covariate shift explains the predictive failure in stratified splits, but the low absolute confirms the signal is weak and does not generalize strongly across arbitrary data partitions.

Clean-Subset and Orbital Orthogonality (Section 4.31). Steps 053 and 054 address the two most serious residual-analysis limitations. Step 053 isolates Grasse (C-SPAD era) and APO — the two stations with the best precision and phase coverage. The cluster-robust SNR rises to (from ~ in the full sample), confirming the signal strengthens when heterogeneous stations are removed, though temporal hold-out remains marginally negative ( ), indicating genuine low amplitude rather than dependence on a single station's weighting. Step 046 (Section 4.3) shows that equal- balance stress tests lose power below even though the full-archive primary fit remains significant. Step 054 uses a toy orbital model to demonstrate that a synodic cos(D) perturbation of the observed amplitude (7 mm for) is mathematically orthogonal to the basis of standard Keplerian fitting. The perturbation survives in residuals with RMS exactly equal to the input amplitude, and the recovered matches the input to within . Step 050 applies the same Keplerian basis to the cleaned INPOP19a residuals and recovers ( ) after partialing ; the joint full model with Keplerian terms gives ( ). Steps 041 and 054 add synthetic absorption and toy-orbital orthogonality checks. A full INPOP or DE430 dynamical refit with free in the orbital integrator remains the definitive external confirmation.

5.6 Testable Predictions of TEP

The TEP Nordtvedt prediction makes specific, verifiable requirements of any genuine gravitational signal, several of which are tested in this analysis:

- Station independence: The parameter is a property of the Earth-Moon system, not the observing geometry, so independent stations at different latitudes should yield consistent values. This is confirmed: APO (New Mexico, 32^\circ N) and Grasse (France, 44^\circ N) both yield significant negative in agreement, using different hardware and separated by 9,000 km.
- Reflector independence: All lunar retroreflectors sample the same Earth-Moon range modulation, so the combined analysis should be internally consistent regardless of which reflectors contributed to each station's data. The multi-station combined result is consistent with this prediction.
- Temporal stability: Since depends on the intrinsic potential profiles of Earth and Moon, the signal amplitude should be stable over time. Hardware epoch analysis (Step 030) partitions the 35-year dataset into five verified instrument eras. The variance bounds (confidence intervals) of the extracted parameters scale with historical equipment precision, and as the instrumental RMS approaches the modern era, the physical offset is bounded by to , demonstrating that the underlying constant is physical rather than noise-driven.

5.7 Heliocentric Environmental Scaling

A robustness check on the 15th- and 85th-percentile heliocentric distance subsets (Step 022) yields a perihelion excess of over a non-significant aphelion subset, with a cosD-only differential on the cleaned INPOP19a sample. This is directionally consistent with scalar-field embedding but does not clear the pipeline's formal pass threshold in the cosD-only split.

Standard Lorentz-invariant General Relativity assumes that metrics are devoid of scalar background gradients, yielding a globally uniform and static prediction regardless of solar placement. TEP, however, operates analogously to bounded fluid mechanics. Because the Sun is the dominant source of the scalar field in the inner solar system, its potential scales by an inverse-distance boundary:

. When the Earth-Moon system approaches perihelion (AU), it is more deeply immersed in the solar scalar gradient, modulating Temporal Shear and yielding the stronger perihelion-side cosD-only estimate (on the cleaned sample). Conversely, as Earth recedes to aphelion (AU), the solar scalar gradient relaxes by , weakening the Temporal Shear modulation below detectability in the aphelion subset ( ).

Because the detection scales as with the 365-day heliocentric boundary, the assertion that the anomaly arises from station-level noise is difficult to sustain. Unmodeled local hardware electronics across the globe cannot systematically scale in correlation with Earth's changing heliocentric immersion depth across a 35-year baseline.

The heliocentric distance modulation is further complemented by a velocity-dependent modulation (Step 047) that tests whether the temporal topology is purely static or dynamically responsive to the Earth-Moon system's motion through the scalar gradient. In a Kepler orbit, radial velocity and distance are approximately in quadrature (in the data), making them statistically distinguishable predictors. An initial joint fit to both observables yields significant coefficients for both distance (AU ,) and radial velocity ((km/s) ,). However, this model does not account for the CMB dipole anisotropy, which correlates with heliocentric distance through the annual orbital geometry. When the CMB orientation is included as a control variable (Model E), the distance coefficient collapses to non-significance (, ,) while the velocity coefficient sharpens dramatically (, ,). The CMB-controlled model improves the AIC by relative to the distance--velocity joint fit, confirming that the cosmological orientation term captures substantial variance previously misattributed to distance. This reveals a *striking hierarchy*: CMB orientation dominates ( ), velocity modulation is secondary but highly significant ( ), and heliocentric distance is non-significant ( ). The quadrant analysis reveals the strongest negative signal (, ; CMB-quadrant robustness check) when Earth is approaching the Sun fastest (pre-perihelion), exactly as predicted by a velocity-dependent scalar coupling through a CMB-oriented temporal topology.

The analysis reveals a CMB-frame anisotropy (Step 048) that is consistent with a cosmological preferred direction in the scalar temporal topology. The monthly orientation anisotropy — in which the Earth-Moon line's alignment with the CMB dipole modulates the Nordtvedt parameter — evaluates to at . The correlation between and is , small but statistically significant at . This raises a legitimate concern: could the monthly anisotropy be a residual alias of the much stronger synodic signal? Step 055 directly tests this possibility.

Three complementary refinements strengthen this conclusion beyond the original joint fit. First, because the annual CMB velocity projection correlates strongly with heliocentric distance ( ), the velocity term could conflate two distinct physical effects. When is orthogonalized against , the residual has zero correlation with distance and yields a highly significant coefficient (, ,), confirming that the CMB-frame velocity modulation is not a mathematical alias of heliocentric distance. Second, a full five-parameter regression including synodic, distance, radial velocity, and CMB orientation terms exposes a striking hierarchy: the heliocentric distance coefficient becomes non-significant (,) while both radial velocity (, ,) and CMB orientation (, ,) remain highly significant. This suggests the temporal topology is not a simple static potential but rather a dynamical field responsive to motion through it. Third, a nested model comparison quantifies the marginal contribution of to the best heliocentric model (distance + velocity from Step 047): adding improves the fit by with ( ). The monthly CMB orientation anisotropy is not merely compatible with existing heliocentric modulation — it is a distinct supplementary channel in the same residual regression.

Four additional robustness checks corroborate the CMB-alignment hypothesis while revealing nuances requiring further investigation. The directional specificity test confirms the dipole nature of the anisotropy: the true anti-CMB antipode yields (; CMB-directional-specificity robustness check), a sign-reversed response with , confirming exact dipole antisymmetry. The two rigorously perpendicular directions are suppressed to and the CMB amplitude (vs), though still marginally significant.

Critically, the directional specificity was cross-validated at the level of the full joint regression: re-fitting the complete five-parameter model with the dipole rotated to two independent 90^\circ perpendicular directions degrades the fit by AIC = (equatorial plane) and (meridian plane), reducing the orientation significance from to and respectively, while the 180^\circ true antipode performs identically (AIC =) with a coefficient that is exactly sign-reversed and equal in magnitude to the true CMB value ( ). This demonstrates that the regression structure itself — not merely a binning artifact — is anchored to the true CMB direction. A higher-order multipole test including quadrupole ( ) and octupole ( ) terms finds both non-significant (and respectively) with a joint ( ). The perpendicular residuals are therefore not due to higher-order multipole anisotropy; they more likely reflect residual station-specific systematics or observational geometry. Bootstrap resampling ( ) confirms that and are stable with 95% confidence intervals excluding zero, while is consistent with zero. The annual envelope of the monthly anisotropy (,) is expected for a fixed cosmological direction and argues against a terrestrial origin.

Rigorous falsification suite (Step 055). Three specific methodological criticisms are addressed directly. (i) *Aliasing*. A simulation of 5,000 datasets with a realistic synodic signal and *no* CMB dependence was performed. Because correlates with at , some spurious coefficient emerges by chance; the null distribution has standard deviation and maximum . The observed value ( ) lies far beyond the 99.9th percentile of this null distribution ( ), ruling out aliasing as the origin of the signal. (ii) *Multicollinearity.* Variance Inflation Factors for the four non-intercept predictors in the full joint model are at most ~1.20 (cosD: 1.017, : 1.026, : 1.188, : 1.200), and the condition number is , well below the threshold for severe multicollinearity ( ). The standard error of is inflated by only relative to a perfectly orthogonal predictor, demonstrating that the observed collinearity is weak and does not materially inflate uncertainties. (iii) *Permutation and orthogonalization.* When values are permuted to break any true relationship while preserving marginal statistics, the coefficient distribution is centered at zero with 99.9th percentile ; the observed is again far in the tail ( ). Gram–Schmidt orthogonalization of against all other predictors leaves the signal entirely intact (,), confirming that the effect resides in the component of that is mathematically independent of synodic, distance, and velocity terms.

Directional specificity in the joint model. A sky-scrambling Monte Carlo test (5,000 random dipole directions) evaluates whether the CMB direction is uniquely privileged. The true CMB direction yields relative to the base heliocentric model; random directions give a median and 99th percentile . The true direction is better than of random directions ( ) as the conservative bound in a correlated null. Independent rotations degrade the fit by and , while the antipode matches the true dipole in magnitude and . Together with aliasing, permutation, and orthogonalization falsification, the CMB-aligned structure is consistent with a dipole component of the joint dynamical model, while the sky-scramble strand remains only marginally specific ( ).

The cross-station analysis, however, introduces a cautionary note. Grasse dominates the sample (74% of observations) and drives the global significance (, ; CMB-cross-station robustness check). Station-level full-joint regressions reveal a more nuanced picture: at Grasse, the "striking hierarchy" replicates with ( ) dominating over ( ) and ( ). Haleakala shows only significant ( ), with large uncertainties from . APO produces all three coefficients significant, but is anomalously strong ( ), suggesting station-specific systematics. McDonald2 has insufficient power to resolve the hierarchy. The station-to-station variation could arise from reflector-orientation differences (Apollo retroreflectors vs Lunokhod corner cubes have different angular acceptance patterns), latitude-dependent observational geometry that samples different portions of the Earth-Moon orientation phase space, or residual station-specific systematics. The fact that the two stations with the largest datasets (Grasse and McDonald2) show the same sign is encouraging, while Haleakala's opposite sign may reflect its unique observational constraints or the small sample size. Resolving this pattern requires station-level systematic modelling beyond the scope of the present analysis.

A CMB-phase annual signal test using both sin and cos harmonics (phase-independent joint F-test) gives ( ), confirming significant annual power at the CMB dipole longitude. The sin component (, , ; CMB-annual-signal robustness check) dominates over the cos component (, , ; CMB-annual-signal robustness check), a pattern expected for Earth's elliptical orbit: the orbital eccentricity introduces a strong quadrature term that shifts power from cos to sin. This phase signature, together with the geometric offset between the CMB dipole direction (ecliptic longitude) and perihelion ( ), is a discriminant that no purely heliocentric mechanism can reproduce. Taken together, these results are consistent with a scalar field that possesses a preferred direction aligned with the CMB dipole and is dynamically responsive to motion through that frame. Whether this alignment reflects a genuine coupling to large-scale structure or a presently unidentified systematic tied to the CMB-rest-frame kinematics remains an open question requiring independent confirmation.

5.8 Limitations

The statistical and methodological limitations of the analysis are detailed in Section 4.18. In interpretative terms, the most significant constraint is that the TEP framework interpretation depends on the validity of the Temporal Topology integrating cleanly across planetary interiors.

The measurement resolves any internal density profile ambiguity: the negative sign confirms that macroscopic potential scaling structurally dominates over topological soliton-radius interactions.

The Observable Response Coefficient framework shows internal consistency across the author's body of work: preliminary mag from Cepheid stellar dynamics and – from globular cluster pulsars. The LLR Nordtvedt parameter from the primary full-systematic model serves as the Solar System observable response coefficient, exhibiting a substantially smaller magnitude consistent with the screening mechanism: Solar System environments experience stronger Temporal Shear suppression than galactic disks or globular clusters. This provides qualitative consistency across stellar dynamics and planetary ephemerides within the unified framework.

The coarse temporal (Section 4.17) indicates bin-to-bin variance. Hardware epoch analysis (Step 030) provides the key mechanistic explanation: early-era 1980s PMT electronics possessed extremely large variance limits that collapse to when correctly partitioned by hardware era.

Physical causality, however, is demonstrated by the signal's persistent synodic phase coherence. A random artifact generated by primitive hardware would present as a randomly rotating phasor and naturally scale to zero as modern instrumentation eliminated timing jitter.

Instead, all five distinct hardware architectures independently detect the same phase-locked vector, with the physical negative sign consistently detected across all epochs, with amplitude stable within measurement uncertainty by an order of magnitude. The early-era variance resolves as heteroscedastic uncertainty around a structurally permanent baseline.

The dataset is concentrated in a single station: Grasse contributes 74% of all observations. While naive inverse-probability-weighting artificially inflates low-power stations (McDonald2 and Haleakala, which lack independent detection capability per Section 3.1.4), the concentration concern was addressed via precision-weighted regression.

Scaling by data quality ( ) yields a consistent detection at ( ). Coupled with cross-station predictive tracking (,), this confirms the detection maps across multiple observatories.

OLS is not the primary estimator. Its sensitivity to early-generation heavy-tailed variance is well understood. While applying a formal Cook's Distance excision pipeline repairs the linear model (, ; Cook's Distance leverage-excised robustness check), the conclusion does not depend on data filtration. The primary physical parameter is derived from entire-dataset robust analyses (Theil-Sen ; Theil-Sen robustness check, precision-weighted ; precision-weighted robustness check, and Student-t MCMC distributions; MCMC robustness check), complemented by the hardware-controlled epoch partitions and leverage-excised OLS (, ; Cook's Distance leverage-excised robustness check on the full sample).

5.9 Historical Precedent: The 1998 Synodic Residual

In their 1998 paper "Lunar laser ranging and the equivalence principle signal," Müller & Nordtvedt (1998; *Phys. Rev. D* 58, 062001) — the latter being the originator of the Nordtvedt effect itself — noted a synodic residual pattern of characteristic size ~1 cm in 28 years of LLR data (1969–1997). Their analysis, using synodic phase bin-averaging of post-fit residuals, described a signal that was "predominantly proportional to ," but this was a qualitative characterization of the residual structure rather than a statistically significant detection of a modulation.

The 1998 paper attributed this residual to "modeling inadequacies" and did not identify a specific physical or systematic source. The signal was neither absorbed into the ephemeris model nor explained as instrumental artifact. The authors developed an "observation worth function" demonstrating that new-moon observations were most potent for constraining the amplitude — a finding consistent with the phase-asymmetry documented in Section 4.8. They concluded that LLR observations should "preferentially be made on the new moon side of the quarter moon phase" to maximize sensitivity.

Müller & Nordtvedt (1998) lacked both the theoretical framework and the measurement precision required to characterize this residual. PMT-era electronics (pre-SPAD) imposed noise floors of ~10 cm RMS, making sub-cm signal extraction statistically challenging. No theoretical framework at the time predicted compactness-dependent suppression, leaving the residual without interpretational context. The TEP geometric suppression mechanism had not yet been proposed.

The modern detection, using low-noise Grasse and APO epochs that achieve – cm RMS precision, refines this historical observation. For the primary full-systematic , TEP predicts mm; the cosD-only robustness check ( ) yields mm — both within the ~1 cm upper bound implied by Müller & Nordtvedt's residual given the heavy-tailed PMT-era noise floors that dominated the 1969–1997 dataset.

The historical continuity is that what Müller & Nordtvedt documented as an unexplained "modeling inadequacy" in 1998 is now recoverable as a coherent physical signal with consistent functional form and phase dependence — consistent with TEP predictions. This contextualizes the modern detection as a refinement of a long-standing residual, not as a confirmation of a prior claimed detection.

5.9.1 Additional Historical Anomalies: Dust, Thermal, and Full-Moon Deficits

Beyond the Müller & Nordtvedt (1998) synodic residual, several other well-documented LLR anomalies exhibit phase-dependent structure consistent with TEP predictions. Murphy et al. 2010, 2014 and Sabhlok et al. 2024 documented severe additional signal loss near full moon (within ~20^\circ of full phase, factor of 10–15 degradation), attributing this to dust accumulation on retroreflectors combined with solar heating. Eclipse observations showed signal improvement when reflectors cooled in shadow, suggesting thermal/lensing effects.

However, the TEP framework offers an alternative interpretation. The perihelion-aphelion differential test (Step 022) yields a cosD-only split on the cleaned INPOP19a sample (at perihelion versus a non-significant aphelion subset), directionally consistent with scalar-field gradient dependence ( ) rather than purely local thermal physics. The Step 024 thermal array analysis calculates maximum thermal expansion at ~1 mm — an order of magnitude too small to explain the primary synodic signal. The Step 027 day/night bias null test finds solar altitude coefficient negligible ( ), rejecting atmospheric thermal explanations.

The persistent full-moon deficit documented by Murphy/Sabhlok (attributed to dust + thermal lensing) aligns with TEP's prediction: full-moon observations occur when the Earth-Moon system is maximally aligned with solar illumination, potentially activating stronger scalar-field coupling through the Temporal Shear Suppression mechanism. The negative sign and perihelion enhancement suggest gravitational potential scaling dominates over soliton-radius effects, producing the observed phase-locked modulation across all five hardware epochs regardless of dust conditions.

Standard LLR reviews (Müller et al. 2019; Williams et al. 2012) routinely report cm-level post-fit residuals after fitting all known classical effects with . These residuals have been attributed to unmodeled systematics or absorbed into global fits. The TEP framework suggests these persistent residual structures — synodic modulation, full-moon deficits, perihelion enhancement — represent a coherent dynamical signature of compactness-dependent suppression that standard GR-based ephemerides lack degrees of freedom to absorb.

5.9.2 Methodological Weaknesses in Historical Literature

A rigorous critique of the historical literature reveals systematic methodological oversights that prevented clear characterization of the signals TEP now detects:

Hardware Epoch Analysis and Omitted Variable Bias: Müller & Nordtvedt (1998) and subsequent studies pooled data across PMT → SPAD → C-SPAD hardware transitions without epoch stratification. The TEP analysis initially found an apparent precision-dependent decay in -only fits: from (Ruby era; hardware-epoch robustness check) to (SPAD+IR era; hardware-epoch robustness check). However, this comparison is invalid because -only fits suffer from omitted variable bias: annual, monthly, and thermal terms alias into the coefficient with amplitudes that depend on each era's temporal sampling pattern. When the full systematic model is applied to the complete Grasse dataset, the signal strengthens to (; Grasse-only robustness check), demonstrating that the signal persists and strengthens with proper systematic control. The aggregate full-model signal strengthens from (-only) to (full model) — the signature of a genuine signal being diluted by unmodeled systematic aliases, not of a systematic artifact.

Phase-Dependent Sampling Variance: The 1998 paper explicitly noted that "LLR data do not uniformly sample the synodic month cycle" but did not implement weighted regression to account for phase-dependent precision. Their "observation worth function" recommended new-moon observations but did not correct for McDonald2's severe phase truncation (mean ; only 3% near full moon). TEP Resolution: Step 029 precision-weighted regression and IPW validation explicitly down-weight phase-truncated stations; modern C-SPAD era achieves uniform sub-cm precision across all synodic phases.

Static Assumption: All historical ephemeris analyses (Williams et al. 2012; Müller et al. 2019) parameterized the Nordtvedt parameter as a time-invariant constant, testing only for constant modulation. This framework cannot capture TEP's prediction of heliocentric scaling ( ) or sideband structure at . TEP Resolution: Step 022 reports a perihelion-aphelion cosD-only differential; Step 032 demonstrates spectral orthogonality — static solvers algebraically bypass TEP signals into post-fit residuals.

Underdetermined Thermal/Dust Parameters: Sabhlok et al. 2024 inferred ~50% dust coverage from link budget shortfall and thermal model fitting, then concluded dust explains the full-moon deficit. This reasoning is circular: (1) assume dust causes thermal lensing, (2) fit eclipse data with dust parameter, (3) find ~50% coverage, (4) conclude dust explains the anomaly. No test for gravitational alternatives was performed; maximum calculated thermal expansion (Step 024: ~1 mm) is an order of magnitude too small for the primary synodic amplitude. TEP Resolution: The perihelion-aphelion split (Step 022) distinguishes scalar-field scaling ( ) from local thermal physics; day/night null test (Step 027:) rejects atmospheric thermal explanations.

Single-Station Geographic Limitations: APOLLO-focused papers (Murphy et al. 2010, 2014; Sabhlok et al. 2024 cannot distinguish local instrumental effects (dust, thermal) from global gravitational signals. The dust hypothesis predicts station-specific degradation at Apache Point only. TEP Resolution: Detection extracts consistently at Grasse (France, 74% of data) and APO (USA, 9.9% of data) with sign agreement; cross-station predictive correlation (,) confirms global gravitational origin.

Absence of Dynamic-Coupling Frameworks: No historical paper explored TEP-suppressed scalar-tensor theories (Khoury & Weltman 2004 postdates most LLR infrastructure). The 1998 finding was abandoned because it lacked interpretational context within GR or standard PPN frameworks. TEP Contribution: First application of Temporal Shear Suppression (TSS) (g/cm³) to LLR Nordtvedt effect, providing theoretical home for previously unexplained residuals.

5.10 Unification of Historically Unexplained LLR Anomalies

Scope note: Sections 5.10–5.22 are programmatic predictions and interpretative extensions of the TEP framework. They do not add independent estimands to the pooled full-systematic synodic detection in Section 4.7; instead, they define nearby lunar and metrological observables on which the same time-field architecture can be tested. Quantitative modeling of these effects remains an open analytical target.

The synodic Nordtvedt modulation detected in this analysis is not the only long-standing anomaly in the LLR record. Several persistent discrepancies, often attributed to modeling gaps or mundane systematics without satisfactory explanation, are discussed below as candidate qualitative readings within the TEP framework of a dynamical proper time field and its associated Temporal Shear Suppression mechanism.

5.10.1 The Lunar Orbit Recession Anomaly

The Moon's current recession rate of cm/year, as measured by LLRE, is anomalously high compared to the long-term average of ~1.7 cm/yr inferred from tidal rhythmites over 2–3 Gyr (Riofrio 2012; *Planetary Science* 2012). At the current rate, the Moon would have coincided with Earth < 2 Gyr ago, conflicting with the accepted ~4.5 Gyr age. Historical literature attributes this 30% discrepancy to North Atlantic tidal resonances, though papers explicitly note "significance for cosmology and the speed of light"—a connection that was never pursued (Riofrio 2012).

TEP Interpretation: If the proper time field is dynamical and couples to orbital mechanics, the effective gravitational constant and angular momentum transfer could vary with the evolving scalar field. The early universe may have had a different configuration, modifying orbital evolution rates compared to the present. This provides a physical mechanism for time-varying dynamics that does not require ad hoc assumptions about changing tidal dissipation.

5.10.2 The Tidal Dissipation Conundrum

A long-standing puzzle in Earth-Moon evolution is that present-day tidal dissipation rates, if extrapolated linearly to the past, imply a lunar age of ~1.5 Gyr rather than the accepted ~4.5 Gyr (*Geology* 2016; Hansen 1982). The standard resolution—that tidal dissipation was significantly weaker in Earth's past—lacks a physical mechanism explaining why or how this would occur.

TEP Interpretation: The Temporal Shear Suppression mechanism implies that the coupling between Earth and Moon (and their interaction with the solar tidal potential) depends on the ambient scalar field . As evolves cosmologically, the effective tidal coupling factor could have been different in the early Earth-Moon system, resolving the age discrepancy without invoking unconstrained parameter changes.

5.10.3 Unification of Persistent Anomalies

The TEP framework provides a single theoretical structure that addresses multiple LLR anomalies that have persisted without satisfactory explanation:

Anomaly	Previous Explanation	TEP Resolution
3.8 cm/yr recession (2.2× historical)	"Tidal resonance" (ad hoc)	Dynamical affects orbital evolution
Tidal dissipation conundrum	"Weaker in past" (no mechanism)	Time-varying from evolution
Synodic sampling bias	"Difficult to extract"	Phase-asymmetry diagnostic (Step 029)
Post-fit residual structure	"Modeling problems"	Physical TEP signals (orthogonal to Keplerian)
1998 ~1 cm residual	Unexplained ( )	Validated (consistency)
Full-moon deficit	Dust + thermal (8.6× mismatch)	Scalar-field activation (perihelion test)

The primary full-systematic detection at , together with secondary Pearson ( ) and cosD-only OLS ( ) diagnostics, its heliocentric scaling and spectral orthogonality properties, motivates a unified TEP reading of these historical anomalies. The sections that follow are not additional detection claims.

5.11 The Secular Eccentricity Anomaly ( )

A particularly robust challenge to standard lunar ephemerides is the unexplained secular increase in orbital eccentricity. Williams & Boggs (2009) reported a residual rate of after accounting for known tidal dissipation in the Earth and Moon. Standard geophysical models, which rely on the tidal quality factor , struggle to reconcile this growth with the observed recession rate; increasing to match the eccentricity gain invariably overestimates the semi-major axis expansion. Consequently, this discrepancy is often categorized as a "modeling gap" in core-mantle coupling.

In the TEP framework, this secular gain is one qualitative reading of a dynamically scaling coupling constant. Unlike standard PPN models where is a static invariant, TEP mandates that the effective coupling scales with the ambient scalar potential, . This environmental modulation (independently validated in Step 022 via the perihelion enhancement) breaks the time-symmetry of the work integral over a closed orbit. The Earth-Moon system accumulates a non-zero net work per anomalistic cycle due to the heliocentric gradient, driving a secular increase in eccentricity. The measured primary parameter is the order of magnitude required by this reading; it is not a separate fitted eccentricity solution.

5.12 The Full Moon Paradox and Eclipse Suppression

A long-standing observational obstacle in LLR is the "Full Moon Curse"—a 10-fold signal deficit near full phase, often with an additional 10-fold loss attributed to "thermal lensing" from dust-obscured retroreflectors (Murphy et al. 2010; Sabhlok et al. 2024). The primary evidence for this thermal interpretation is the "Eclipse Recovery": the rapid return of signal strength when the Moon enters Earth's shadow. Standard models posit that shadowing cools the front faces of the corner-cube prisms, restoring the optical coherence lost to solar-induced temperature gradients.

However, the quantitative mismatch is stark: the Step 026 analysis indicates that worst-case thermal expansion of the Apollo housing is ~1 mm—an order of magnitude too small to explain a 100-fold signal collapse. TEP offers a more fundamental resolution: the Earth's shadow during an eclipse acts as a scalar flux shield. The Earth's bulk (kg) interrupts the primary solar scalar stream, momentously relaxing the Temporal Shear Suppression state within the lunar reflectors. The recovery of signal strength is thus a "suppression-state transition" in the dynamical time field, providing a geometric explanation for the coherence restoration that thermal diffusion alone struggles to satisfy on the observed 15-minute timescales.

5.13 The Secular AU Increase ( )

Astrometric analyses of planetary ephemerides have identified a secular increase in the Astronomical Unit of approximately 7–15 cm/yr (Krasinsky & Brumberg 2004; Standish 2005). Within standard General Relativity, this is often dismissed as a modeling artifact or attributed to solar mass loss. However, solar mass loss ( ) predicts a recession rate of only ~1 cm/yr—leaving more than 85% of the signal unexplained.

The TEP framework reinterprets the AU increase as a direct tracking of the global cosmological evolution of the proper time field, . This is not a "recession" of the Earth from the Sun in the gravitational sense ( ), but rather a consequence of the evolving relationship between the matter metric ( ) and the background geometry. The expansion represents a measurable change in the relationship between atomic measurement standards and the underlying gravitational potential well. TEP thus identifies the observed increase in the AU as a signature of the field's macroscopic dynamics.

5.14 The Synodic Sampling Blind-Spot

A foundational methodological weakness in LLR-based tests of GR is the existence of the "solar avoidance" gap. Because ground stations cannot range to the Moon near New Moon ( ), standard ephemeris fits for are mathematically over-determined by data from the "quiet" quarter-moon phases. Standard solvers thus "dead-reckon" the orbit across the New Moon region based on the assumption of .

This creates a circularity where TEP signatures are naturally absorbed or "dismissed" because they operate most strongly exactly where the data density is zero. This detection of a significant signal in the sparse residuals that persist near New Moon (, Step 025) constitutes a direct falsification of the ephemeris's assumption. By analyzing the residuals of an model precisely where it lacks constraints, the analysis unveils the physical reality that standard multi-parameter fits are structurally blind to.

5.15 Residual Phase Lags in Lunar Libration

A persistent discrepancy in lunar geodesy concerns the "unexplained" phase lags in the Moon's rotation and physical libration. Williams et al. (2001, 2014) identified dissipative signatures larger than predicted for a solid Moon. To resolve this, researchers have invoked a fluid lunar core and complex core-mantle boundary (CMB) interactions. While these models can be tuned to fit the data, they remain sensitive to the assumption of a static gravitational background.

TEP provides a non-dissipative alternative. The synchronization holonomy implies that any body rotating through a non-uniform scalar gradient accumulates a path-dependent time-transport delay. For the Moon, rotating every 27.3 days through the Earth's potential well, this holonomy manifests as a localized "phase lag" in the measurement of its orientation. By reinterpreting these 13.6-day and 27.3-day residuals as holonomy signatures rather than purely internal dissipation, TEP offers a solution that unifies the Moon's rotational dynamics with its orbital Nordtvedt modulation, without requiring ad-hoc geophysical tuning.

5.16 The Solution Uniqueness Challenge: Parameter Masking

The standard LLR analysis pipeline fits up to 100+ parameters simultaneously, including the lunar initial state, reflector coordinates, tidal , and core properties. A critical, yet often unstated, assumption is that the underlying gravitational physics is perfectly described by General Relativity ( ). The analysis suggests that if a genuine TEP signal exists, the multi-parameter fitting process "partitions" the signal's energy across these geophysical degrees of freedom.

This "masking" effect implies that the current "consensus" values for the Moon's interior—such as its core radius or mantle viscosity —may be biased by the accidental absorption of the scalar field's dynamic variance. The TEP framework challenges the uniqueness of these geophysical solutions, asserting that a non-zero is not merely a small correction to the LLR fit, but a foundational shift that requires a re-evaluation of the entire lunar interior model. TEP thus moves the conversation from "fitting residuals" to "redefining the baseline," providing a more robust and physically consistent account of the complete LLR dataset.

5.17 The Spectral "Fingerprint" of Dynamic Coupling

To move the TEP interpretation beyond plausible reinterpretation into the realm of mathematical necessity, the spectral consequences of environmental suppression must be examined. Standard LLR ephemerides analyze the Nordtvedt parameter as a static invariant. However, the TEP framework mandates a coupling that scales with the ambient scalar potential, yield a time-varying parameter , where is the Sun's mean anomaly.

Analytically, the expansion of the periodic range perturbation generates a specific tripartite spectrum:

This "sideband leakage" creates a distinctive spectral fingerprint. Because standard solvers typically parameterize the central line without allocating independent degrees of freedom matched to the sidebands, much of that sideband power remains in the post-fit residuals under static- fits. Residual structure at 32.13 days ( ), in the geophysical quiet space documented in Step 032, is therefore difficult to mimic with standard tidal or thermal templates alone; it motivates the dynamical TEP sideband interpretation pending full integrator refits with and sideband-capable physics enabled at source level.

5.18 Analytical Convergence of Independent Error Paths

The case for TEP is further fortified by the numerical convergence of independent observational channels. In precision metrology, the "gold standard" for a discovery is the extraction of consistent physical parameters from disparate phenomena. The TEP framework demonstrates convergence across multiple astrophysical regimes through Observable Response Coefficients that reflect domain-specific screening and astrophysical amplification:

1. The Orbital Channel: The Nordtvedt parameter from the primary full-systematic model ( ), with cosD-only and leverage-excised checks at and .
2. The Galactic Channel: Preliminary Observable Response Coefficients mag (Cepheids) and – (pulsars) from related work in the same framework reflect the unscreened regime response.
3. The Rotational Channel: Residual phase lags in physical libration (deviation from Williams et al. 2001) provide additional evidence for differential coupling.

The fact that these distinct channels all reveal evidence for TEP effects— albeit with different Observable Response Coefficients reflecting their respective screening environments—suggests internal coherence within the framework. This cross-domain comparison is contextual and does not constitute an independent replication of the pooled LLR synodic estimand.

5.19 Bayesian Evidence Ratios

The strength of the detection can be formalized using Bayesian model comparison. The Bayes Factor ( ) compares the TEP hypothesis (, dynamic suppressed Nordtvedt) against the Null hypothesis (, General Relativity + Tidal resonances). Given the Pearson correlation diagnostic ( ) and the Step 022 perihelion-aphelion cosD-only differential ( ), the evidence ratio is computed via the Savage-Dickey density ratio as:

The nominal KDE estimate is large, but Step 016 records substantial bandwidth sensitivity across the tested kernels (range – ; robustness flag false). The Bayes factor is therefore treated as a secondary diagnostic whose stable component is the posterior exclusion of zero and BIC comparison, not as the headline discovery statistic. The primary quantitative claim remains the full-systematic OLS estimand at .

5.20 Synthesis with Coordinated Lunar Time (LTC)

Scope note: The following LTC discussion is a programmatic implication of the TEP framework and is not part of the pooled synodic detection estimand.

The contemporary relevance of the TEP framework is underscored by the 2024 White House directive to establish "Coordinated Lunar Time" (LTC). Current relativistic models (Ashby & Patla 2024; Turyshev et al. 2024) acknowledge that clocks on the lunar surface tick approximately 58.7 microseconds faster per day than their terrestrial counterparts. While General Relativity attributes this drift to the difference in static gravitational potential, TEP reinterprets the discrepancy as a suppression state transition.

Because the Moon operates in a lower-density regime ( ) and possesses a shallower potential well than Earth, it resides at a different coordinate in the dynamic proper time field . The 58.7 s/day "offset" is the integrated local manifestation of the same synchronization holonomy the analysis has detected in the LLR synodic residuals. By recognizing that LTC is not merely a coordinate correction but an emergent property of the scalar matter-metric, TEP provides the first comprehensive physical justification for the world’s new cislunar navigation standards. The establishment of LTC constitutes the first involuntary operational implementation of the Temporal Equivalence Principle in human civilization.

5.21 Causal Safety and the GW170817 Constraint

A rigorous covariant theory must respect the multi-messenger constraints of modern astrophysics, including the arrival-time coincidence of GW170817 and GRB 170817A. This measurement enforced a world-leading bound on the fractional difference between the speed of gravity ( ) and light ( ): . Many scalar-tensor theories were discarded because their disformal couplings ( ) predicted a varying that contradicted this observation.

TEP preserves causal safety via its conformal Temporal Shear Suppression mechanism. In the late-universe limit and on cosmological distance scales where the scalar potential is uniform, the disformal term in the matter metric is naturally suppressed. Because gravitational waves and photons share the same null geodesics of the background in suppressed environments, the TEP framework inherently satisfies the GW170817 bound. The detected LLR anomalies, which arise from high-order directional gradients near local potential wells, represent "near-field" modulations that do not propagate into the far-field speed-of-gravity constraints.

5.22 Cross-Domain Universality of Observable Response Coefficients

The Jakarta v0.8 framework introduces Observable Response Coefficients ( ) that quantify domain-specific astrophysical responses while preserving theoretical unity. Different environments exhibit different effective responses due to varying degrees of Temporal Shear suppression and astrophysical amplification. Table 4 summarizes this internal consistency across the author's body of work.

The screening ratio quantitatively demonstrates that LLR operates in a strongly screened Solar System regime, while the preliminary galactic coefficients from related work (Papers 10 and 11) reflect weakly screened environments. This systematic variation with environmental density is a predicted feature of the Temporal Shear Suppression mechanism: the same microscopic coupling (constrained by Cassini to) produces different observable responses depending on the screening environment, exactly as predicted by the TSS mechanism.

The internal consistency across the framework—from Solar System ephemerides to galactic stellar dynamics—motivates the TEP framework as a coherent cross-domain hypothesis. The fact that LLR, Cepheids, pulsars, and wide binaries all show TEP-compatible signals, albeit with different Observable Response Coefficients reflecting their respective screening environments, is contextual support rather than an independent replication of the pooled LLR synodic estimand.

System / Observatory	Primary Observable	Observable Response Coefficient	Statistical Significance
LLR (This Work)	Synodic Nordtvedt Modulation	(Solar System screened regime)	(full-systematic primary)
Cepheids	Period-Luminosity Distance Bias	mag (preliminary)	Planck tension resolution
Globular Cluster Pulsars	Spin-Down Excess	– (preliminary)	
GNSS Timing	Synchronization Holonomy ( )	Screened regime response	Independent Signatures
Gaia DR3 Wide Binaries	Wide Binary Dynamics		Internal Consistency Only

5.23 The Path to Falsification: The Triangle Test

In accordance with the highest academic standards, the manuscript concludes by defining the prerequisite for falsifying the TEP-LLR results. While LLR provides a powerful probe of orbital dynamics, the definitive experimental test for synchronization holonomy is the triangle test. This proposed experiment involves three atomic clocks on a closed, multi-leg time-transfer loop (e.g., at the 1,000–3,000 km scale). Standard General Relativity predicts that the residual holonomy around the closed loop, after subtracting all known relativistic effects, should be zero.

TEP predicts a non-zero, invariant measure of non-integrability at the fractional level. Independent verification through precision clock networks would provide falsification or confirmation of the current TEP interpretation of LLR anomalies. Detection of the predicted holonomy at the anticipated magnitude would provide independent confirmation of the TEP interpretation.

5.24 Statistical Hardening and Birge Ratio Scaling

The pipeline computes the Birge Ratio for each regression (Step 003). When , formal uncertainties are scaled upward by ; when , no downward rescaling is applied. For the primary full-systematic model, and , so the headline detection is not Birge-inflated. The conservative over-dispersion policy remains available for regressions with .

6. Conclusion

The analysis of the 35-year Lunar Laser Ranging (LLR) record reveals a synodic modulation in the post-fit residuals consistent with the Temporal Equivalence Principle. The primary estimand is the pooled full-systematic model (cosD + annual + monthly + thermal): (; cluster-robust). The signal strengthens from (cosD-only) to (full model), consistent with a synodic component diluted by unmodeled annual, monthly, and thermal aliases. Section 4.0 consolidates the independent synodic spine; the sections that follow develop station pooling, regression detail, systematics, dynamical modulation, and orthogonality without re-opening the headline claim.

Station pooling with proper systematic control supports a common negative signal: a mixed model with station-specific systematics yields (; ,). Step 056 recovers the same negative by linearized post-fit extraction on the published INPOP19a and DE430 archives (and). Full-model AR(1) GLS ( ) remains consistent with the primary result after temporal autocorrelation is addressed.

Mandatory external closure now defines the decisive falsification arc beyond the pooled residual channel: source-level INPOP or DE430 integrator refits with free, together with independent range-level reductions on archived CRD normal points, must reproduce or refute the synodic amplitude extracted here. Frequency-domain orthogonality tests and the ephemeris-absorption simulation (Steps 032, 041, 054) indicate why that channel is structurally favored for this TEP component when static Keplerian solvers assume ; they do not replace those integrator-level checks.

Conditioned on the primary synodic fit, joint regressions on the same residual channel (Steps 047–048, 055) provide strong corroborative CMB orientation evidence (,) while heliocentric distance is non-significant ( ). Step 055 excludes aliasing, multicollinearity, and permutation artifacts; directional rotations anchor the joint model to the CMB dipole, while conservative sky-scramble specificity remains marginal ( ). Section 4.24.2 reports dipole antisymmetry, multipole hierarchy, and bootstrap stability for this extension. These dynamical results reinforce the same scalar-field architecture as the primary estimand; they are corroborative structure within the residual channel rather than a replacement numerical estimator.

These results provide substantial empirical support for a dynamical proper time field in the sense of a nonzero synodic residual-channel Nordtvedt parameter. Within the TEP framework, this elevates invariance statements traditionally treated as global postulates—most notably the universality of light propagation—from axioms to emergent local constraints of the field equations: the universal speed of light becomes a local theorem of the dynamical time-field environment rather than an externally imposed global invariant. The same synodic footprint supplies the physical anchor for the Müller & Nordtvedt (1998) post-model residual, for long-standing LLR anomaly threads developed in Section 5, and for the scalar-field interpretation of coordinated lunar-time standards discussed in Section 5.20.

6.1 Synthesis of Results

Section 4.0 and the read-order guide in Section 4 consolidate the synodic spine before station critiques, multiplicity control, and extended diagnostics (Sections 4.20–4.31, especially Section 4.15 for robustness and Section 4.3 for balance stress tests). Universal sign and pooling are assessed jointly through the common- mixed model (,), not through equal per-station thresholds.

DE430 (short baseline, outlier sensitivity) is a supplementary cross-ephemeris strand, while the CMB-oriented joint layer is treated as strong corroborative dynamical evidence whose most conservative sky-scramble specificity is . Neither reopens the headline pooled estimand when read in the order of Section 4.

6.2 Interpretative Framework

The extracted limit of resolves the theoretical suppression ambiguity in TEP. The negative sign supports the interpretation that gravitational potential scaling (Temporal Shear suppression) dominates over soliton-radius scaling in the Earth-Moon system.

The Observable Response Coefficient framework shows internal consistency across the program: preliminary mag from Cepheid stellar dynamics and – for globular cluster pulsars from related work. The LLR Nordtvedt parameter serves as the Solar System observable response coefficient, exhibiting a substantially smaller magnitude consistent with stronger screening in dense environments. A complementary weak-field test reports a saturation amplitude in Gaia DR3 wide binaries, reflecting the specific dynamics of that regime.

While TEP does not make a precise quantitative prediction for from a single universal coupling constant due to suppression model uncertainties and domain-specific astrophysical amplification, the observed Nordtvedt variance is within the order-of-magnitude parameter range expected from the formal TEP framework (to) and provides qualitative geometric parity across stellar dynamics, high-z galaxies, and planetary ephemerides.

The Temporal Shear Suppression mechanism operates through the continuous spatial profile of the time field (Temporal Topology), where Earth's substantially deeper compactness ( ) relative to the Moon ( ) suppresses

Temporal Shear more strongly, naturally producing the observed differential coupling and the measured within this framework.

The detection of a non-zero has implications for gravity theories. In General Relativity, exactly. The result therefore demands a decisive fork in interpretation under external closure tests: either (1) a modification to GR in the form of a scalar-tensor theory such as TEP, or (2) a residual-channel artefact introduced by unmodeled systematics in the LLR reduction and ephemeris processing chain. TEP provides a concrete scalar-tensor framework that explains the observed signal while remaining consistent with all other precision tests through TEP geometric suppression and Yukawa suppression mechanisms.

6.3 Reproducibility

This analysis is designed to be fully reproducible. The public repository contains the end-to-end analysis code needed to regenerate the manuscript tables, figures, and archived outputs; execution instructions are provided in the repository README.

6.4 Data Availability

The manuscript source, complete analysis code, generated figures, intermediate outputs, and the raw and processed data are available on GitHub and archived on Zenodo for long-term reproducibility.

- Analysis repository: complete analysis code, reproducible outputs, and build instructions.
- Input data: INPOP19a residuals from IMCCE — publicly available through the Paris Observatory.
- Processed outputs: All intermediate and final data products (`interim/` , `outputs/` , `figures/`) are version-controlled and reproducible from the input data.
- Documentation: `README.md` provides installation instructions, a dependency list (`requirements.txt`), and a quick-start guide.

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Additional papers in the TEP program (Papers 2–5, 7–8, 11–13) are catalogued in the Supplementary Information (Section 8).

Data Availability & Reproducibility

This work follows open-science practices. All empirical results are fully reproducible from raw data using the documented pipeline. Step 000 verifies the five INPOP19a station residual files and the DE430 cross-ephemeris residual file against `data/raw/data_manifest.json` SHA-256 checksums before any downstream analysis runs. Monte Carlo, injection, and toy-model steps are labeled in their JSON outputs and are not treated as independent observational detections.

Repository & Code

The repository contains a deterministic, version-controlled analysis pipeline with analysis steps for LLR data processing and statistical analysis. All canonical steps are orchestrated by `scripts/steps/run_all_steps.py` with comprehensive logging.

Repository Structure

```
TEP-LLR/
├── data/
│   ├── raw/                    # Raw LLR residuals (INPOP19a, DE430)
│   ├── interim/                # Intermediate processing results
│   └── processed/              # Final processed datasets
├── scripts/
│   ├── steps/                  # Analysis pipeline steps and run_all_steps.py
│   └── utils/                  # Utility functions (parsing, plotting)
├── results/
│   └── outputs/                # JSON/CSV analytical outputs
├── logs/                       # Per-step execution logs
├── site/
│   └── components/             # Manuscript HTML sections
├── requirements.txt            # Python dependencies
└── README.md                   # Documentation
```

Data Provenance

Data Source	Provider	Access Method	Size	Location
INPOP19a Residuals	IMCCE (Paris Observatory)	Public download	~2 MB	`data/raw/`

Data Source	Provider	Access Method	Size	Location
DE430 Residuals	JPL	Public download	~0.5 MB	`data/raw/`
APOLLO Data	Apache Point Observatory	Public download	~0.3 MB	`data/raw/apollo11/`, `apollo14/`, `apollo15/`

Pipeline Architecture

The analysis pipeline comprises 57 sequential steps organized into logical groups. Each step is a standalone Python script in `scripts/steps/` that produces JSON outputs and detailed logs in `logs/step_*.log`. Manuscript step labels in Results and Discussion follow the `step_XXX` script prefix wherever possible (e.g. Step 050 `step_050_corrected_tep_analysis.py`). Where a script's internal docstring uses a different ordinal, the manuscript cites the script prefix and the Results subsection given in the mapping table below.

Complete Step Inventory & Runtime

The canonical pipeline comprises 57 sequential steps organized into logical groups. The core statistical analysis (steps 000–003) provides the primary detection, while steps 004–055 perform extended systematic checks, robustness validation, and theoretical consistency tests. Recent enhancements include:

- Step 004: Temporal autocorrelation analysis with Durbin-Watson statistic and significance testing
- Step 008: Data-driven systematic error budget computed directly from residuals (cross-ephemeris scatter, seasonal variation, station means, tidal harmonics, diurnal amplitude)
- Outlier detection: Enhanced documentation of MAD threshold with methodological justification and statistical rationale

Manuscript Step ↔ Pipeline Script Map

Selected analyses where the manuscript label, script prefix, and internal docstring diverge:

Manuscript step	Pipeline script	Primary JSON	Re se
010–013	`step_010` – `step_013`	`step_010` – `step_013` outputs	§4 4.1
029 (day/night)	`step_027_day_night_thermal_bias.py`	`step_027_day_night_thermal_bias.json`	§4
029 (station power / WLS)	`step_029_station_power_analysis.py`	`step_029_station_power_analysis.json`	§4 4.2
030 (geometric elongation)	`step_028_geometric_elongation.py`	`step_028_geometric_elongation.json`	§4 (S 03
032 (hardware epochs)	`step_030_hardware_epoch_analysis.py`	`step_030_hardware_epoch_analysis.json`	§4
034 (orthogonality)	`step_032_ephemeris_orthogonality_proof.py`	`step_032_ephemeris_orthogonality_proof.json`	§4
045 (matched-window)	`step_045_independent_validation.py`	`step_045_independent_validation.json`	§4 §5
050 (primary extraction)	`step_050_corrected_tep_analysis.py`	`step_050_corrected_tep_analysis.json`	§4
055 (CMB falsification)	`step_055_cmb_rigorous_falsification.py`	`step_055_cmb_rigorous_falsification.json`	§4 / §
056 (linearized post-fit extraction)	`step_056_dynamical_integrator_eta_refit.py`	`step_056_dynamical_integrator_eta_refit.json`	§4

Step	Script	Description	Runtime
000	`step_000_llr_data_ingestion.py`	Downloads and parses INPOP19a residual data from Paris Observatory	~0.2s
001	`step_001_data_preprocessing.py`	Computes Moon-Sun elongation, synodic phase, and performs data quality validation	~8.9s
002	`step_002_de430_preprocessing.py`	DE430 ephemeris preprocessing and comparison	~0.9s
003	`step_003_statistical_analysis.py`	Computes Pearson correlation, linear regression, and differential analysis	~1.2s
004	`step_004_detection_analysis_advanced.py`	Advanced robust analysis with multiple independent analysis families (bootstrap, permutation, robust regression, outlier detection, station-by-station, temporal stability, phase-binned, systematic error modeling, sensitivity, and cross-validation)	~5.4s
006b	`step_006b_de430_outlier_robustness.py`	DE430 outlier robustness: threshold sweep (– MAD), phase-bin chi-square, bootstrap CI, permutation test	~15s
005–039	`step_005_*.py` through `step_039_*.py`	Extended validation, systematic controls, physical probes, and defensibility analyses through dust sensitivity	~1m 20s total
040–043	`step_040_*.py` through `step_043_*.py`	Unified results table, ephemeris absorption simulation, multiple-testing correction, temporal-bin variation	~25s total
044–055	`step_044_*.py` through `step_056_*.py`	Systematic projection, independent validation, station-balanced TEP, velocity modulation, CMB anisotropy, EOP systematics, corrected TEP primary extraction, cross-validation, station distribution, clean subset, toy orbital perturbation, CMB falsification suite, and linearized dynamical integrator refit	~2m total

Total Runtime Summary

Component	Steps	Runtime
Core Statistical Analysis	4 (000-003)	~25.6s
Extended Systematic Analysis	52 (004–055, including 006b)	~6m 27s
Total	**57**	**~6m 53s**

Reproduction Instructions

Quick Start (Full Reproduction)

```
# 1. Clone repository
git clone https://github.com/matthewsmawfield/TEP-LLR.git
cd TEP-LLR

# 2. Install dependencies
pip install -r requirements.txt

# 3. Run full pipeline (generates JSON outputs and logs)
python scripts/steps/run_all_steps.py

# 4. Results will be in:
#    - results/outputs/   (JSON analytical outputs)
#    - logs/              (Detailed execution logs)
#    - site/dist/         (Built manuscript site)

# 5. Regenerate manuscript Markdown and static site from site/components/
node site/build.js
```

System Requirements

Component	Minimum	Recommended
CPU	2 cores	8+ cores (for multiprocessing)
RAM	4 GB	8 GB
Storage	100 MB	500 MB
Runtime	~7m	~6m 53s (with multiprocessing)

Key Analysis Outputs

- `results/outputs/step_003_statistical_analysis.json` — Pearson correlation, linear regression, and differential analysis results
- `results/outputs/step_004_detection_analysis_advanced.json` — Comprehensive analysis with bootstrap, permutation, robust regression, outlier detection, station-by-station, temporal stability, phase-binned, systematic error modeling, sensitivity, and cross-validation
- `results/outputs/step_001_data_preprocessing.json` — Data preprocessing statistics and validation results
- `data/processed/INPOP19a_all_stations_residuals.csv` — Processed INPOP19a dataset with computed elongation phases
- `data/processed/[station]_residuals.csv` — Individual station processed datasets

Log Files

Each step produces detailed logs:

- `logs/step_000_llr_data_ingestion.log` — Data ingestion log
- `logs/step_001_data_preprocessing.log` — Data preprocessing and validation log
- `logs/step_002_de430_preprocessing.log` — DE430 ephemeris preprocessing log
- `logs/step_003_statistical_analysis.log` — Statistical analysis log
- `logs/step_004_detection_analysis_advanced.log` — Advanced 13-method analysis log

Software Dependencies

Package	Version	Purpose
Python	3.8+	Language runtime
NumPy	1.20+	Numerical computing
SciPy	1.5+	Statistical functions
Pandas	1.2+	Data manipulation
Matplotlib	3.3+	Visualization

All dependencies are specified in `requirements.txt`.

Validation & Testing

The pipeline includes comprehensive validation:

- Data Integrity: Checks for missing or malformed residuals
- Phase Consistency: Verifies synodic phase calculations against ephemeris data
- Station Independence: Validates that results are consistent across independent observatories
- Cross-Ephemeris Validation: DE430 dataset from JPL shows complex behavior; full dataset shows no significant correlation (,), but after removing 37 phase-specific outliers (0.8% of data) that cluster at – elongation, yields at with bootstrap validation (95% CI on :) and permutation test ( ). Chi-square test confirms outliers are not uniformly distributed across phases (,), indicating genuine measurement errors at specific phases. The primary detection relies on INPOP19a ephemeris.

Reproducibility Checklist

To verify successful reproduction:

1. All 63 canonical steps complete with "PASS" status in pipeline output
2. JSON files in `results/outputs/` (step_000 through step_061, including 006b)
3. Key result: Primary full-systematic OLS ( ); cluster-robust
4. Key result: Pearson correlation on the 6σ-cleaned primary sample ( ): ( ),
5. Key result: Cook's Distance leverage-excised robustness check (SNR =)
6. Key result: Naïve OLS robustness check
7. Key result: MCMC robustness check (SNR =)
8. Key result: Full-model AR(1) GLS robustness check (SNR =)
9. Key result: cosD-only AR(1) GLS robustness check (SNR = cluster-robust)
10. Key result: AR(1) parameter (full model), (cosD-only); significant temporal autocorrelation
11. Key result: Precision-weighted robustness check (SNR =)
12. Key result: Bootstrap 95% CI = [-0.0411, -0.0197] (excludes zero)
13. Key result: Permutation test $p < 10^{-4}$ (no permutation achieved observed)
14. Key result: Station consistency: Grasse shows strongest individual detection (SNR =), all stations show consistent negative direction except Haleakala
15. Key result: Cross-validation (Step 051): Predictive is negative in temporal and station-stratified splits due to severe covariate shift (KS on predictor distribution); random 5-fold CV achieves for the full-systematic model. No coefficient instability (for epoch-dependence test)

Supplementary Information: TEP Program Series Context

The Temporal Equivalence Principle (TEP) program comprises multiple peer-reviewed and preprint investigations testing the same theoretical framework across distinct observational channels. The present paper (Paper 17) is self-contained; the table below is provided for readers wishing to trace the full program.

Paper	Title / Focus	Role relative to Paper 17
0	Temporal Equivalence Principle: Dynamic Time & Emergent Light Speed. Framework, axioms, two-metric structure, and falsifiability.	Defines the scalar-tensor formalism, conformal/disformal sectors, and screening ontology.
1–3	Global Time Echoes (GNSS clock-network correlations, 25-year CODE analysis, raw-RINEX consistency).	Empirical motivation for treating precision timing residuals as probes of Temporal Topology.
4	Temporal-Spatial Coupling in Gravitational Lensing (dark-matter reinterpretation).	Demonstrates TEP's expression in lensing observables.
5	Global Time Echoes: Empirical Synthesis (GNSS phenomenology consolidation).	Argues Solar System compatibility with suppressed Temporal Shear.
6	Universal Critical Density (cross-scale consistency of $\rho_T \approx 20$ g/cm³).	Supplies the screening saturation scale used in Section 2.3.
7	The Soliton Wake (RBH-1 as a temporal topology candidate).	Astrophysical application of the ρ_T framework.
8	Optical-Domain Consistency Test via Satellite Laser Ranging.	Complementary two-way optical ranging at terrestrial baselines.
9	What Do Precision Tests of GR Actually Measure? (measurement taxonomy).	Distinguishes direct-fit, residual-channel, and clock-sector observables.
10	Suppressed Density Scaling in Globular Cluster Pulsars.	Galactic-scale Observable Response Coefficient (unscreened regime).
11	The Cepheid Bias (Hubble tension resolution).	Galactic-scale Observable Response Coefficient (unscreened regime).
12	JWST High-Redshift Anomalies (unified resolution).	Cosmological application of TEP.
13	Temporal Shear Recovery in Gaia DR3 Wide Binaries.	Weak-field, weakly screened complement to the compactness-suppressed LLR test.
17	LLR Tests of the Strong Equivalence Principle (this work).	Screened Solar-System Nordtvedt test using INPOP19a and DE430 residuals.